Combinatorial and Algebraic Aspects of Polynomials over Finite Fields

by

Daniel Nelson Panario Rodriguez

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Abstract

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This thesis investigates several algebraic algorithms that deal with univariate polynomials over finite fields. Our main focus is the factorization problem, but we also consider polynomial irreducibility tests, and constructions of both irreducible polynomials and finite fields. A methodology based on generating functions and their asymptotic analysis allows us to study the behavior of the algorithms in question. This analysis reveals parameters of intrinsic interest for understanding the behavior of the algorithms. Moreover, they provide insight into the structure of polynomials over finite fields in a broader sense.

First, we summarize and extend known results on counting random polynomials. We investigate properties such as the average number of irreducible factors whose degrees lie in an interval, the average number of polynomials with no factors or with all factors having degrees in an interval, among others. We briefly state problems where these results give useful information. We also survey the known algorithms for factoring polynomials.

Second, we give the first average-case analysis of a complete polynomial factorization algorithm. We fully analyze the classical method for factoring polynomials over finite fields consisting of three stages: squarefree factorization, distinct-degree factorization, and equal-degree factorization, including some variants for each stage. Given a partition

of the possible degrees of the irreducible factors of a polynomial, we study how they split into subintervals. This naturally relates with recent factorization algorithms. It turns out that more computational work is needed when a subinterval contains the degrees of more than one irreducible factor. We study several properties of such intervals.

Finally, the methodology is applied to other problems involving polynomials over finite fields. We give two variations of existing algorithms for testing the irreducibility of polynomials. In addition, we provide a construction of very sparse irreducible polynomials. We conclude by improving lower bounds for the Euler function for polynomials and for the density of normal elements.

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Chapter 1

Introduction

1.1 Overview of the thesis

This thesis deals with algebraic algorithms that work with univariate polynomials over finite fields. The problems we are interested in include polynomial factorization, irreducibility tests, and constructions of both irreducible polynomials and finite fields. The main goal of this research is to analyze and develop efficient algebraic algorithms that work with polynomials over finite fields. By efficient algorithms we mean fast methods in both theoretical and practical terms. We consider both average-case analysis and worst-case analysis for these efficient algorithms.

The main contribution of the thesis is the adaptation of a known combinatorial framework to the analysis of algorithms for polynomials over finite fields. This framework, based on generating functions and their asymptotic analysis, allows us to analyze the behavior of the algorithms in question. A complete analysis of algorithms for polynomials reveals parameters that are of intrinsic interest for understanding polynomials over finite fields. Such a task has not been undertaken systematically before, except for (mostly worst-case) big-Oh bounds. We systematically apply this methodology and achieve two types of results. On the one hand, results of more mathematical interest give information on the structure of random polynomials over finite fields. On the other hand, results of more computational flavor explain the behavior of the algorithms on random inputs.

We summarize the results of the thesis at a more concrete level. In Chapter 2 we

revisit and generalize most of the known results for random polynomials over finite fields. This shows the potential and succinctness of our methodology. We study properties of polynomials of degree n with the degrees of some of its irreducible factors lying in a fixed interval (independent of n) contained in $[1, n]$. We investigate properties such as the average number of irreducible factors whose degrees lie in an interval, the average number of polynomials with no factors or with all factors having degrees in an interval, among others. We briefly state problems where these results give useful information. Some of the results in this section are known in terms of number-theoretical functions. We give new proofs for them that rely only on the size of the field, and on the degree of the polynomials.

A central topic of this thesis is the factorization of polynomials over finite fields. Chapter 3 gives a survey on this problem. Several of the algorithms presented in this survey are studied in the chapters that follow. The main topics of this chapter are contained in von zur Gathen & Panario (1996).

The average-case analysis of polynomial factorization algorithms is presented in Chapter 4. We fully analyze the classical method for factoring polynomials over finite fields consisting of three stages: squarefree factorization, distinct-degree factorization, and equal-degree factorization (see the definitions in Chapter 3). We give the probability that a random polynomial has no repeated factors, and the expected total degree of its repeated factors. From this we deduce that the cost of the first stage, squarefree factorization, is dominated by the computation of a greatest common divisor (gcd) of polynomials. We fully analyze the second stage, distinct-degree factorization, proving that it dominates the total cost of the factorization algorithm. We then study the probability that, after this stage, the polynomial has been completely factored. We also show that the expected degree of the remaining reducible factor (the input for the third stage) is asymptotic to $\log n$, as $n \to \infty$. Finally, the third stage, equal-degree factorization, is analyzed by means of a refinement process similar to the analysis of digital trees. For each stage, we include the analysis of several variants. The main contribution of this chapter is a framework for the average case analysis of a general factorization algorithm. An extended abstract containing the main results of this chapter appeared in Flajolet,

Gourdon & Panario (1996).

In order to reduce the cost of the distinct-degree factorization stage, factorization algorithms of the 1990's perform this task in the following three steps: break the interval $[1, n]$ into several subintervals; isolate the factor containing the product of all the irreducible factors with degrees in each subinterval; then, perform a distinct-degree factorization for each of these products. This process results in a complete distinct-degree factorization (see the algorithms of Chapter 3). To understand the behavior of these algorithms and to analyze them, one has to study how the degrees of the irreducible factors of a polynomial split into given subintervals. This is the focus of Chapter 5. It turns out that more computational work is needed when a subinterval contains the degrees of more than one irreducible factor, since the corresponding product of these irreducible factors requires a nontrivial distinct-degree factorization. We call such an interval a *multi-factor* interval for that polynomial. We study properties such as the probability of an interval being multi-factor for a polynomial, the average number of multi-factor intervals for a polynomial, the average number of irreducible factors of a polynomial whose degrees lie in any of its multi-factor intervals, the expected total degree of irreducible factors of a polynomial whose degrees lie in any of its multi-factor intervals, among others. We include a discussion of the relationship between these properties and recent factorization algorithms.

In Chapter 6, we turn our attention to other problems, also involving polynomials over finite fields, that share an underlying problem, namely the construction of extension fields. A fundamental problem in finite fields, arising in the construction of extension fields but also being of independent interest, is finding irreducible polynomials. We give two variations of existing algorithms for testing the irreducibility of polynomials. We present an experimental comparison of several methods for testing irreducibility. The most important contribution of this section is an efficient algorithm for the problem, and a theoretical study of the algorithm's behavior. The latter includes the study of the probability that a random polynomial of degree n has no irreducible factors of degree up to $O(\log n)$ (this extends results of Chapter 2). In addition, we provide a construction of infinite families of very sparse irreducible polynomials. Then we focus on the problem

of finding normal elements in a finite field. We improve the lower bound for the Euler totient function for polynomials. This is the analogue of the Euler totient function, $\phi(n)$, for integers, i.e., for a monic polynomial f, $\Phi(f)$ is the number of polynomials of smaller degree than the degree of f that are relatively prime to f. This lower bound is useful when solving systems of linear equations over finite fields. Moreover, in the case $f = x^n - 1$, $\Phi(f)$ provides the density of normal elements. The existence of these elements permits another way of constructing extension fields. The main result of this section is a positive lower bound on the density of normal elements for infinitely many fields. This ensures that only a constant number of samplings is required in randomized constructions of normal elements, and hence, extension fields. A large part of the results in this chapter are taken from the joint papers Gao & Panario (1995, 1997).

Finally, Chapter 7 states several directions for further research, including possible improvements to the problems studied in this thesis, as well as different problems for which our general framework may be useful.

The rest of this chapter is organized as follows. In Section 1.2, we briefly discuss bases for finite fields, and present the costs of computing several arithmetical operations in finite fields under these bases. The algebraic algorithms in this thesis use as elementary steps basic polynomial operations such as products, divisions, gcds, powers of one polynomial modulo another, etc. These costs are therefore needed for the analysis of the algorithms. In Section 1.3, we present the methodology that allows a unified treatment of probabilistic properties (mean, variance, distribution) of important parameters related to random polynomials over finite fields. We make extensive use of this methodology throughout the thesis. Finally, in Section 1.4 we state several basic mathematical results needed in later chapters.

We conclude this section with important comments regarding attribution of results to their originators. Throughout this thesis, we only give proofs for theorems that are either original or are new versions of known results. Results from other authors are stated where they are needed with a reference to their proofs.

We assume basic knowledge of finite fields. However, when needed, we state without proof results about them. Our main reference is the excellent book by Lidl & Niederreiter

(1983). Other very good sources in finite fields are Berlekamp (1968), McEliece (1987), Jungnickel (1993), and Menezes *et al.* (1993).

1.2 Arithmetic in finite fields

For a prime power q and an integer $n \geq 2$, let $\mathbb{F}_q$ and $\mathbb{F}_{q^n}$ be the finite fields with q and q^n elements, respectively. We use the number of operations in $\mathbb{F}_q$ as the cost measure of an algorithm. We assume that the arithmetic in $\mathbb{F}_q$ is given. In general, we use "big-Oh" notation to represent the asymptotic behavior of functions. Sometimes, we will use the "soft O" notation to ignore logarithmic factors: $g = \tilde{O}(n)$ means that $g = O(n(\log n)^\ell)$ for some constant ℓ.

The extension field $\mathbb{F}_{q^n}$ of degree n over $\mathbb{F}_q$ can be viewed as a vector space of dimension n over the field $\mathbb{F}_q$. Two well-known general approaches for producing extension fields are the polynomial-basis construction, and the normal-basis construction. The following sections discuss the cost of computing several basic polynomial operations such as multiplication, division, and exponentiation under both bases.

1.2.1 Polynomial bases

Let $f \in \mathbb{F}_q[x]$ be an irreducible polynomial of degree n, i.e., a polynomial that cannot be written as a nontrivial product of two polynomials in $\mathbb{F}_q[x]$ of lower degrees. In this case, the ring of polynomials modulo f, $\mathbb{F}_q[x]/(f)$, is a finite field with q^n elements. The theorem of existence and uniqueness of finite fields ensures that for every positive integer n and every finite field $\mathbb{F}_q$, there is a unique (up to isomorphism) extension $\mathbb{F}_{q^n}$ of $\mathbb{F}_q$ with q^n elements. Therefore, $\mathbb{F}_{q^n} \cong \mathbb{F}_q[x]/(f)$. A *polynomial basis* for $\mathbb{F}_{q^n}$ over $\mathbb{F}_q$ is a basis of the form $\{1, x, x^2, \cdots, x^{n-1}\}$. To implement arithmetic in $\mathbb{F}_{q^n}$, one can use the ring of polynomials over $\mathbb{F}_q$ with arithmetic modulo f. Indeed, operations involving elements in $\mathbb{F}_{q^n}$ can be computed via their associated polynomials modulo f. Therefore, finding irreducible polynomials in finite fields is a crucial step in the construction of polynomial basis.

In polynomial bases, a multiplication of two polynomials of degree at most n can be

done in $O(n^2)$ operations in $\mathbb{F}_q$ using "classical" arithmetic, or in $O(n \log n \log\log n)$ operations in $\mathbb{F}_q$ using "fast" arithmetic (Schönhage & Strassen 1971, Schönhage 1977, Cantor & Kaltofen 1991). A division with remainder can be computed with $O(n^2)$ operations in $\mathbb{F}_q$ using classical methods, or with $O(n \log^2 n \log\log n)$ operations in $\mathbb{F}_q$ using Fast Fourier Transform (FFT) methods. The cost of computing a gcd of two polynomials of degree at most n is $O(n^2)$ operations in $\mathbb{F}_q$ using classical methods, or $O(n \log^2 n \log\log n)$ operations in $\mathbb{F}_q$ using FFT methods (Aho *et al.* 1974, §8.9). An operation that is needed in many algorithms of this thesis is the computation of $h^q \bmod g$ for polynomials h and g of degree at most n. This exponentiation $h^q \bmod g$ can be done by means of the classical *repeated squaring* method (Knuth 1981, p. 441–442). The cost of computing $h^q \bmod g$ by this method is $O(\log q)$ polynomial products, i.e., $O(n^2 \log q)$ operations in $\mathbb{F}_q$ using straightforward methods, or $O(n \log q \log n \log\log n)$ operations in $\mathbb{F}_q$ using FFT based methods. In practical applications, an important operation is the exponentiation α^e for $\alpha \in \mathbb{F}_{q^n}$ and $e \in \mathbb{N}$. First, by Fermat's little theorem (see Theorem 1.4.3), $\alpha^{q^n} = \alpha$, so we may assume $e < q^n$. Then, the cost of the repeated squaring algorithm is $O(n^3 \log q)$ operations in $\mathbb{F}_q$ using straightforward methods, or $O(n^2 \log q \log n \log\log n)$ operations in $\mathbb{F}_q$ using FFT-based methods. Brickell *et al.* (1992) show that in a polynomial basis, exponentiation can be computed with $O(n/\log n)$ multiplications in $\mathbb{F}_{q^n}$, and thus $O(n^2 \log\log n)$ operations in $\mathbb{F}_q$. Their algorithm seems to require a preprocessing stage of about n multiplications in $\mathbb{F}_{q^n}$, and, more important from a practical point of view, storage for $O(n/\log n)$ elements from $\mathbb{F}_{q^n}$.

It was widely believed that fast polynomial arithmetic was not practical for computer algebra problems; however, Shoup (1993) showed that this is not true. Indeed, his experiments give a crossover point for the superiority of fast arithmetic, as polynomials of degree 25 modulo a 100-bit prime (see also Shoup 1996). However, no comparison between fast methods and the algorithm of Karatsuba & Ofman (1962) seems to have been done for a general field $\mathbb{F}_q$. For comparisons over $\mathbb{F}_2$, see Reischert (1995) and von zur Gathen & Gerhard (1996).

Let ω be an achievable exponent for matrix multiplication, so that we can multiply two $n \times n$ matrices with $O(n^\omega)$ operations in $\mathbb{F}_q$. Then systems of linear equations can

be solved in $O(n^\omega)$ operations in $\mathbb{F}_q$. Classical linear algebra methods yield $\omega = 3$, and the current record is $\omega < 2.376$ (Coppersmith & Winograd 1990).

1.2.2 Normal bases

An element $\alpha \in \mathbb{F}_{q^n}$ is *normal* over $\mathbb{F}_q$ if and only if its conjugates $\alpha, \alpha^q, \ldots, \alpha^{q^{n-1}}$ are linearly independent over $\mathbb{F}_q$. When α is normal, $(\alpha, \alpha^q, \ldots, \alpha^{q^{n-1}})$ is a basis, and it is called the *normal basis* generated by α. Hensel (1888) proves that normal elements exist for any field and any degree of extension (see also Schönemann 1846, and Eisenstein 1850).

An important property of a normal basis representing $\mathbb{F}_{q^n}$ is that the qth power of an element is just a cyclic shift of its coordinates. Agnew *et al.* (1988), Stinson (1990), and von zur Gathen (1991) prove that, using any normal basis, exponentiation can be computed with $O(n/\log_q n)$ multiplications in $\mathbb{F}_{q^n}$, and with storage for $O(n/\log_q^2 n)$ elements of $\mathbb{F}_{q^n}$. Therefore, normal bases have some advantage over polynomial bases when computing exponentiations. Since this operation is fundamental in applications such as cryptography, many recent hardware implementations of finite fields use normal basis (Massey & Omura 1981, Onyszchuk *et al.* 1988, Agnew *et al.* 1991, 1993). The question is how to perform multiplication in a fast way using normal bases.

Using normal bases, a multiplication in $\mathbb{F}_{q^n}$ can be computed with $O(n^3)$ operations in $\mathbb{F}_q$ using a naive approach, or $O(n^2)$ operations in $\mathbb{F}_q$ using optimal normal bases (Mullin *et al.* 1989). A division operation can be performed in $O(n^3)$ operations in $\mathbb{F}_q$, and exponentiation in $O(n^2 \log\log n)$ operations in $\mathbb{F}_q$ (von zur Gathen 1991).

Gao, von zur Gathen & Panario (1995a, 1995b) prove that multiplication and division in $\mathbb{F}_{q^n}$ when using normal bases can be computed within the same time as when using polynomial bases. More precisely, multiplication in $\mathbb{F}_{q^n}$ can be computed in $O(n \log n \log\log n)$ operations in $\mathbb{F}_q$, division in $O(n \log^2 n \log\log n)$ operations in $\mathbb{F}_q$, and exponentiation in $O(n^2 \log\log n)$ operations in $\mathbb{F}_q$. This proves that, given current knowledge, normal bases offer some asymptotic computational advantage over polynomial bases. However, we do not claim that our method is practical. Since we do not have an implementation, it is therefore not clear at what input sizes this method becomes more efficient in practice

than other approaches.

Several cryptographic applications use a fixed element α of high order and compute α^e for many random large integers e. In applications like Diffie & Hellman's (1976) key exchange, ElGamal's (1985) public-key cryptosystem, and Blum & Micali's (1984) pseudorandom bit generator, α is chosen such that exponentiation of α is easy. Gao, von zur Gathen & Panario (1996) give an algorithm for computing exponentiations of certain elements in $\mathbb{F}_{q^n}$, called Gauss periods of type (n,k) over $\mathbb{F}_q$, in $O(n^2)$ operations in $\mathbb{F}_q$, for fixed q and k (no storage is required). This is faster than any known algorithms for exponentiation of an arbitrary element in $\mathbb{F}_{q^n}$. Using this algorithm, we provide experimental results on the multiplicative order of Gauss periods showing that they have high order and are often primitive (self-dual) normal elements in finite fields. Furthermore, an efficient pseudorandom bit generator based on exponentiation of Gauss periods of type (n,k) in $\mathbb{F}_{2^n}$ over $\mathbb{F}_2$ is provided. We do not include these results in the thesis; the interested reader is directed to Gao, von zur Gathen & Panario (1996) for precise definitions and results.

Roughly speaking, when the most important operations are products, divisions, and gcds, polynomial bases should be used; when exponentiation is the most common operation, normal bases are better.

1.3 Basic methodology

This section describes basic tools for the general framework we use to analyze parameters of random polynomials. This methodology allows us to analyze the average-case behavior of several algorithms dealing with polynomials over finite fields, as well as to obtain useful information about them.

Our framework consists of two steps. The first one is based on the determination of generating functions (either univariate or multivariate) to count polynomials over finite fields with special characteristics. The second step consists of techniques to extract the asymptotic behavior of the coefficients in our generating functions by studying their singularities.

1.3.1 Generating functions

A *class of combinatorial structures* is a set on which a *size* function is defined. Let $\mathcal{C}$ be such a class. Throughout this thesis, we denote the size of an element $\alpha \in \mathcal{C}$ by $|\alpha|$, the number of elements in $\mathcal{C}$ of size n by c_n, and the set of elements in $\mathcal{C}$ of size n by C_n. In this thesis, we can assume that all C_n are finite. As an aside, one can formally define a combinatorial class as a pair $(\mathcal{C}, |\cdot|)$ where $\mathcal{C}$ is a collection of objects and the function $|\cdot| : \mathcal{C} \to \mathbb{N}$ is such that the inverse image of any integer is finite.

The *counting sequence* of $\mathcal{C}$ is the sequence of integers $\{c_n\}$ for $n \geq 0$. The *generating function* (also called *ordinary generating function*) and the *exponential generating function* of $\mathcal{C}$ are respectively defined by

$$C(z) = \sum_n c_n z^n \qquad \text{and} \qquad \hat{C}(z) = \sum_n c_n \frac{z^n}{n!}.$$

Observing that the term z^n appears as many times as there are objects in $\mathcal{C}$ of size n, we can write

$$C(z) = \sum_{\alpha \in \mathcal{C}} z^{|\alpha|} \qquad \text{and} \qquad \hat{C}(z) = \sum_{\alpha \in \mathcal{C}} \frac{z^{|\alpha|}}{|\alpha|!}.$$

The nth coefficient of $C(z)$ is denoted by $[z^n]C(z)$ (analogously, for $\hat{C}(z)$).

In this thesis, we restrict our discussion to generating functions for polynomials over a finite field $\mathbb{F}_q$. Let $\mathcal{P}$ be the collection of all monic polynomials over $\mathbb{F}_q$, and let $\mathcal{I}$ be the collection of all monic irreducible polynomials over $\mathbb{F}_q$. The size $|\omega|$ of an irreducible polynomial $\omega \in \mathcal{I}$ is the degree of ω. The formal identity

$$\frac{1}{1-\omega} = 1 + \omega + \omega^2 + \cdots$$

expresses arbitrary repetitions of the irreducible polynomial ω. The product, with repetitions allowed, of elements taken from $\mathcal{I}$ generates the collection $\mathcal{P}$. Therefore, $\mathcal{P}$ can be represented by the expression

$$\mathcal{P} = \prod_{\omega \in \mathcal{I}} (1-\omega)^{-1}. \tag{1.1}$$

In this context, $\mathcal{I}$ may be identified with

$$\mathcal{I} = \sum_{\omega \in \mathcal{I}} \omega.$$

Let z be a formal variable. The substitution $\omega \mapsto z^{|\omega|}$ produces the generating function

$$I(z) = \sum_{\omega \in \mathcal{I}} z^{|\omega|} = \sum_n I_n z^n,$$

where I_n is the number of monic irreducible polynomials of degree n, and the last equality holds by collecting all irreducible factors of the same degree. The same substitution applied to $\mathcal{P}$ yields the generating function

$$P(z) = \prod_{\omega \in \mathcal{I}} (1 - z^{|\omega|})^{-1} = \prod_{n=1}^{\infty} (1 - z^n)^{-I_n}. \tag{1.2}$$

The coefficient $P_n = [z^n]P(z)$ evaluates to the number of monic polynomials of degree n. Obviously, $P_n = q^n$, and therefore we have

$$P(z) = \frac{1}{1 - qz}. \tag{1.3}$$

It is possible to implicitly determine I_n from Eq. (1.2) and Eq. (1.3). Indeed, a well-known derivation using the Moebius inversion formula (Carlitz 1932, p. 41; Berlekamp 1968, Theorem 3.43, p. 84) yields

$$I_n = \frac{1}{n} \sum_{k|n} \mu(k) q^{n/k}. \tag{1.4}$$

In particular, we have the following important consequence.

Theorem 1.3.1 *The number I_n of monic irreducible polynomials of degree n over a finite field $\mathbb{F}_q$ satisfies*

$$I_n = \frac{q^n}{n} + O\left(\frac{q^{n/2}}{n}\right). \tag{1.5}$$

This identity shows that a fraction very close to $1/n$ of the polynomials of degree n over any finite field $\mathbb{F}_q$ is irreducible. This result was first proven by Gauss for the case of finite prime fields. It appeared in the posthumous book Gauss (1889, pp. 602-629); see also Schönemann (1846) and Dedekind (1857).

Lower and upper bounds for I_n are well-known (Lidl & Niederreiter 1983, p. 142, Ex. 3.26 & 3.27), and are summarized in the following theorem.

Theorem 1.3.2 *The number I_n of irreducible polynomials of degree n over a finite field $\mathbb{F}_q$ satisfies*

$$\frac{q^n}{n} - \frac{q(q^{n/2} - 1)}{(q-1)n} \leq I_n \leq \frac{q^n - q}{n}. \tag{1.6}$$

1.3.2 Parameters

Sometimes we want to study properties (that we will call *parameters*) of polynomials of certain degree n. Formally, a parameter is a function from the class of monic polynomials over finite fields to the natural numbers. For instance, one may want to study the number of monic polynomials of degree n having k irreducible factors. We may define a parameter that maps a monic polynomial to the number of its irreducible factors. This allows us to study the number of polynomials of degree n that have k irreducible factors, and the average number of factors in a random polynomial of degree n.

These kind of problems can be naturally analyzed by means of bivariate generating functions. In these functions, we have two variables: the first one for the degree of the polynomials we are interested in, and the second one for the value of the parameter.

The *bivariate generating function* of a sequence of integers $\{a_{nk}\}$ is defined by

$$A(z,u) = \sum_{n\geq 0}\sum_{k\geq 0} a_{nk} z^n u^k.$$

We denote a_{nk} by $[z^n u^k]A(z,u)$, $\sum_{k\geq 0} a_{nk}u^k$ by $[z^n]A(z,u)$ (also by $a_n(u)$), and $\sum_{n\geq 0} a_{nk}z^n$ by $[u^k]A(z,u)$ (also by $a_k(z)$).

For the collection $\mathcal{P}$ of monic polynomials, let χ be some integer-valued parameter on $\mathcal{P}$. Denote by $|\pi|$ the degree of the polynomial $\pi \in \mathcal{P}$. Consider the bivariate generating function

$$P(z,u) = \sum_{\pi\in\mathcal{P}} z^{|\pi|}u^{\chi(\pi)} = \sum_{n\geq 0}\sum_{k\geq 0} p_{nk} z^n u^k.$$

Then, the coefficient $[z^n u^k]P(z,u)$ represents the number of polynomials of degree n with χ–parameter equal to k.

Bivariate generating functions contain all the information related to the distribution of the parameter. Averages and standard deviation can be obtained by successive differentiations of these bivariate generating functions with respect to u (the variable marking the parameter), and then setting $u = 1$. We will extensively use the following well-known formulas for moments from a bivariate generating function (see for instance, Sedgewick & Flajolet 1996, Table 3.6, p. 138):

$$\text{average: } \frac{[z^n]\frac{\partial P(z,u)}{\partial u}\Big|_{u=1}}{[z^n]P(z,1)} = \frac{p_n'(1)}{p_n(1)}, \qquad \text{variance: } \frac{p_n''(1)}{p_n(1)} + \frac{p_n'(1)}{p_n(1)} - \left(\frac{p_n'(1)}{p_n(1)}\right)^2. \tag{1.7}$$

1.3.3 Asymptotic analysis

Generating functions encode exact information in their coefficients. In many cases, the extraction of the coefficients from a generating function is a difficult task. Fortunately, there are powerful methods that allow us to determine the asymptotic form of the coefficients of complicated generating functions directly from their singularities. In particular, it is well-known that the behavior near a dominant positive singularity (one with the smallest modulus) is an important source of coefficient asymptotics.

These methods give conditions under which the asymptotic behavior of the coefficients can be determined using a local asymptotic expansions near a dominant singularity. In other words, these methods give conditions for which the following implication is valid

$$f(z) \sim \sigma(z) \Rightarrow [z^n]f(z) \sim [z^n]\sigma(z),$$

where $f(z)$ is the generating function to be studied and $\sigma(z)$ is its approximation near the singularity.

Most of the generating functions $f(z)$ studied in this thesis are singular at $z = 1/q$ with an isolated singularity of algebraic-logarithmic type. In these cases, we can apply the following result from Flajolet & Odlyzko (1990).

Theorem 1.3.3 *Let $f(z)$ be a function analytic in a domain*

$$\mathcal{D} = \{z : |z| \leq z_1, |\ Arg\ (z - 1/q)| > \frac{\pi}{2} - \varepsilon\},$$

where $z_1 > \frac{1}{q}$, and ε are positive real numbers. Let $k \geq 0$ be any integer, and α a real number with $\alpha \neq 0, -1, -2, \ldots$. If in a neighborhood of $z = 1/q$, $f(z)$ has an expansion of the form

$$f(z) = \frac{1}{(1 - qz)^\alpha} \left(\log \frac{1}{1 - qz}\right)^k (1 + o(1)), \tag{1.8}$$

then the coefficients satisfy, asymptotically,

$$[z^n]f(z) = q^n \frac{n^{\alpha-1}}{\Gamma(\alpha)} (\log n)^k (1 + o(1)). \tag{1.9}$$

We often find generating functions of the form $p(z)f(z)$ in which $p(z)$ is a polynomial and $f(z)$ satisfies the condition of Theorem 1.3.3. In such cases, if $h(z) = p(z)f(z)$, then

$$[z^n]h(z) = q^n p(1/q) \frac{n^{\alpha-1}}{\Gamma(\alpha)} (\log n)^k (1 + o(1)). \tag{1.10}$$

The translation from Eq. (1.8) to Eq. (1.9) or to Eq. (1.10) is achieved by the so-called transfer lemmas that require analytic continuation of $f(z)$ outside its circle of convergence. Such a condition is usually verified by inspection.

However, there are some situations in which generating functions do not satisfy the hypothesis of Theorem 1.3.3. For instance, some of the generating functions in Chapter 4 have a natural boundary at $|z| = 1$ (each point at the unit circle is singular). Darboux's method can be used in these cases (see Darboux 1878; Comtet 1974, Theorem 8.4; Odlyzko 1996). In Section 4.4, we will need a Darboux theorem. The version we use is the following (see Flajolet & Sedgewick 1993b, p. 91).

Theorem 1.3.4 *Assume that $f(z)$ is continuous in the closed disk $|z| \leq 1$, and is in addition k times continuously differentiable ($k \geq 0$) on $|z| = 1$. Then,*

$$[z^n]f(z) = o(1/n^k).$$

Alternatively, we could use the following Tauberian theorem (Hardy & Littlewood 1914, Flajolet & Sedgewick 1993b, Odlyzko 1996).

Theorem 1.3.5 *Let $f(z) = \sum_{n\geq 0} f_n z^n$ be a power series with radius of convergence 1 satisfying, for $0 \leq z < 1$,*

$$f(z) \sim \frac{1}{(1-z)^\alpha} L\left(\frac{1}{1-z}\right),$$

as $z \to 1^-$, where $\alpha > 0$ and L is a function varying slowly at infinity. Furthermore, assume the side condition that the coefficients $f_n = [z^n]f(z)$ are all non-negative. Then,

$$\sum_{k=1}^{n} f_k \sim \frac{n^\alpha}{\Gamma(\alpha+1)} L(n).$$

If $\{f_n\}_{n\geq 0}$ is monotonic, then

$$f_n \sim \frac{n^{\alpha-1}}{\Gamma(\alpha)} L(n),$$

in agreement with singularity analysis.

1.3.4 The permutation model

It is well-known that, as the cardinality q of the field goes to infinity (n staying fixed), the distribution of the degrees of the irreducible factors for a random polynomial of degree

n converges to the distribution of the length of cycles for a random permutation of size n. Generating functions for polynomials over a finite field $\mathbb{F}_q$ often have a dominant singularity at $z = 1/q$ while generating functions for permutations have a dominant singularity at $z = 1$. This means that when q goes to infinity, generating functions of random polynomials at z/q converge to generating functions for permutations. For instance, the generating function of all monic polynomials, when normalized with the change $z \leftarrow z/q$, is

$$P\left(\frac{z}{q}\right) = \frac{1}{1-z} = \sum_{n=1}^{\infty} n! \frac{z^n}{n!},$$

which is obviously the exponential generating function of all permutations. Similarly, it can be proven that

$$I\left(\frac{z}{q}\right) \underset{(q\to\infty)}{\to} \ln\frac{1}{1-z} = \sum_{n=1}^{\infty}(n-1)!\,\frac{z^n}{n!},$$

the exponential generating function of cyclic permutations. In addition, we can see that $P(z)$ is related to the unique decomposition of permutations into cycles (Goulden & Jackson 1983) by

$$\frac{1}{1-z} = \exp\left(\ln\frac{1}{1-z}\right).$$

This gives rise to a useful observation: probabilistic properties of the decomposition of polynomials over finite fields into irreducible factors frequently resemble the corresponding properties of the cycle decomposition of permutations, when $q \to \infty$. An instance of this permutation model is mentioned in Greene & Knuth (1990) in connection with the probability that a random polynomial admits factors of distinct degrees which for large q and large n is found to approach $e^{-\gamma}$. This result is useful for several polynomial factorization algorithms (Section 4.4 discusses this example in detail). Many times in this thesis we will mention the permutation model.

We conclude this section with a list of relevant references. Many books deal with generating functions but few focus on generating functions for polynomials over finite fields. Chapter 3 of Berlekamp's book (1968) is a remarkable exception. We should also point out the exercise section 4.6.2 of Knuth's book (1981). A complete presentation of (ordinary and multivariate) generating functions will appear in Flajolet & Sedgewick (1996; preprint in Flajolet & Sedgewick 1993a). The paper by Flajolet and Odlyzko (1990) is

our main reference for singularity analysis (also Flajolet & Sedgewick 1993b). For an excellent survey on asymptotic methods, with many more tools that we will use in this thesis, see Odlyzko (1996). In particular, it is our main reference for Tauberian theorems (see also Greene & Knuth 1990).

1.4 Mathematical results

In this section, we present some miscellaneous results needed in the thesis.

Theorem 1.4.1 *For $m \to \infty$, the harmonic sum satisfies*

$$H_m = \sum_{i=1}^{m} 1/i = \ln m + \gamma + \frac{1}{2m} - \frac{1}{12m^2} + \frac{1}{120m^4} + O\left(\frac{1}{m^6}\right),$$

where $\gamma = 0.577216\ldots$ is the Euler constant (see for instance, Graham et al. *1994, p. 452).*

Theorem 1.4.2 *The following inequalities hold:*

(a) For $n \geq 1$,

$$2 \leq \left(1 + \frac{1}{n}\right)^n \leq e.$$

(b) For $n \geq 2$,

$$e \leq \left(1 - \frac{1}{n}\right)^{-n} \leq 4.$$

Theorem 1.4.3 *(a) (Fermat's Little Theorem.)*
Let p be a prime number. For all $a \in \mathbb{Z}$ not divisible by p

$$a^{p-1} \equiv 1 \bmod p.$$

(b) (A generalization for any finite field.) For all $a \in \mathbb{F}_q$,

$$a^q = a.$$

Theorem 1.4.4 *(Chinese Remainder Theorem) Let $f_1, \ldots, f_k \in \mathbb{F}_q[x]$ be pairwise relatively prime polynomials, and let $f = f_1 \cdots f_k$. Then, we have the isomorphism*

$$\mathbb{F}_q[x]/(f) \cong \mathbb{F}_q[x]/(f_1) \times \cdots \times \mathbb{F}_q[x]/(f_k).$$

Now that we have presented the mathematical preliminaries, we are ready to move on to the core material of the thesis.

Chapter 2

Counting Polynomials over Finite Fields

In this Chapter, we focus on counting polynomials over finite fields whose irreducible factors have degrees with special characteristics. Some of the results are known and some are new. Here these results are stated for the first time within a common framework.

The expected number of irreducible factors of a random polynomial is reviewed. This result is generalized to express the expected number of irreducible factors whose degrees lie in a given interval. The number of polynomials over finite fields which have no irreducible factors in a given interval, and which have all of their irreducible factors in an interval are analyzed. These quantities were previously studied by other authors who give the results depending on number-theoretical functions. Our proofs do not rely on such functions, and our results depend only on the size of the field and the degree of the polynomial. As corollaries, we obtain several results such as the number of polynomials with all of its irreducible factors of the same fixed degree, with no irreducible factor of some fixed degree, and so on.

2.1 Motivation and results

This chapter uses the basic methodology introduced in Section 1.3. The goal is at the same time to review several known results on counting polynomials, to show the machin-

ery of generating functions/asymptotic analysis for polynomials over finite fields at work, and to provide new results based on generalizations of these known results. In addition, several of the techniques that we extensively use in this thesis are introduced. This exposition has the advantage of putting together known and new results into a common framework. We are not aware of another presentation with this generality.

In Section 2.2, we study the number of irreducible factors of a random polynomial. First, we state well-known results on this parameter. Then, we analyze the mean value and variance of the number of irreducible factors whose degrees lie in a fixed interval. From this study, we revisit results on the mean value and variance of the number of roots and of irreducible factors of any fixed degree.

Car (1987) analyzes the number of polynomials without irreducible factors having degrees in a given interval. These results depend on number-theoretical functions that appear when studying the analogous problem of integers without small and large primes. More precisely, her results depend on bounds for the Dickman and Buschstab functions (see Tenenbaum 1995). These bounds are obtained using the saddle point method. In Section 2.3, we study these quantities without resorting to such functions. Our results are explicitly expressed in terms of the degree of the polynomials and of the number of elements in the field under consideration. When the size of the field goes to infinity, results analogous to the cycle structure of permutations are obtained. For a constant m, the results for the important special case of polynomials without factors in the interval $[1, m]$ are stated. These polynomials are useful for the analysis of some irreducibility tests for polynomials over finite fields (see Chapter 6).

In Section 2.4, we study the number of polynomials with all irreducible factors in a fixed interval. We then restrict our attention to polynomials with all factors in the interval $[1, m]$ for a constant m. These polynomials are naturally related to the analysis of some algorithms for computing discrete logarithms over finite fields. Finally, we explicitly give the number of polynomials formed only by linear factors.

There are some classical results on counting polynomials with some particular characteristics that we will not cover in this thesis. The study of the factorization pattern of polynomials can be found in Cohen (1969), Car (1982), Knopfmacher *et al.*

(1994), among others. Polynomials in several indeterminates are studied in Carlitz (1963, 1965), and Cohen (1968). The number of polynomials with preassigned coefficients have been largely studied by Dickson (1911), Uchiyama (1955), Hayes (1965), Cohen (1972), among others. A starting point for the reader interested in all the above topics are the outstanding notes at the end of Chapters 3 and 4 in Lidl & Niederreiter (1983).

Very sparse irreducible polynomials, a particular case of polynomials with preassigned coefficients, have received renewed interest due to their applications to the construction of extension fields. An up-to-date account on these results is in Menezes *et al.* (1993). We deal with constructions of sparse irreducible polynomials in Section 6.3.

2.2 The number of irreducible factors of a polynomial

One of the most widely studied parameters related to polynomials over finite fields is the number of irreducible factors of a random polynomial. We start this section reviewing these results without proving them. Pointers to the papers containing the proofs are included. Then we focus on the main result of this section, namely, a study of the number of irreducible factors of a random polynomial with degrees in a fixed interval. Finally, as a corollary, we derive the number of factors of fixed degree in a random polynomial.

The next theorem provides a summary of results associated with the number of irreducible factors of a random polynomial of degree n.

Theorem 2.2.1 *Let Ω_n be a random variable counting the number of irreducible factors of a random polynomial of degree n over $\mathbb{F}_q$, where each factor is counted with its order of multiplicity.*

1. *The mean value of Ω_n is asymptotic to $\log n + O(1)$, or more precisely, to the harmonic sum: $H_n = \sum_{i=1}^{n} 1/i$.*

2. *The variance of Ω_n is asymptotic to $\log n + O(1)$.*

3. *For any two real constants* $\lambda < \mu$,

$$Pr\left\{\log n + \lambda\sqrt{\log n} < \Omega_n < \log n + \mu\sqrt{\log n}\right\} \to \frac{1}{\sqrt{2\pi}}\int_{\lambda}^{\mu} e^{-t^2/2}dt.$$

4. *The distribution of* Ω_n *admits exponential tails.*

5. *A local limit theorem holds.*

Remarks:

1. The average number of irreducible factors of a random polynomial of degree n appears in Berlekamp (1968, Ex. 3.6, p. 86). Then, it also appears in Knuth (1981, Ex. 4.6.2.5, p. 437), and with more details in Mignotte & Nicolas (1983), Flajolet & Soria (1990) and Knopfmacher & Knopfmacher (1993).

2. The variance is sketched in Flajolet & Soria (1990), and is given with more terms in Knopfmacher & Knopfmacher (1993). The latter also covers the case of *distinct* factors.

3. For any two real constants $\lambda < \mu$, if

$$\Pr\left\{\log n + \lambda\sqrt{\log n} < \Omega_n < \log n + \mu\sqrt{\log n}\right\} \to \frac{1}{\sqrt{2\pi}}\int_{\lambda}^{\mu} e^{-t^2/2}dt,$$

then it is said that Ω_n satisfies a central limit theorem or that a Gaussian limit distribution holds. The existence of this limit distribution provides information on the distribution near the mean value. The central limit for Ω_n is in Flajolet & Soria (1990, Corollary 1).

4. Exponential tails essentially indicate that large deviations from the mean are unlikely. In the case of the number of irreducible factors of random polynomials, this is proven in Flajolet & Soria (1993).

5. Local limit theorems basically deal with density functions. They are studied in depth in Bender (1973) and Bender & Richmond (1983). For our particular case, the local limit theorem holds as a consequence of the results in Gao & Richmond (1992).

In this thesis, we will neither study central and local limit theorems nor exponential tails. For precise definitions and results, the reader is referred to the above papers.

2.2.1 The number of irreducible factors of a polynomial with degrees in an interval

Theorem 2.2.1 shows that the average number of irreducible factors of a random polynomial of degree n is asymptotic to $\log n$ with a standard deviation of $\sqrt{\log n}$. A natural variation is to consider the same parameter but for an interval $[a, b] \subseteq [1, n]$. In this section, we study this variant. More advanced results related to the distribution of irreducible factors with degrees lying in partitions of the interval $[1, n]$ are presented in Chapter 5.

Theorem 2.2.2 *Let $\Omega_n^{[a,b]}$ be a random variable counting the number of irreducible factors of a random polynomial of degree n over $\mathbb{F}_q$ with degrees belonging to a fixed interval $[a, b]$, where each factor is counted with its order of multiplicity.*

- *The mean value of $\Omega_n^{[a,b]}$ is asymptotic to $\sum_{k=a}^{b} \frac{I_k}{q^k(1-q^{-k})}$.*
- *The variance of $\Omega_n^{[a,b]}$ is asymptotic to $\sum_{k=a}^{b} \frac{I_k}{q^k(1-q^{-k})^2}$.*

PROOF. Let $\mathcal{I}$ be the collection of all monic irreducible polynomials in $\mathbb{F}_q$, $\omega \in \mathcal{I}$ any irreducible factor, and u a marking variable. Formally, the collection of polynomials marking the irreducible factors with degree lying in the interval $[a, b]$ can be written as

$$\prod_{\omega,\, |\omega| \in [a,b]} \frac{1}{1 - u\,\omega} \cdot \prod_{\omega,\, |\omega| \notin [a,b]} \frac{1}{1 - \omega},$$

where $|\omega|$ is the degree of $\omega \in \mathcal{I}$. Let z be a formal variable. The substitution $\omega \mapsto z^{|\omega|}$ produces the bivariate generating function

$$P(z, u) = \prod_{k<a} (1 - z^k)^{-I_k} \prod_{k=a}^{b} (1 - uz^k)^{-I_k} \prod_{k>b} (1 - z^k)^{-I_k}.$$

As usual (see Section 1.3.2), the coefficient $[z^n u^k] P(z, u)$ gives the number of polynomials of degree n with k irreducible factors whose degrees belong to the interval $[a, b]$.

In order to find the mean value of the number of irreducible factors with degree in the interval, we differentiate $P(z,u)$ with respect to u and evaluate at $u = 1$. Taking logarithms, we have

$$\frac{\partial P(z,u)}{\partial u} = \exp\left(\sum_{k\notin[a,b]} (-I_k)\ \log(1-z^k) + \sum_{k\in[a,b]} (-I_k)\ \log(1-uz^k)\right)\left(\sum_{k=a}^{b} \frac{I_k z^k}{1-u\,z^k}\right).$$

Let $T(z)$ be the value of $\frac{\partial P(z,u)}{\partial u}$ in $u = 1$. Eqs (1.2) and (1.3) imply that

$$\exp\left(\sum_{k\notin[a,b]} (-I_k)\ \log(1-z^k) + \sum_{k\in[a,b]} (-I_k)\ \log(1-z^k)\right) = \frac{1}{1-qz}.$$

Therefore,

$$[z^n]T(z) = [z^n]\frac{1}{1-qz}Q(z), \qquad Q(z) = \sum_{k=a}^{b} \frac{I_k z^k}{1-z^k}. \tag{2.1}$$

The mean value of $\Omega_n^{[a,b]}$ is given by

$$\frac{1}{q^n}[z^n]\left.\frac{\partial P(z,u)}{\partial u}\right|_{u=1} = \frac{1}{q^n}[z^n]\frac{1}{1-qz}\sum_{k=a}^{b}\frac{I_k z^k}{1-z^k}. \tag{2.2}$$

$T(z)$ presents singularities in $z = 1/q$, and on the kth roots of unity for $k = a,\ldots,b$. Since the number of singularities at kth roots of unity is finite, we can immediately apply singularity analysis for the dominant singularity in $z = 1/q$. Near the dominant singularity, $q^{-n}T(z)$ is equivalent to

$$\frac{1}{q^n}\frac{1}{1-qz}\sum_{k=a}^{b}\frac{I_k}{q^k(1-q^{-k})}. \tag{2.3}$$

The transfer lemma given by Theorem 1.3.3 together with Eqs. (2.1), and (2.3) implies that the mean value of $\Omega_n^{[a,b]}$ is asymptotic to

$$Q\left(\frac{1}{q}\right) = \sum_{k=a}^{b}\frac{I_k}{q^k(1-q^{-k})}. \tag{2.4}$$

The variance can be computed using an analogous approach. Indeed, differentiating $P(z,u)$ again with respect to u we obtain

$$\frac{\partial^2 P(z,u)}{\partial u^2} = \exp\left(\sum_{k\notin[a,b]} (-I_k)\ \log(1-z^k) + \sum_{k\in[a,b]} (-I_k)\ \log(1-uz^k)\right) \cdot \left(\left(\sum_{k=a}^{b}\frac{I_k z^k}{1-u\,z^k}\right)^2 + \sum_{k=a}^{b}\frac{I_k z^{2k}}{(1-u\,z^k)^2}\right).$$

The variance is then derived using formula (1.7). When $z \to 1/q$, this computation gives

$$\sum_{k=a}^{b} \frac{I_k}{q^k} \frac{1}{1-q^{-k}} + \sum_{k=a}^{b} \frac{I_k}{q^{2k}} \frac{1}{(1-q^{-k})^2} = \sum_{k=a}^{b} \frac{I_k}{q^k (1-q^{-k})^2}.$$

□

To conclude our study, we give estimates for the mean value and variance in the above theorem. In order to make these estimates meaningful, we assume for the rest of this section that b is not a constant with respect to n. Considering Eq (2.4),

$$\begin{aligned} Q\left(\frac{1}{q}\right) &= \sum_{k=a}^{b} \frac{I_k}{q^k} \frac{1}{1-q^{-k}} = \sum_{k=a}^{b} \frac{I_k}{q^k} \left(\sum_{j \geq 0} \left(\frac{1}{q^k} \right)^j \right) \\ &= \sum_{k=a}^{b} \frac{I_k}{q^k} \left(1 + O\left(\frac{1}{q^k} \right) \right). \end{aligned}$$

From Eq (1.3.1), $I_k = q^k/k + O\left(\frac{q^{k/2}}{k}\right)$, it follows that

$$Q\left(\frac{1}{q}\right) = \sum_{k=a}^{b} \frac{1}{k} \left(1 + O\left(\frac{1}{q^{k/2}} \right) \right).$$

For the error term,

$$\sum_{k=a}^{b} \frac{1}{k\, q^{k/2}} \leq \sum_{k \geq 1} \frac{1}{k\, (\sqrt{q})^k} = -\ln(1-\sqrt{q}) = \ln \frac{1}{1-\sqrt{q}}.$$

Since in our analysis q is fixed, the error term is $O(1)$, and finally we obtain

$$Q\left(\frac{1}{q}\right) = \sum_{k=a}^{b} \frac{1}{k} + O(1).$$

One can consider the harmonic sum approximation in Theorem 1.4.1, and therefore conclude

$$Q\left(\frac{1}{q}\right) = \begin{cases} \ln b + O(1) & \text{if } a = 1, \\ \ln \frac{b}{a-1} + O(1) & \text{otherwise.} \end{cases}$$

Thus, since the total number of monic polynomials of degree n is q^n, the mean value of the number of irreducible factors with degrees belonging to an interval $[a, b]$ for a random polynomial in $\mathbb{F}_q[x]$ of degree n is asymptotic as $n \to \infty$ to $\sum_{k=a}^{b} \frac{1}{k} + O(1)$.

By analogous computations to the ones above for estimating the mean value of $Q\left(\frac{1}{q}\right)$, the variance of the number of irreducible factors of a random polynomial of degree n whose degrees belong to an interval $[a, b]$ is therefore

$$\sum_{k=a}^{b} \frac{1}{k} + O(1).$$

The above estimate for $\sum_{k=a}^{b} \frac{I_k}{q^k(1-q^{-k})}$ is not very tight. One could have proven the much stronger result $\sum_{k=a}^{b} \frac{I_k}{q^k(1-q^{-k})} = \Theta\left(\sum_{k=a}^{b} \frac{1}{k}\right)$. The proof of such an estimate would follow closely the ones in Theorem 5.3.2 and Theorem 6.2.3.

In addition to the results in Theorem 2.2.2, for any two real constants $\lambda < \mu$ and $T_{a,b} = \sum_{k=a}^{b} \frac{1}{k}$, we could prove that

$$\Pr\left\{T_{a,b} + \lambda\sqrt{T_{a,b}} < \Omega_n^{[a,b]} < T_{a,b} + \mu\sqrt{T_{a,b}}\right\} \to \frac{1}{\sqrt{2\pi}} \int_{\lambda}^{\mu} e^{-t^2/2} dt.$$

Therefore a central limit theorem holds. Moreover, the distribution of $\Omega_n^{[a,b]}$ admits exponential tails, and a local limit theorem holds exactly as in the case of Theorem 2.2.1 (see Flajolet & Soria 1993, and Gao & Richmond 1992).

2.2.2 The number of factors of fixed degree in a random polynomial

Two important particular cases of intervals containing degrees of irreducible factors are the interval $[1, 1]$, which corresponds to the linear factors, and the interval $[r, r]$, for any fixed r, which correspond to the irreducible factors of degree r.

The number of linear factors, that is the number of roots, of a random polynomial seems to be first studied by Zsigmondy (1894) for the prime field case. The case of polynomials with no roots is interesting when studying the distinct values that a polynomial can take. This is related with permutation polynomials and was studied by Uchiyama (1955). Later, Knopfmacher & Knopfmacher (1990) presented a detailed analysis including variance.

The number of irreducible factors of specified degree in polynomials of degree n was given by Williams (1969). A detailed analysis including variance appears in Knopfmacher & Knopfmacher (1993).

Corollary 2.2.3 *Let r be a positive integer, and let Ω_n^r be a random variable counting the number of irreducible factors of degree r in a random polynomial of degree n over $\mathbb{F}_q$, where each factor is counted with its order of multiplicity.*

- *The mean value of Ω_n^r is asymptotic to $\frac{I_r}{q^r(1-q^{-r})}$.*

- *The variance of* Ω_n^r *is asymptotic to* $\frac{I_r}{q^r(1-q^{-r})^2}$.

PROOF. The mean value follows from Eq. (2.4), with $a = b = r$ and $z = 1/q$. Therefore, as $n \to \infty$, the mean value approaches

$$\frac{I_r}{q^r(1-q^{-r})}.$$

The variance also follows from Theorem 2.2.2. We have

$$\frac{I_r}{q^r}\frac{1}{1-q^{-r}} + \frac{I_r}{q^{2r}}\frac{1}{(1-q^{-r})^2} = \frac{I_r}{q^r}\frac{1}{1-q^{-r}}\left(1 + \frac{1}{q^r(1-q^{-r})}\right) = \frac{I_r}{q^r(1-q^{-r})^2}.$$

□

Obviously, the case $r = 1$ in the above corollary gives the average number and variance for the number of roots in a polynomial of degree n over $\mathbb{F}_q$.

Theorem 2.2.2 and Corollary 2.2.3 refer to the case when the multiplicity of factors is being counted. Analogous results hold for the case of *distinct* factors. Since the proofs would be similar to the ones presented above, we do not include these analogous results. However, in Chapters 4 and 5, the number of distinct irreducible factors of specified degree in a random polynomial will be needed (see this result in Theorem 4.5.1). As we will see in this thesis, this study is useful when analyzing several polynomial factorization algorithms.

2.3 The number of polynomials without factors with degrees in a fixed interval

Several authors have studied the number of polynomials with no irreducible factors of certain fixed degree. Zsigmondy (1894) studied the number of monic polynomials of degree n over a prime field without irreducible factors of degree r for $1 \leq r \leq n$. The number of monic polynomials with no roots was studied by Uchiyama (1955) (see also Cohen 1973).

A natural generalization of the above results is the study of the number of monic polynomials of degree n without irreducible factors whose degree belongs to an interval

$[a,b] \subseteq [1,n]$. To the best of our knowledge, the only results in this direction are due to Car (1987). She relates our problem to the classical number-theoretical problem of counting integers without prime factors in an interval. Therefore, her results depend on number-theoretical functions. In addition, no precise estimation in terms of q and the interval values $[a,b]$ is given. Our main interest in this section is to show that such estimation is possible by means of methods from analytical combinatorics.

Theorem 2.3.1 *The number of monic polynomials of degree n over $\mathbb{F}_q$ with no irreducible factor of degree in a fixed interval $[a,b]$ is asymptotic, as $n \to \infty$, to*

$$q^n \prod_{k=a}^{b} \left(1 - \frac{1}{q^k}\right)^{I_k} \underset{q\to\infty}{\sim} e^{-\sum_{k=a}^{b} 1/k}.$$

PROOF. Let $\mathcal{I}$ be the collection of all monic irreducible polynomials in $\mathbb{F}_q$. Formally, the collection of all monic polynomials with all irreducible factors without degrees between a and b is

$$\mathcal{P}_{[a,b]} = \prod_{\omega\in\mathcal{I},\ |\omega|<a} (1+\omega+\omega^2+\cdots) \cdot \prod_{\omega\in\mathcal{I},\ |\omega|>b} (1+\omega+\omega^2+\cdots) = \prod_{\omega\in\mathcal{I},\ |\omega|\notin[a,b]} (1-\omega)^{-1}.$$

Let z be a formal variable, and $|\omega|$ the degree of $\omega \in \mathcal{I}$. The substitution $\omega \mapsto z^{|\omega|}$ produces the generating function $P_{[a,b]}(z)$ of polynomials with all irreducible factors with degree outside the interval $[a,b]$, that is,

$$P_{[a,b]}(z) = \prod_{\omega\in\mathcal{I},\ |\omega|\notin[a,b]} \left(1 - z^{|\omega|}\right)^{-1} = \prod_{k\notin[a,b]} \left(1 - z^k\right)^{-I_k} = \frac{1}{1-qz} \prod_{k=a}^{b} (1-z^k)^{I_k}.$$

As usual, $[z^n]P_{[a,b]}(z)$ represents the coefficient of z^n in $P_{[a,b]}(z)$.

When the interval $[a,b]$ is fixed, $P_{[a,b]}(z)$ presents a dominant singularity in $z = \frac{1}{q}$. Let $Q_{[a,b]}(z)$ be $\prod_{k=a}^{b}(1-z^k)^{I_k}$. Applying the transfer lemma in Theorem 1.3.3, the number of monic polynomials of degree n over $\mathbb{F}_q$ with no irreducible factors having degrees in the interval $[a,b]$ is

$$q^n\, Q_{[a,b]}\left(\frac{1}{q}\right) = q^n \prod_{k=a}^{b} (1-q^{-k})^{I_k}.$$

Since the total number of monic polynomials of degree n over $\mathbb{F}_q$ is q^n, the probability that all irreducible factors of a random polynomial in $\mathbb{F}_q[x]$ of degree n have degrees

outside the interval $[a, b]$ is asymptotically, as $n \to \infty$,

$$\prod_{k=a}^{b} \left(1 - \frac{1}{q^k}\right)^{I_k}.$$

In the following, we estimate the constant: $Q_{[a,b]}(\frac{1}{q}) = \prod_{k=a}^{b} \left(1 - \frac{1}{q^k}\right)^{I_k}$. Taking logarithms and using Theorem 1.3.1 to approximate I_k by $q^k/k + O\left(\frac{q^{k/2}}{k}\right)$, we have

$$\begin{aligned} Q_{[a,b]}\left(\frac{1}{q}\right) &= \exp\left(\log\left(\prod_{k=a}^{b}\left(1 - \frac{1}{q^k}\right)^{I_k}\right)\right) \\ &= \exp\left(\sum_{k=a}^{b} I_k \left(-\sum_{j\geq 1}\frac{1}{jq^{jk}}\right)\right) \\ &\sim \exp\left(\sum_{k=a}^{b}\left(-\frac{1}{k} - \frac{1}{2kq^k} - \frac{1}{3kq^{2k}} - \cdots\right)\right). \end{aligned}$$

When $q \to \infty$

$$\prod_{k=a}^{b} \left(1 - \frac{1}{q^k}\right)^{I_k} \sim e^{-\sum_{k=a}^{b} 1/k},$$

that is precisely the probability of permutations with no cycles of length in $[a, b]$. Moreover, considering the approximation of the harmonic sum in Theorem 1.4.1, one could prove

$$Q_{[a,b]}\left(\frac{1}{q}\right) \asymp \begin{cases} \frac{1}{b}\, e^{-\gamma} & \text{if } a = 1 \\ \frac{a-1}{b}\, e^{-\gamma} & \text{otherwise} \end{cases}$$

□

In the above proof, we take $q \to \infty$ instead of considering either a big-Oh estimate or a Θ-estimate as before. One can prove a Θ-estimate for $\prod_{k=a}^{b} \left(1 - \frac{1}{q^k}\right)^{I_k}$, and indeed, we will do so in Theorem 6.2.3. In addition, making $q \to \infty$ allowed us to show an instance of the permutational model (see Section 1.3.4).

A particular case of interest for us is the number of polynomials with no irreducible factors of degrees from 1 to a fixed value m. This study is useful in relation to some algorithms for testing irreducibility. This relation together with the analysis of the more technical case when m is not fixed is presented in Section 6.2.2.

Corollary 2.3.2 *Denote by $P_q(n, m)$ the probability that a random monic polynomial of degree n over $\mathbb{F}_q$ contains no irreducible factor of degree less than or equal to m. Then,*

when $n \to \infty$,

$$P_q(n,m) = \prod_{k=1}^{m} \left(1 - \frac{1}{q^k}\right)^{I_k} \underset{q\to\infty}{\sim} e^{-\sum_{k=1}^{m} 1/k} \underset{m\to\infty}{\sim} \frac{e^{-\gamma}}{m}.$$

2.4 The number of polynomials with all factors in a fixed interval

In this section, we study the number of polynomials with all of their irreducible factors with degree belonging to an interval $[a,b] \subseteq [1,n]$. The particular case of polynomials of degree n with all of their irreducible factors of degree up to certain value m has received special attention. These polynomials, called *smooth* polynomials, are particularly useful in the index calculus method for computing discrete logarithms over extensions of finite fields. For the range $n^{\frac{1}{100}} < m < n^{\frac{99}{100}}$, Odlyzko (1985) estimates the number of m-smooth polynomials by means of the saddle point method (the interested reader is referred to Odlyzko 1996 for an explanation of the saddle point method, and Odlyzko 1994 for an actual account of the relation between smooth polynomials and discrete logarithms over finite fields).

To the best of our knowledge, the only result that deals with a general interval $[a,b]$ appeared in Car (1987). Again, these results are given in terms of number-theoretical functions related to the problem of counting integers with all of its prime factors in an interval. No precise estimation in terms of q and the interval values $[a,b]$ is given. We show that such estimation is possible.

Theorem 2.4.1 *The number of monic polynomials of degree n over $\mathbb{F}_q$ with all irreducible factors having degrees in the fixed interval $[a,b]$ for $a \neq b$ is asymptotic, as $n \to \infty$, to*

$$\binom{n - 1 + \sum_{k=a}^{b} I_k}{n}.$$

PROOF. Let $\mathcal{I}$ be the collection of all monic irreducible polynomials in $\mathbb{F}_q$. Formally, the collection of all monic polynomials with all irreducible factors with degrees between a and b is

$$\mathcal{P}_{[a,b]} = \prod_{\omega\in\mathcal{I},\ |\omega|\in[a,b]} (1 + \omega + \omega^2 + \cdots) = \prod_{\omega\in\mathcal{I},\ |\omega|\in[a,b]} (1 - \omega)^{-1}.$$

As usual, let z be a formal variable, and $|\omega|$ the degree of $\omega \in \mathcal{I}$. The substitution $\omega \mapsto z^{|\omega|}$ produces the generating function $P_{[a,b]}(z)$ of polynomials with all irreducible factors with degree in the interval $[a, b]$, that is,

$$P_{[a,b]}(z) = \prod_{\omega\in\mathcal{I},\ |\omega|\in[a,b]} \left(1 - z^{|\omega|}\right)^{-1} = \prod_{k\in[a,b]} \left(1 - z^k\right)^{-I_k}. \tag{2.5}$$

The singularities in (2.5) are the kth roots of unity for $k = a, \ldots, b$. Among these singularities, only the one with larger growth has to be considered; this singularity provides the dominant contribution to the coefficient. Indeed, the singularity at the real value 1 is dominant since $|P_{[a,b]}(z)|$ grows much faster when $z \to 1^-$ than when z tends to the other singularities in the unit circle. In other words, we are considering a well-known phenomenon: the largest contribution is given by the singularity with the greatest multiplicity. Observe that by hypothesis $a \neq b$, and thus, there exists such a singularity. Therefore,

$$\prod_{k\in[a,b]} \left(1 - z^k\right)^{-I_k} \underset{z\to 1}{\sim} \left(\frac{1}{1-z}\right)^{\sum_{k=a}^{b} I_k}.$$

Applying the so-called transfer lemma (Theorem 1.3.3), we obtain

$$[z^n] P_{[a,b]}(z) \sim [z^n] \left(\frac{1}{1-z}\right)^{\sum_{k=a}^{b} I_k}.$$

Using the expansion (see for instance, Sedgewick & Flajolet 1996, Table 3.1, p. 83)

$$\left(\frac{1}{1-z}\right)^j = \sum_{r=0}^{\infty} \binom{r-1+j}{r} z^r,$$

we have

$$[z^n] \left(\frac{1}{1-z}\right)^{\sum_{k=a}^{b} I_k} = \binom{n-1+\sum_{k=a}^{b} I_k}{n}.$$

□

Before studying the case of intervals $[a, b]$ with $a = b$, we briefly comment on a generalization and on a particularly important instance of the above theorem.

In the above proof we consider a fixed interval. A similar result could be proven when the upper extreme b of the interval depends on n but grows slowly with respect to n, say $b < O(\log \log n)$. We will show this technique for a different generating function in

Theorem 6.2.5. For faster growth of b with respect to n, methodologies like the saddle point method are required.

A particular case of interest is the number of polynomials with all irreducible factors of degrees from 1 to a fixed value m, that is, the smooth polynomials.

Corollary 2.4.2 *Denote by $P_q(n,m)$ the probability that a random monic polynomial of degree n over $\mathbb{F}_q$ contains all of its irreducible factors of degree less than or equal to m. Then, when $n \to \infty$,*

$$P_q(n,m) \sim \frac{1}{q^n} \binom{n-1+\sum_{k=1}^m I_k}{n} \sim \frac{1}{q^n} \frac{n^{\sum_{k=1}^m I_k - 1}}{(\sum_{k=1}^m I_k - 1)!}$$

PROOF. The only part we need to prove is the last equivalence. The kth *falling factorial* of a real number r is defined as $r^{\underline{k}} = r(r-1)\cdots(r-k+1)$. Writing the binomial coefficient in terms of the falling factorial, we obtain

$$\begin{aligned}\binom{n-1+\sum_{k=1}^m I_k}{n} &= \frac{1}{(\sum_{k=1}^m I_k - 1)!} \left(n + \sum_{k=1}^m I_k - 1\right)^{\underline{\sum_{k=1}^m I_k - 1}} \\ &= \frac{1}{(\sum_{k=1}^m I_k - 1)!} \left(n + \sum_{k=1}^m I_k - 1\right)\left(n + \sum_{k=1}^m I_k - 2\right)\cdots(n+1).\end{aligned}$$

Then, when $n \to \infty$, we have

$$\binom{n-1+\sum_{k=1}^m I_k}{n} \sim \frac{n^{\sum_{k=1}^m I_k - 1}}{(\sum_{k=1}^m I_k - 1)!}.$$

□

The complement of the m-smooth polynomials are the polynomials that have at least one factor in the interval $[m+1, n]$. Lower bounds on the number of such polynomials are useful in algorithms for computing Smith normal forms (the interested reader is referred to Giesbrecht 1995). We observe that the study of smooth polynomials automatically gives results for these polynomials.

For the rest of this section, we focus on the case of intervals $[a,b]$ with $a = b$. This case needs to be treated independently. When $a = b$, the generating function $P_{[a,b]}(z)$ reduces to

$$\left(\frac{1}{1-z^a}\right)^{I_a}.$$

In this case, all singularities have the same multiplicity. Therefore, each singularity gives a substantial contribution to the coefficient. However, a straightforward combinatorial argument reveals the exact coefficients. Indeed, if a divides n, then the number of factors of degree a is n/a, which have to be choosen from I_a elements with repetition allowed. Therefore, the number of polynomials of degree n formed only by irreducible factors of degree a is given by

$$\begin{cases} \binom{\frac{n}{a}-1+I_a}{\frac{n}{a}} & \text{if } a \text{ divides } n\ , \\ 0 & \text{otherwise.} \end{cases}$$

We conclude this section with an example for the simple and interesting case of number of polynomials formed by products of linear factors.

Example 2.4.3 *Let us consider the number of monic polynomials of degree n over $\mathbb{F}_q$ such that all of their irreducible factors are linear. In this case, $a = b = 1$, and $P_{[1,1]}(z)$ contains only one singularity at $z = 1$. Thus,*

$$[z^n]\left(\frac{1}{1-z}\right)^{I_1} = \binom{n-1+I_1}{n} = \binom{n-1+q}{n}.$$

For instance, let $q = 2$ and $n = 3$. Then, the number of polynomials of degree 3 over $\mathbb{F}_2$ having only linear factors is $\binom{3-1+2}{3} = 4$. In fact, the 4 polynomials of degree 3 over $\mathbb{F}_2$ having only linear factors are: x^3, $x^2(x-1)$, $x(x-1)^2$, and $(x-1)^3$. In general, for every positive integer n, the number of polynomials of degree n over $\mathbb{F}_2$ having only linear factors is $\binom{n-1+2}{n} = n+1$. The $n+1$ polynomials are x^n, $x^{n-1}(x-1), \ldots, (x-1)^n$. This process is the same in general for any field $\mathbb{F}_q$ since $\binom{n-1+q}{n}$ counts precisely the number of ways one can choose n elements with repetitions from a set with q elements.

Chapter 3

Factoring Polynomials over Finite Fields

Average-case analysis of polynomial factorization algorithms is a central topic of this thesis. In this chapter, we survey several algorithms for the factorization of univariate polynomials over finite fields. An earlier version of this survey was presented by the author as a part of his PhD qualifying exam (Panario 1994). This chapter is an extended version of that paper (see von zur Gathen & Panario 1996). Previous surveys on the factorization problem for general fields have appeared in Kaltofen (1982, 1990, 1992).

3.1 Introduction

Let us start by defining the problem.

> Given a monic univariate polynomial $f \in \mathbb{F}_q[x]$, find the complete factorization $f = f_1^{e_1} \cdots f_k^{e_k}$, where $f_1, \ldots, f_k$ are pairwise distinct monic irreducible polynomials and $e_1, \ldots, e_k$ are positive integers.

Finding the factorization of a polynomial over a finite field is of interest not only by itself but also for many applications in computer algebra, algebraic coding theory, cryptography, and computational number theory. Polynomial factorization over finite fields is used as a subproblem in algorithms for factoring polynomial over the integers (Zassenhaus 1969, Collins 1979, Knuth 1981, Lenstra *et al.* 1982), for constructing cyclic redundancy

codes (Berlekamp 1968), for decoding BCH codes (MacWilliams & Sloane 1977, van Lint 1981), for designing public key cryptosystems (Chor & Rivest 1988, Lenstra 1991), for computing discrete logarithms (Odlyzko 1985), and for estimating the number of points on elliptic curves (Buchmann 1990).

Major improvements have been done in the polynomial factorization problem during this decade both in theory and in practice. From a theoretical point of view, many asymptotically faster algorithms and variants have been proposed. However, these advances are yet more striking in practice where variants of the asymptotically fastest algorithms allow to factor polynomials over finite fields in reasonable amounts of time that were not possible few years ago. Our purpose in this survey is to underline the basic ideas behind these methods, to overview experimental results, as well as to give a comprehensive up-to-date bibliography of the problem.

The general goal is to develop algorithms with running time bounded by a polynomial in the input size, that is, the degree of the polynomial to be factored and the logarithm of q.

All results in this chapter are given for asymptotic worst-case behavior. Unless stated, we consider fast polynomial arithmetic and classical linear algebra for the computations. We recall from Section 1.2.1 the cost of the basic operations in the algorithms. A multiplication of two polynomials of degree at most n can be done in $O(n \log n \log\log n)$ operations in $\mathbb{F}_q$ (Schönhage & Strassen 1971, Schönhage 1977, Cantor 1989, Cantor & Kaltofen 1991, von zur Gathen & Gerhard 1996). A division with remainder can be computed with $O(n \log n \log\log n)$ operations in $\mathbb{F}_q$. The cost of a gcd between two polynomials of degree at most n can be taken as $O(n \log^2 n \log\log n)$ (Aho *et al.* 1974, §8.9). For polynomials h, g of degree at most n, the exponentiation $h^q \bmod g$ can be done by means of the repeated squaring method (Knuth 1981, p. 441–442), with $O(\log q)$ polynomial products, i.e., $O(n \log q \log n \log\log n)$ operations in $\mathbb{F}_q$ using fast methods. For linear algebra, let ω be an achievable exponent for matrix multiplication, so that we can multiply two $n \times n$ matrices with $O(n^\omega)$ operations in $\mathbb{F}_q$. Then, systems of linear equations can be solved in $O(n^\omega)$ operations in $\mathbb{F}_q$. Classical linear algebra methods yield $\omega = 3$, and the current record is $\omega < 2.376$ (Coppersmith & Winograd 1990).

We organize this survey as follows. In Section 3.2, we present a general factoring algorithm. In Section 3.3, we discuss the Berlekamp algorithm. In Section 3.4, we summarize the probabilistic and deterministic algorithms that exist for factoring polynomials over finite fields. In Section 3.5, we briefly introduce average-case analysis for polynomial factorization algorithms.

3.2 A general factoring algorithm

Many algorithms for factoring polynomials over finite fields comprise the following three stages:

SFF *squarefree factorization* replaces a polynomial by squarefree ones which contain all the irreducible factors of the original polynomial with exponents reduced to 1;

DDF *distinct-degree factorization* splits a squarefree polynomial into a product of polynomials whose irreducible factors all have the same degree;

EDF *equal-degree factorization* factors a polynomial whose irreducible factors have the same degree.

The algorithms for the first and second part are deterministic, while the fastest algorithms for the third part are probabilistic.

3.2.1 Squarefree factorization

Some factoring algorithms require that the input polynomial have no repeated factors. A polynomial $f \in \mathbb{F}_q[x]$ is *squarefree* if and only if for any $h, g \in \mathbb{F}_q[x]$ with $f = gh^2$ we have $h \in \mathbb{F}_q$. Thus, a polynomial is squarefree if it has no proper square divisors. If f is not squarefree, a factor can be found quickly by computing gcd (f, f'). In addition, we can find the *squarefree factorization* of a polynomial f of degree n, i.e., monic squarefree pairwise relatively prime polynomials $g_1, \ldots, g_n \in \mathbb{F}_q[x]$ such that $f = g_1 g_2^2 \cdots g_n^n$. Thus, g_i is the product of those monic irreducible polynomials in $\mathbb{F}_q[x]$ that divide f exactly to the power i.

An algorithm for finding the squarefree factorization of a polynomial is given below (see Yun 1976, and Knuth 1981, Exercise 4.6.2-36). The running time of the algorithm is $O(n \log^2 n \log\log n + n \log q)$ or $O^{\sim}(n \log q)$, and thus, both in theory and in practice, we can consider this a trivial step.

Algorithm *Squarefree factorization (SFF)*

Input: *A monic polynomial $f \in \mathbb{F}_q[x]$ of degree n, where char $(\mathbb{F}_q) = p$.*

Output: *Monic squarefree pairwise relatively prime polynomials $g_1, \ldots, g_n$, and integers $e_1, \ldots, e_n \geq 1$ such that $f = \prod_{i=1}^{n} g_i^{e_i}$.*

u := gcd (f, f');
if $u = 1$, then return $(f, 1)$;
if $1 \leq$ deg $u < n$, then recursively compute SFF(u) and SFF(f/u).
    Merge the output lists, and return the merged list.
if $u = f$, then $f := \sum_{0 \leq i \leq n/p} f_{ip} x^{ip}$ with $f_0, f_p, \ldots, f_n \in \mathbb{F}_q$.
    Set $h := \sum_{0 \leq i \leq n/p} f_{ip}^{1/p} x^i$, and compute SFF($h$) with
    output $(h_1, \ldots, h_n; d_1, \ldots, d_n)$. Return $(h_1, \ldots, h_n; pd_1, \ldots, pd_n)$.

3.2.2 Distinct-degree factorization

The second step of the general factorization method is to find the *distinct-degree factorization*, that is, to split a squarefree polynomial into polynomials whose irreducible factors all have the same degree. Let $f \in \mathbb{F}_q[x]$ of degree n be the polynomial to be factored. The algorithm below is based on the following theorem (see Lidl & Niederreiter 1983, p. 91, Theorem 3.20).

Theorem 3.2.1 *For $i \geq 1$, the polynomial $x^{q^i} - x \in \mathbb{F}_q[x]$ is the product of all monic irreducible polynomials in $\mathbb{F}_q[x]$ whose degree divides i.*

Algorithm *Distinct-degree factorization (DDF)*

Input: *A monic squarefree polynomial $f \in \mathbb{F}_q[x]$ of degree n.*

Output: *The set of pairs (g, d), where g the product of all monic irreducible factors of f of degree d with $g \neq 1$.*

```
i := 1;  S := ∅;  f* := f;
while deg f* ≥ 2i do
    g := gcd(f*, x^{q^i} − x mod f*);
    if g ≠ 1, then S := S ∪ {(g, i)};
    f* := f*/g;
    i := i + 1;
endwhile;
if f* ≠ 1, then S := S ∪ {(f*, deg f*)};
return S.
```

The correctness of the above algorithm follows from Theorem 3.2.1. The number of operations in $\mathbb{F}_q$ is $O^{\sim}(n^2 \log q)$ using the repeated squaring method, with a space requirement of $O(n)$ elements in $\mathbb{F}_q$. This algorithm was found by Gauß around 1798 and appears in his Nachlaß (1889). It was rediscovered several times (Galois 1830, without explicitly mentioning the removal of factors; Serret 1866, Arwin 1918, Cantor & Zassenhaus 1981). The special case $d = 1$ is in LeGendre (1785). It is interesting to note that the first algorithm for equal-degree factorization (e.g., to compute the roots of a polynomial all of whose irreducible factors are linear) with running time polynomial in $\log q$ took over almost two centuries longer to be discover (Berlekamp 1970).

This way of computing powers can be improved using the "iterated Frobenius" method (von zur Gathen & Shoup 1992, Algorithm 3.1). If $R = \mathbb{F}_q[x]/(f)$, the *Frobenius map* on R is defined by

$$\begin{aligned} \Phi: \quad R &\longrightarrow R, \\ \alpha &\longmapsto \alpha^q. \end{aligned}$$

Computing iterations of the Frobenius map $\alpha, \alpha^q, \ldots, \alpha^{q^n}$ for $\alpha \in R$ is a basic component for distinct-degree factorization and several other problems in finite fields. Given $m \leq n$ computations $\alpha, \alpha^q, \ldots, \alpha^{q^m}$, and $h \in \mathbb{F}_q[x]$ the canonical representative of α^{q^m}, the iterated Frobenius method obtains the next m values $\alpha^{q^{m+1}}, \ldots, \alpha^{q^{2m}}$ using a fast multipoint evaluation algorithm to compute $g(\alpha^{q^i}) = \alpha^{q^{m+i}}$, for $1 \leq i \leq m$. Using the iterated Frobenius method, the above algorithm for distinct-degree factorization has running time

$O^{\sim}(n^2 + n \log q)$, with a space requirement of $O(n^2)$ elements in $\mathbb{F}_q$.

In von zur Gathen & Shoup (1992, §6), a distinct-degree factorization based on the following blocking strategy is given. They divide the interval $1, \ldots, n$ into about $\sqrt{n}$ intervals of size $\sqrt{n}$, and then for each interval, they compute the joint product of the irreducible factors whose degree lies in that interval. A complete distinct-degree factorization is obtained with the help of the distinct-degree algorithm for each interval, and iterated Frobenius for computing powers. The running time of this algorithm is still $O^{\sim}(n^2 + n \log q)$, but the space requirement drops to $O(n\sqrt{n})$.

A new distinct-degree factorization algorithm is given in Kaltofen & Shoup (1995). They present a family of algorithms for this stage using fast matrix multiplication, and parametrized by β with $0 \leq \beta \leq 1$, that uses $O(n^{(\omega+1)/2+(1-\beta)(\omega-1)/2} + n^{1+\beta+o(1)} \log q)$ operations in $\mathbb{F}_q$. Taking $\omega = 2.376$ and minimizing the exponent of n, they get an algorithm that uses $O(n^{1.815} \log q)$ operations in $\mathbb{F}_q$. This is the first subquadratic-time algorithm for the distinct-degree factorization step for small q. Since the algorithm uses fast matrix multiplication, its practicality is not clear, but they show how to adapt their technique to derive a practical version of this algorithm.

The state of the art in the problem is described in Figure 3.1 (from Kaltofen & Shoup 1995). The asymptotically fastest algorithms are compared considering the dependence between n and $\log q$, and using a fast matrix multiplication method. As a simplified picture, the asymptotically fastest running time algorithms are: for $\log q < n^{0.4545}$, Kaltofen & Shoup (1995); for $n^{0.4545} < \log q < n^{1.376}$, von zur Gathen & Shoup (1992); and for larger fields, Berlekamp (1970).

Major advances have happened in the distinct-degree factorization problem over the last years not only from a theoretical point of view but also from a practical one. Shoup (1996) presents an implementation of a practical version of the algorithm in Kaltofen & Shoup (1995) using $O(n^{2.5} + n^{1+o(1)} \log q)$ operations in $\mathbb{F}_q$, with a space requirement of $O(n^{1.5})$ elements in $\mathbb{F}_q$. In addition, the author proposes a set of benchmarks for polynomial factorization algorithms consisting of factoring polynomials of degree n over an n-bit prime finite field (see also von zur Gathen 1992, and Monagan 1993). For instance, Shoup's algorithm factored a polynomial of degree 1024 modulo a 1024-bit

prime in about 51 hours on a 20-MIPS SUN-4 workstation.

An implementation over $\mathbb{F}_2$ of a variant of the distinct-degree factorization algorithm is presented in von zur Gathen & Gerhard (1996). As an example of the capability of this algorithm, it took 2 days to completely factor a pseudorandom polynomial of degree 262143 over $\mathbb{F}_2$ on two SPARC Ultra 1 workstations.

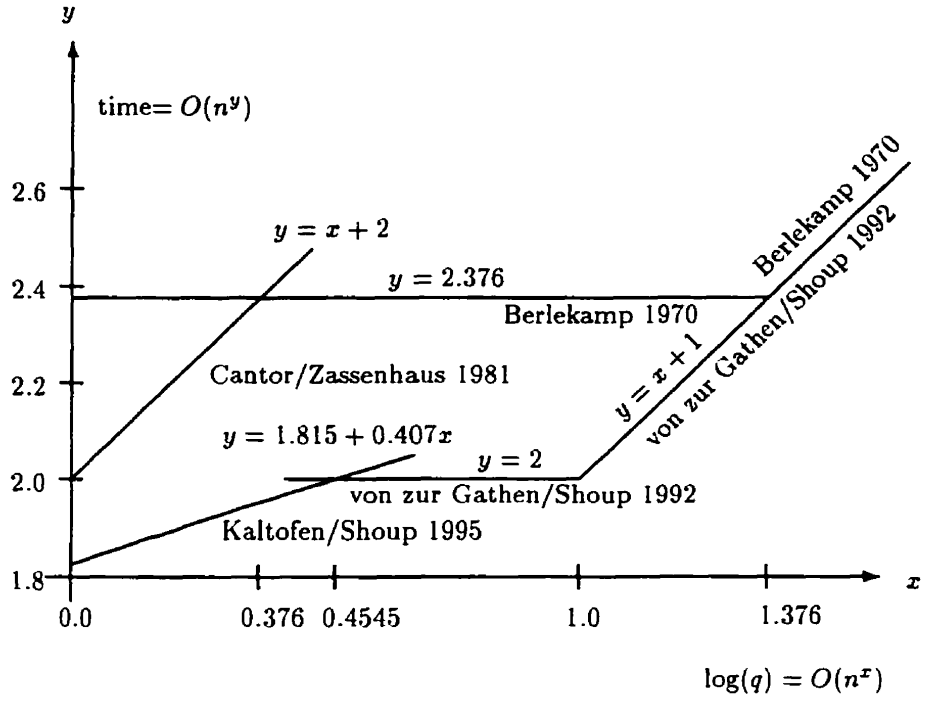


Figure 3.1: Running time comparison of some DDF algorithms.

3.2.3 Equal-degree factorization

We concentrate on algorithms for factoring a monic squarefree univariate polynomial f over a finite field $\mathbb{F}_q$ of degree n with $r \geq 2$ irreducible factors $f_1, \ldots, f_r$, each of degree d. The algorithms that we present here are probabilistic.

First, we describe a simple variant of the algorithm in Cantor & Zassenhaus (1981). Since $f_1, \ldots, f_r$ are pairwise relatively prime, the Chinese Remainder Theorem given in Theorem 1.4.4 provides the isomorphism:

$$\begin{aligned} \mathbb{F}_q[x]/(f) &\longrightarrow \mathbb{F}_q[x]/(f_1) \times \cdots \times \mathbb{F}_q[x]/(f_r), \\ h \bmod f &\longmapsto (h \bmod f_1, \ldots, h \bmod f_r). \end{aligned}$$

Let us write $R = \mathbb{F}_q[x]/(f)$, and $R_i = \mathbb{F}_q[x]/(f_i)$ for $1 \leq i \leq r$. Then R_i is a field with q^d elements and so contains $\mathbb{F}_q$

$$\mathbb{F}_q \subseteq \mathbb{F}_q[x]/(f_i) = R_i \cong \mathbb{F}_{q^d} \quad \text{for } 1 \leq i \leq r.$$

For $h \in \mathbb{F}_q[x]$, f_i divides h if and only if $h \equiv 0 \bmod f_i$, that is, if and only if the ith component of $h \bmod f$ is zero. We immediately see that if $h \in \mathbb{F}_q[x]$ is such that $(h \bmod f_1, \ldots, h \bmod f_r)$ has some zero components and some nonzero components, i.e., $h \bmod f$ is a nonzero zerodivisor in R, then $\gcd(h, f)$ is a nontrivial factor of f, and we call h a "splitting polynomial". Therefore, we look for polynomials with this property.

First consider q odd. Take $m = (q^d - 1)/2$ and an r-tuple $(h_1, \ldots, h_r)$, $h_i \in R_i^\times = \mathbb{F}_{q^d}^\times = \mathbb{F}_{q^d} \setminus \{0\}$. In $\mathbb{F}_{q^d}^\times$, half of the elements are quadratic residues and the other half are quadratic nonresidues. Thus, $h_i^m = \pm 1$, with the same probability for both values when h_i is chosen randomly. Now, choose at random (uniformly) a polynomial $h \in \mathbb{F}_q[x]$, with $\deg h < n$, and let us assume that $\gcd(h, f) = 1$ (otherwise we have already found a partial factorization). The components $(h_1, \ldots, h_r)$ of its image under the Chinese remainder isomorphism are independently and uniformly distributed random elements in $R_i^\times = \mathbb{F}_{q^d}^\times$. Since $h_i^m = +1$ or $h_i^m = -1$ each with probability $1/2$, the probability that $\gcd(h^m - 1, f)$ is not a proper factor f, i.e., all the components in $(h_1^m - 1, \ldots, h_r^m - 1)$ are equal, is $2 \cdot \left(\frac{1}{2}\right)^r = \left(\frac{1}{2}\right)^{r-1} \leq 1/2$. Running the algorithm l times ensures a probability of failure less than $\left(\frac{1}{2}\right)^l$.

After producing a factorization $f = g_1 g_2$, one may proceed in two ways: either by applying the algorithm recursively to g_1 and g_2, or by "refining" an already calculated factorization $f = \prod_{i=1}^{s} u_i$ by $\gcd(u_i, h^m - 1)$ for random h. For any $0 < \varepsilon < 1$, with $2\lceil \log \frac{r^2}{\varepsilon} \rceil$ such random choices, one obtains the complete factorization of f with probability at least $1 - \varepsilon$. We give this algorithm below.

Algorithm *Equal-degree factorization (EDF)*

Input: *A monic squarefree polynomial $f \in \mathbb{F}_q[x]$ of degree $n = rd$, with $r \geq 2$ irreducible*

factors each of degree d, and a confidence parameter ε.

Output: *The set of monic irreducible factors of f.*

```
Factors := {f};  k := 1;  t := 2 ⌈log r²/ε⌉;
while k ≤ t do
   Choose h ∈ F_q[x] with deg h < n at random;
   g := gcd(h, f);
   if g = 1, then
      g := h^((q^d − 1)/2) − 1 (mod f)
   endif;
   for each u ∈ Factors with deg u > d do
       if gcd(g, u) ≠ 1 and gcd(g, u) ≠ u, then
          Factors := Factors \ {u};
          Factors := Factors ∪ {gcd(g, u)} ∪ {u/gcd(g, u)};
       endif;
   if Size(Factors) = r, then return Factors;
   k := k + 1;
endwhile;
return 'Failure'.
```

The difference between the above algorithm and the one in Cantor & Zassenhaus (1981) is that the latter algorithm does not require that $h \in \mathbb{F}_q[x]$ with $\deg h < n$ satisfies $\gcd(h, f) = 1$. In that case, biased probabilities appear (see this algorithm in Section 4.5, and in Cantor & Zassenhaus 1981).

The running time of the above algorithm is $O^{\sim}(n^2 \log q \log^{-1} \varepsilon)$. In the case q even, a primitive third root of unity is considered, and the factorization is carried in the quadratic extension of $\mathbb{F}_q$ (see Cantor & Zassenhaus 1981).

Other probabilistic algorithms for equal-degree factorization are due to Ben-Or (1981) and von zur Gathen & Shoup, (1992, §3, Algorithm 3.6). These algorithms are based on trace computations of random elements in $\mathbb{F}_q[x]/(f)$. Choose $h \in \mathbb{F}_q[x]/(f)$ at random and compute its trace $g = \sum_{i=0}^{d-1} h^{q^i}$. The trace function has image $\mathbb{F}_q$, so raising g to the

$(q-1)/2$ power in the case of odd characteristic, or computing $\sum_{i=0}^{k-1} g^{2^i}$ when $q = 2^k$, leads to a non-trivial factorization of f in a similar way, and with similar probabilities, as in Cantor & Zassenhaus (1981).

The running time of Ben-Or's algorithm is the same as that of Cantor & Zassenhaus (1981). A variant in the procedure for computing traces leads to the asymptotically fastest algorithm for the equal-degree problem, using $O^{\sim}(n^{(\omega+1)/2} + n \log q)$ operations in $\mathbb{F}_q$ (von zur Gathen & Shoup 1992, §5).

As things now stand, equal-degree factorization can be done faster than distinct-degree factorization using randomized algorithms.

3.3 Algorithms based on linear algebra

The pioneering modern algorithms for the factorization of polynomials over finite fields are from Berlekamp (1967, 1968, 1970). Let $f \in \mathbb{F}_q[x]$ be a monic squarefree univariate polynomial of degree n, and $f_1, \ldots, f_r \in \mathbb{F}_q[x]$ its irreducible monic factors that we want to compute. If $R = \mathbb{F}_q[x]/(f)$ and $R_i = \mathbb{F}_q[x]/(f_i)$ for $1 \le i \le r$, then $R \cong R_1 \times \cdots \times R_r$ by the isomorphism of the Chinese Remainder Theorem (see Theorem 1.4.4).

Recall from Section 3.2.2 the definition of the Frobenius map Φ on R. Using Fermat's Little Theorem (Theorem 1.4.3), it can be shown that Φ is $\mathbb{F}_q$-linear. Consider the fixed points of Φ, i.e., the kernel of the mapping $\Phi - I$, where I is the identity function from R to itself. Let $B = \{h \in R : h^q = h\}$. For $a \in \mathbb{F}_{q^n}$, we have $a^q = a$ if and only if $a \in \mathbb{F}_q$, and thus $B \cong \mathbb{F}_q \times \cdots \times \mathbb{F}_q$, r times. If the image of $h \in R$ under this isomorphism is $\left(h^{(1)}, \ldots, h^{(r)}\right)$, we write: $h = [h^{(1)}, \ldots, h^{(r)}]$, and we call $h^{(i)}$ the ith component of h. We have

$$B = \{h \in R : h^{(i)} \in \mathbb{F}_q,\ i = 1, \ldots, r\}.$$

As in Camion (1980), we call B the *Berlekamp subalgebra* of R. From the above comments, it follows that B is an $\mathbb{F}_q$-vector space whose dimension is r, the number of irreducible factors of f.

The Berlekamp subalgebra of R helps us to factor the polynomial f. Indeed, Fermat's

Little Theorem implying that $h^q - h = \prod_{\alpha \in \mathbb{F}_q}(h - \alpha)$ for $h \in \mathbb{F}_q[x]$, shows that

$$f = \prod_{\alpha \in \mathbb{F}_q} \gcd(f, h - \alpha), \tag{3.1}$$

for $h \in \mathbb{F}_q[x]$ with $\overline{h} = (h \bmod f) \in B$ (see Berlekamp 1968, Theorem 6.11). If $\overline{h}$ is a constant, this product degenerates into the trivial factorization $f \cdot 1$. However, since $\deg \overline{h} < n$, if $\deg \overline{h} \geq 1$ then $\prod_{\alpha \in \mathbb{F}_q} \gcd(f, h - \alpha)$ gives a non trivial factorization of f. Therefore, to have a partial factorization of f, it is sufficient to consider $\overline{h} \in B$ with $\deg \overline{h} \geq 1$, and compute the product in Eq. (3.1). In general, this computation will not lead to the complete factorization of the polynomial. To achieve this factorization, we introduce the following notion.

Definition 3.3.1 *A set $S = \{h_1, \ldots, h_m\}$ is a* separating set *for f, if for any two distinct irreducible factors f_i and f_j of f, there exists $h_k \in S$ such that $\prod_{\alpha \in \mathbb{F}_q} \gcd(f, h_k - \alpha)$ separates f_i from f_j (i.e., f_i and f_j appear as factors of different gcds in the previous decomposition).*

Berlekamp (1968) proves that if $\{1, v_2, \ldots, v_r\}$ is a basis of B, then it is a separating set.

We can compute a basis for B by forming the matrix of the mapping $\Phi - I$ with respect to the basis $1, x, \ldots, x^{n-1}$ of R. Using Gaussian elimination, we obtain a basis of $\ker(\Phi - I)$, which is precisely B. From this basis, we have the number of irreducible factors of f and, incidentally, an irreducibility test. Indeed, we can test for polynomial irreducibility by computing a basis of B and checking if it has dimension one.

Let $(1, v_2, \ldots, v_r)$ be a basis for B (without lost of generality we can take 1 as one of the elements in the base). This algorithm factors f computing

$$f = \prod_{\alpha \in \mathbb{F}_q} \gcd(f, v_2 - \alpha),$$

and refining the partial factorization successively for $v_3, \ldots, v_r$ until the complete factorization of f is obtained.

The running time of this algorithm is $\tilde{O}(n^3 + qn^2)$ operations in $\mathbb{F}_q$, using Gaussian elimination with classical arithmetic to find a basis of B. The algorithm is practical only for small q.

As an aside, parts of the above method were known before Berlekamp. For instance, the above matrix construction appears in Petr (1937), Butler (1954) and Schwarz (1956), but it was Berlekamp who put all the elements together.

A randomized version of the above algorithm appears in Berlekamp (1970). For $h \in \mathbb{F}_q[x]$, let $\overline{h} = (h \bmod f) \in B$ with $\overline{h} = [h^{(1)}, \ldots, h^{(r)}]$, and take q odd (the even case can be treated with the trace function as in the equal-degree algorithm at the end of Section 3.2.3). Since $h^{(i)} \in \mathbb{F}_q$ for $1 \leq i \leq r$, each component of $g = h^{(q-1)/2} \bmod f$ under the isomorphism of the Chinese Remainder Theorem is 0, +1 or −1. This implies that

$$f = \gcd(f, g-1)\ \gcd(f, g)\ \gcd(f, g+1),$$

separates f_i from f_j if and only if $h^{(i)} \neq h^{(j)}$. Therefore, unless all components are equal, $\gcd(f, h^{(q-1)/2} \bmod f)$ yields a nontrivial factorization of f.

Berlekamp adapts this idea choosing $v \in B$ randomly, and computing $u = v^{(q-1)/2} \bmod f$. If $u \in B \backslash \mathbb{F}_q$, i.e., not all of the components of u are equal, then either $\gcd(f, u)$ or $\gcd(f, u-1)$ gives a nontrivial factor of f. As in the equal-degree factorization process, we have a probability greater than $1/2$ of having a non-trivial factor of f. Therefore, the expected running time is $O^{\sim}(n^3 + n \log q)$ operations in $\mathbb{F}_q$.

For problems of large size, the basis computation in the Berlekamp algorithm can be done faster using Wiedemann's sparse linear system solver (Wiedemann 1986). Kaltofen (1992) improves the running time of the algorithm in Berlekamp (1970) to $O^{\sim}(n^2 \log q)$ in this way, and Kaltofen & Lobo (1994) give an implementation of Berlekamp's algorithm that runs in $O^{\sim}(n^2 + n \log q)$ arithmetic operations. They use randomization and a Wiedemann parallel block linear system solver (Wiedemann 1986, Kaltofen 1993a, 1993b, Coppersmith 1994) for finding nonzero elements of $\ker(\Phi - I)$. In 1993, this algorithm factored polynomials of degree 10001 over $\mathbb{F}_{127}$ in less than 4 days. The network used was composed of eight Sun 4 workstations with 32Mbytes of memory each.

A new deterministic algorithm, also based on linear algebra, is presented in Niederreiter (1993a, 1993b); comparisons to Berlekamp's algorithm are in Fleischmann (1993), Gao & von zur Gathen (1994), Lee & Vanstone (1995). A randomized version of Niederreiter's algorithm using Wiedemann's approach is in Gao & von zur Gathen (1994).

3.4 Polynomial factorization algorithms

In this section, we give a comprehensive list of factoring algorithms from Berlekamp (1967) to the present. Berlekamp was the first to give a general algorithm for the problem. Partial results prior to Berlekamp can be found in Lidl & Niederreiter (1983), at the end of Chapter 4. Although deterministic algorithms are the special case of probabilistic algorithms that make no use of their probabilistic choices, for simplicity, we classify the algorithms as either deterministic or probabilistic.

3.4.1 Probabilistic algorithms

The first probabilistic algorithm for factoring polynomials over finite fields appeared in Berlekamp (1970). Berlekamp's paper was a pioneering result on probabilistic algorithms, whose huge success took off only later, after the work of Solovay & Strassen (1977) and Gill (1977). Today, probabilistic choice is used routinely in the many algorithmic applications where it is profitable.

In Sections 3.2 and 3.3, we presented the main ideas of the algorithms listed in the table below. In addition, other probabilistic algorithms for the problem are found in Calmet & Loos (1980), Lazard (1982) and Camion (1983b). For efficient factorization using fewer random bits see Bach & Shoup (1990). Finally, two recent randomized algorithms are in Gao & von zur Gathen (1994) and Kaltofen & Lobo (1994).

Authors	Running time
Berlekamp (1970)	$O^{\sim}(n^{2.376} + n \log q)$
Rabin (1980)	$O^{\sim}(n^3 \log q)$
Cantor & Zassenhaus (1981)	$O^{\sim}(n^2 \log q)$
Ben-Or (1981)	$O^{\sim}(n^2 \log q)$
von zur Gathen & Shoup (1992)	$O^{\sim}(n^2 + n \log q)$
Kaltofen & Shoup (1995)	$O(n^{1.815} \log q)$

Probabilistic algorithms for the special problem of finding roots of polynomials over

finite fields can be found in Berlekamp (1970), Rabin (1980), Ben-Or (1981), and van Oorschot & Vanstone (1989).

3.4.2 Deterministic algorithms

Berlekamp (1967) gives the first deterministic algorithm. We discussed it in detail in Section 3.3. Its running time is $O^{\sim}(n^3 + qn^2)$, so it is not a polynomial-time algorithm in $n \log q$. The major open problem in this area is to find a deterministic polynomial-time algorithm for the problem.

Deterministic algorithms are given in McEliece (1969), Camion (1983a), Menezes *et al.* (1988), Niederreiter (1993a, 1993b), Rothstein & Zassenhaus (1994), and von zur Gathen & Shoup (1992, §9). The latter paper gives the currently fastest algorithm with $O^{\sim}(n^2 + n^{3/2}k + n^{3/2}k^{1/2}p^{1/2})$ operations in $\mathbb{F}_q$, where $q = p^k$. Shoup (1990) gives a deterministic algorithm for the case of a prime field $\mathbb{F}_p$, with running time $O^{\sim}(\sqrt{p}\ \log p\ n^2)$ (Shparlinski 1992 removed the $\log p$ factor). This algorithm factors "almost all" polynomials in polynomial time (see Shoup 1990 and Shparlinski 1992, §1.1). Recently, Evdokimov (1994) gives an algorithm with time $(n^{\log n} \log q)^{O(1)}$, under the extended Riemann hypothesis (ERH) (see also Gao 1995).

Under the ERH, deterministic polynomial-time factoring algorithms are known for some special cases. For factorization of special polynomials, we can cite: Schoof (1985), Rónyai (1987, 1989a), Huang (1984, 1985, 1991a, 1991b). For special fields see: Moenck (1977), von zur Gathen (1987), Mignotte & Schnorr (1988), Rónyai (1989b), Thiong Ly (1989), Shoup (1991a, 1991b), Bach *et al.* (1992), and Menezes *et al.* (1992).

3.5 Average-case analysis

All algorithms we have referred to in this chapter have been analyzed from a worst-case perspective. In general, big-Oh estimates for the worst case are known. For probabilistic algorithms (all of Las Vegas type), expected running times with probability of failure are also known.

Few results are known in terms of average-case analysis of polynomial factorization

algorithms. Shoup (1990) studied his deterministic algorithm using estimates for the number of solutions of equations over finite fields and Weil's bounds (for a similar analysis see Ben-Or 1981, and for background on these issues see Schmidt 1976).

In Flajolet, Gourdon & Panario (1996), we present for the first time a complete average-case analysis of the general algorithm in Section 3.2. This analysis and the ones of several variants of the methods presented in this chapter are given in detail in Chapters 4 and 5.

Chapter 4

Average-case Analysis of Polynomial Factorization Algorithms

This chapter derives basic properties of random polynomials that are of interest in the study of polynomial factoring algorithms. We show that the most important characteristics can be treated systematically by the methodology introduced in Section 1.3. Based on the use of generating functions and singularity analysis, a complete study of the general factoring algorithm of Section 3.2 is given. Parts of this chapter already appeared in Flajolet, Gourdon & Panario (1996). We provide detailed proofs of the theorems given in that extended abstract. In addition, we briefly mention some variants for the algorithms in this chapter.

4.1 Introduction

Chapter 3 examined several algorithms for factoring univariate polynomials over finite fields. A slight variant of the general factorization algorithm of Section 3.2 is given by the following three stages:

ERF *elimination of repeated factors* changes a polynomial into a squarefree form that contains all the irreducible factors of the original polynomial with exponents reduced to 1;

DDF *distinct-degree factorization* splits a squarefree polynomial into a product of polynomials whose irreducible factors all have the same degree; and

EDF *equal-degree factorization* factors a polynomial whose irreducible factors all have the same degree.

Observe that we slightly changed the first stage of the general algorithm above. This simplifies the analyses at no extra cost, as we will show in Section 4.2.

We refer to the general factorization algorithm as a *factorization chain* in order to emphasize that the output from one stage is the input for the next stage. The top-level code of our factorization chain (in pseudo-Maple) is given below.

```
procedure factor(f:polynomial);
     1: a:=ERF(f);
     2: b:=DDF(a);
        F:=1;
     3: for k from 1 to n do
              F:=F.EDF(b[k],k);
        od;
        c:=factor(f/a);
return(F.c);
end;
```

Figure 4.1: The complete factorization chain that serves as our pilot algorithm.

To motivate the chapter, we choose and fully analyze a factorization chain as above. The methodology used is also applicable to a variety of algorithms for each step in the above general scheme, and with similar proofs. The analysis of these algorithms completely reveals interesting parameters that are of general interest for polynomial factoring at large.

All analyses are expressed as asymptotic forms in n, the degree of the polynomial to be factored. We conduct the analysis for a fixed field $\mathbb{F}_q$ with $q = p^m$ and consider $\mathbb{F}_q[x]$.

We work with the following assumptions: the cost of a basic field operation is $O(1)$; the cost of a sum is $O(n)$, and the cost of a product or a gcd is $O(n^2)$, when applied to polynomials of degree $\leq n$. Since our interest is in *dominant asymptotics*, we can freely restrict our attention to the costs of products and gcd's that we take to be $\tau_1 n^2$ and $\tau_2 n^2$, for constants τ_1 and τ_2, respectively.

4.1.1 An application of factoring random polynomials

Section 3.1 presented several applications of factoring polynomials over finite fields. In the following, we discuss in some detail an application to cryptography of factoring *random* polynomials over a finite field $\mathbb{F}_q$.

Before describing the application, some notation needs to be specified. Let q be equal to p^m for some prime p and integer $m \geq 2$. Let f be an irreducible polynomial of degree m over $\mathbb{F}_p$. We represent the elements in $\mathbb{F}_q$ as polynomials of degree less than m over $\mathbb{F}_p$ via the isomorphism $\mathbb{F}_{q^m} \cong \mathbb{F}_q[x]/(f)$. Let g be a polynomial in $\mathbb{F}_q$ of degree less than m that is a primitive element of $\mathbb{F}_q$. Therefore, for any element $b \in \mathbb{F}_q$, $b \neq 0$, there exists an integer x, $0 \leq x \leq q-2$, such that $b = g^x$. We call x the *discrete logarithm of* b in the base g. Then, the *discrete logarithm problem* is to find a computationally feasible algorithm to compute the discrete logarithm of b for any $b \in \mathbb{F}_q$, $b \neq 0$. The security of many public-key cryptosystems is based on the intractability of the discrete logarithm problem (for large q).

The *index calculus method* for computing discrete logarithms over $\mathbb{F}_q$ consists of two parts: a construction of a large database of logarithms, and the computation of individual logarithms.

Index calculus algorithm.

1. Find the logarithms of all irreducible polynomials of degree at most ℓ, where ℓ is a fixed positive integer.

2. For computing the logarithm of an element $b \in \mathbb{F}_q$, $b \neq 0$, take a random integer a,

$1 \leq a \leq q-1$, and compute $h \equiv bg^a \bmod f$. Factor h as

$$h = \prod_{i=1}^{t} p_i^{e_i}.$$

If each irreducible factor p_i is such that $\deg p_i \leq \ell$, then

$$\log b \equiv \sum_{i=1}^{t} e_i \log p_i - a \bmod (q-1),$$

which can be easily evaluated by looking up $\log p_i, 1 \leq i \leq t$ in the database. If not all p_i have degree $\leq \ell$, then generate another integer a and repeat.

Since a is taken at random in $1 \leq a \leq q-1$ and g is primitive, h behaves like a random polynomial. Thus, the discrete logarithm problem contains as a subproblem the factorization of random polynomials. (For an extensive treatment of the discrete logarithm problem see Odlyzko 1985.)

4.1.2 Summary of results

A random polynomial of degree n over $\mathbb{F}_q$ is irreducible with probability tending to 0 (Theorem 1.3.1), and has close to $\log n$ factors on average and with high probability (Theorem 2.2.1). Thus, the factorization of a random polynomial over a finite field is almost surely nontrivial.

The first phase *ERF* of our factorization chain starts with the elimination of repeated factors, a simplified form of squarefree factorization described in Section 4.2. The ERF stage returns the *squarefree part* of the original polynomial, that is a polynomial in which each irreducible factor of the original polynomial appears exactly once. The remaining factors of the original polynomial form what we call *the non-squarefree part.* Theorem 4.2.3 gives the cost of the ERF stage. It shows that up to smaller order terms, the expected cost is dominated by a single gcd between the polynomial f to be factored and its derivative f', so that it is $O(n^2)$ on average. In a precise technical sense, most of the factorization cost results from the subsequent phases (DDF and EDF), since the non-squarefree part has average degree $O(1)$.

The second phase *DDF* that is described in Section 4.3 splits the squarefree part a of the polynomial to be factored into a product $a = b_1 \cdot b_2 \cdots b_n$, where each b_k is the

product of the irreducible factors of a that have degree k. This phase has the highest computational cost, namely $O(n^3)$ on average. Theorems 4.3.1, 4.3.2, and 4.3.3 provide a precise comparison of three strategies: the naïve rule, the "half-degree" rule, and the "early abort" rule whose costs are found to be in the approximate proportion $1 :: \frac{3}{4} :: \frac{2}{3}$. Thus, a saving of about one third results from controlling the DDF phase by the early abort strategy. At the end of this phase, the factorization is complete with a probability ranging asymptotically between 0.56 and 0.67, see Theorem 4.4.1.

The third phase *EDF* can be exactly analyzed and it is found that its expected cost is comparatively small, being $O(n^2)$, see Theorems 4.5.2 and 4.5.5 for precise statements. For each nontrivial factor b_k, it involves a recursive refinement process based on properties of finite fields. The analysis is close to that of digital trees known as "tries" (Knuth 1973a) but under a biased probability model.

Precise statements are given in the next sections with an explicit dependency on the field cardinality q. Some of them involve number-theoretic functions that can be both evaluated and estimated easily. A simplified picture is as follows. The ERF phase involves with high probability little more than a single polynomial gcd. The DDF phase, having cost $O(n^3 \log q)$, is the most intensive computationally, where control by the "early-abort" strategy is expected to bring gains close to 36% at no extra cost. The last phase of EDF is executed less than 50% of the time and its cost is again small compared to that of DDF.

We close this chapter with some comments on variants for these algorithms and a comparison between worst-case and average-case cost for each step.

4.2 Elimination of repeated factors (ERF)

The first step in the factorization chain of a polynomial is the elimination of its repeated factors. In characteristic 0, this is achieved by a gcd between f and its derivative f'. In $\mathbb{F}_q$, $q = p^m$ with p a prime number, additional control is needed in order to deal with pth powers whose derivatives are 0. The corresponding process for ERF is given by the following procedure.

```
procedure ERF(f:polynomial);  [Elimination of repeated factors]
        g := gcd(f,f'); h := f/g; k := gcd(g,h);
        while k<>1 do
              g := g/k; k := gcd(g,h)
        od;
        if  g<> 1 then h := h*ERF(g^(1/p)) fi;
        return(h);
end;
```

The first line of the algorithm collects in h one copy of each of the irreducible factors of f, except the ones whose multiplicity is a multiple of p. The while loop stores in g the factors whose multiplicity is a power of p, without eliminating their repetitions. The last part of the algorithm adds to h the factors in g with repetitions eliminated. The auxiliary computation of pth roots, $g^{1/p}$, is performed in the classical way as described, for example, in Geddes *et al.* (1992 p. 344). Indeed, the `if` command in this procedure treats the case when the polynomial g can be written in the form $\sum_i a_i x^{pi}$. In this case, $g = \left(\sum_i a_i^{1/p} x^i\right)^p$, and so we eliminate repeated factors considering $g^{1/p} = \sum_i a_i^{1/p} x^i = \sum_i a_i^{p^{m-1}} x^i$. The computation of $g^{1/p}$ consists of computing the polynomial $\sum_i a_i^{p^{m-1}} x^i$. When r is the degree of g, its cost is $O(r^2 \log q)$ using classical arithmetic. The total cost of the procedure is $O(n^2 \log q)$. Since q is fixed, the cost is $O(n^2)$.

We observe that if a full squarefree factorization is wanted, it is sufficient to recursively call ERF with the input polynomial f/h. Moreover, there are other algorithms for the full squarefree factorization (see Sections 3.2.1 and 4.6). However, as Theorem 4.2.2 below shows, the reduction of degree induced by the additional computational effort is only $O(1)$. Therefore, the rest of the analysis regarding dominant asymptotics is not affected for the algorithm chosen for the first stage. Thus, we choose elimination of repeated factors, a simpler algorithm to implement than squarefree factorization.

Next we summarize a classical result about the number of squarefree polynomials of certain degree in a finite field. Apparently, this result appeared for the first time in Carlitz (1932). His technique is based on equating coefficients of equal power series for

studying several arithmetical functions for polynomials over finite fields. It is remarkable that the author avoided the use of the associated number-theoretical functions:

> *The method used... However, it seems to be the natural one for the subject, and there seems to be little point to recasting the proofs in "arithmetical" shape.* Leonard Carlitz, December 30, 1930

Theorem 4.2.1 *(i) The number Q_n of squarefree polynomials of degree n in $\mathbb{F}_q[x]$ is*

$$Q_0 = 1, \quad Q_1 = q, \quad Q_n = q^{n-1}(q-1) \quad (n \geq 2).$$

(ii) A polynomial of degree $n \geq 2$ in $\mathbb{F}_q[x]$ has a probability $1 - 1/q$ to be squarefree.

PROOF. For part (i), consider the expression

$$Q = \prod_{\omega \in \mathcal{I}} (1 + \omega), \tag{4.1}$$

A formal algebraic expansion produces a sum of monomials. In this way Q generates the collection of polynomials that are the product of distinct elements taken from $\mathcal{I}$, each polynomial being represented by its corresponding monomial. In other words, Q corresponds to the collection of all monic squarefree polynomials. The usual substitution $\omega \mapsto z^{|\omega|}$ applied to Q yields the series denoted by $Q(z)$ that satisfies

$$Q(z) = \prod_{\omega \in \mathcal{I}} (1 + z^{|\omega|}) = \prod_{k=1}^{\infty} (1 + z^k)^{I_k}.$$

Then, the coefficient $Q_n = [z^n]Q(z)$ evaluates to the number of monic squarefree polynomials of degree n. Recall from Eq. (1.3) that $P(z) = 1/(1 - qz)$. The formula $1 + z = (1 - z^2)/(1 - z)$ applied to the infinite products for $P(z), Q(z)$ entails

$$Q(z) = \prod_{k=1}^{\infty} (1 + z^k)^{I_k} = \prod_{k=1}^{\infty} \left(\frac{1 - z^{2k}}{1 - z^k} \right)^{I_k} = \frac{P(z)}{P(z^2)} = \frac{1 - qz^2}{1 - qz}. \tag{4.2}$$

Therefore, using Eq. (4.2), we obtain

$$[z^n]Q(z) = [z^n]\frac{1 - qz^2}{1 - qz}, \qquad \text{and} \qquad Q_n = \begin{cases} 1 & \text{if } n = 0; \\ q & \text{if } n = 1; \\ q^{n-1}(q-1) & \text{if } n \geq 2. \end{cases} \tag{4.3}$$

Part (ii) is an immediate consequence of Eq. (4.3). □

Theorem 4.2.2 *The non-squarefree part of a random polynomial of degree n has degree tending to*

$$C_q = \sum_{k\geq 1} \frac{kI_k}{q^{2k} - q^k}.$$

We have $C_q \sim 1/q$ as $q \to \infty$.

PROOF. The bivariate generating function of the degree of the non-squarefree part of monic polynomials in $\mathbb{F}_q[x]$ is

$$P(z,u) = \prod_{k\geq 1} \left(1 + \frac{z^k}{1 - u^k z^k}\right)^{I_k}.$$

The average degree of the non-squarefree part is obtained by setting $u = 1$ in the derivative of $P(z,u)$ with respect to u. The result follows by singularity analysis (Theorem 1.3.3).

The asymptotic value of C_q as $q \to \infty$ is obtained using the expansion $k\,I_k = q^k + O(q^{k/2})$. □

We observe that a geometrically decaying probability tail holds. This could be shown since the generating function $P(z, 3/2)$ is dominated by $P(z)$ near its dominant singularity.

Theorem 4.2.2 has important consequences for the recursive structure of the `factor` procedure. First, the overall cost of the recursive calls (Step 4 in the top-level procedure) remains $O(1)$ on average. Next, alternative strategies giving the full squarefree factorization have asymptotically equivalent costs. Finally, the ERF phase has a cost dominated by its first gcd.

Theorem 4.2.3 *The expected cost of the ERF phase applied to a random polynomial of degree n is asymptotically that of a single gcd,*

$$\overline{ERF}_n \sim \tau_2 n^2.$$

4.3 Distinct-degree factorization (DDF)

The second stage of our general method is to find the *distinct-degree factorization*, that is, to split a squarefree polynomial a into polynomials whose irreducible factors have the

same degree. This means expressing a in the form $b_1 \cdot b_2 \cdots b_n$ where b_k is the product of irreducible factors of degree k. This algorithm is based on Theorem 3.2.1, that we restate below (see Lidl & Niederreiter 1983, p. 91, Theorem 3.20). The algebraic principle is that the polynomial $x^{q^k} - x \in \mathbb{F}_q[x]$ is the product of all monic irreducible polynomials in $\mathbb{F}_q[x]$ whose degree divides k.

Theorem. *For $i \geq 1$, the polynomial $x^{q^i} - x \in \mathbb{F}_q[x]$ is the product of all monic irreducible polynomials in $\mathbb{F}_q[x]$ whose degree divides i.*

In this section, we analize the algorithm DDF below and some variants.

4.3.1 The basic algorithm

```
procedure DDF(a:polynomial);        [distinct-degree factorization]
        n := deg(a);
        g := a;                     [a is assumed squarefree]
        h := x;
        for k := 1 to n do
1.            h := h^q mod g;
2.            b[k] := gcd(h-x,g);
3.            g := g/b[k];          [a without irred factors of deg <=k]
4.            if b[k]<>1 then h := h mod g fi;
        od;
        return(b[1].b[2]...b[n]);
end;
```

Let $f \in \mathbb{F}_q[x]$ be a random polynomial of degree n. We study the average cost of the basic DDF algorithm on the output of the ERF algorithm. In section 4.2 we showed that the expected value of the degree of the non-squarefree part of f is $O(1)$ with a geometrically decaying probability tail. This implies that the cost of the basic DDF algorithm on the

non-squarefree part of f is $O(1)$, and thus the dominant term comes from the squarefree part of f.

We briefly define constants τ_1, τ_2 and $\lambda(q)$, which will be used in this section. The cost of multiplying two polynomials of degree less than n module f is $\tau_1 n^2$, and the cost of a gcd between f and a polynomial of degree less than n is $\tau_2 n^2$. The number of products needed to compute h^q mod f is denoted by $\lambda(q)$. Using the repeated squaring process (Knuth 1981, p. 441-442), and denoting by $\nu(q)$ the number of ones in the binary representation of q, the number of products needed to compute $h^q \pmod g$ is

$$\lambda(q) = \lfloor \log_2 q \rfloor + \nu(q) - 1. \tag{4.4}$$

In the following theorem, we estimate the cost of the DDF stage.

Theorem 4.3.1 *The expected cost of the basic DDF algorithm is asymptotic to*

$$\overline{DDF}_n \sim \frac{5}{12}\left(\lambda(q)\tau_1 + \tau_2\right) n^3 \qquad \textit{where} \quad \lambda(q) = \lfloor \log_2 q \rfloor + \nu(q) - 1.$$

PROOF. Let C_j be the cost of line number j of the basic algorithm, and $\overline{C_j}$ its expected cost. Thus, the cost of the basic DDF is $C_1 + C_2 + C_3 + C_4$. Since the mean number of factors of f is $O(\log n)$ (Theorem 2.2.1), we have $\overline{C_3} + \overline{C_4} = O(n^2 \log n)$.

Now, we concentrate on C_1 and C_2. Let d_k denote the degree of polynomial g when the kth iteration of the main loop starts. From the algorithm, it is clear that the parameter d_k is also the sum of the degrees of the distinct factors of f with degree $\geq k$. The quantity $C_1 + C_2$ is equal to $(\lambda(q)\tau_1 + \tau_2)\sum_{k\geq 1} d_k^2$. We denote

$$C = \sum_{k\geq 1} d_k^2. \tag{4.5}$$

The bivariate generating function associated with parameter d_k can be computed by means of the basic decompositions. Indeed, let $\omega \in \mathcal{I}$ be any monic irreducible polynomial in $\mathbb{F}_q$, and u be a marking variable. Formally, the collection of polynomials marking the degree of the distinct irreducible factors of degree $j \geq k$ can be written as

$$\prod_{\omega, |\omega|<k} \frac{1}{1-\omega} \cdot \prod_{\omega, |\omega|=j\geq k} \left(1 + u^j\,\omega + u^j\,\omega^2 + \cdots\right),$$

where $|\omega|$ is the degree of $\omega \in \mathcal{I}$. Let z be a formal variable. The substitution $\omega \mapsto z^{|\omega|}$ produces the bivariate generating function

$$P_k(z,u) = \prod_{j<k} \left(\frac{1}{1-z^j}\right)^{I_j} \prod_{j\geq k} \left(1 + u^j \frac{z^j}{1-z^j}\right)^{I_j}.$$

As usual (see Section 1.3.2), the coefficient $[z^n u^\ell] P(z,u)$ gives the number of polynomials of degree n with total degree ℓ among the distinct irreducible factors whose degrees are not less than k. We observe that the counting of *distinct* factors is needed since in the factorization chain, the input polynomial for the DDF procedure is the output polynomial of the ERF procedure.

The expected value of $C = \sum_{k\geq 1} d_k^2$ is given by

$$\frac{1}{q^n}[z^n]Q(z), \qquad Q(z) = \sum_{k\geq 1} \left(\frac{\partial^2 P_k}{\partial u^2}(z,1) + \frac{\partial P_k}{\partial u}(z,1)\right).$$

Recalling from Eq. (1.3) that $P(z) = \prod_{j\geq 1} (1-z^j)^{-I_j}$, and denoting

$$Q_1(z) = \sum_{j\geq 1} j^3 I_j (z^j - z^{2j}) \qquad \text{and} \qquad Q_2(z) = \sum_{k\geq 1} \left(\sum_{j\geq k} j I_j z^j\right)^2,$$

we can write $Q(z) = P(z)\,(Q_1(z)+Q_2(z))$. The function $Q_1(z)$ can be explicitly expressed in terms of $I(z) = \sum_{j\geq 1} I_j\, z^j$ and the operator $\Theta = z\,\frac{d}{dz}$, as

$$Q_1(z) = \Theta^3 I(z) - \Theta^3 I(z^2).$$

We now show that $I(z) = \sum_{k=1}^{\infty} I_k z^k \sim \ln \frac{1}{1-qz}$. Indeed, using Eq. (1.4),

$$\begin{aligned} I(z) &= \sum_k I_k z^k = \sum_k \left(\frac{1}{k} \sum_{\ell|k} \mu(\ell)\, q^{k/\ell}\right) z^k \\ &= \sum_\ell \frac{\mu(\ell)}{\ell} \left(\sum_{k,\,\ell|k} \frac{\ell}{k} q^{k/\ell} z^k\right) \\ &= \sum_\ell \frac{\mu(\ell)}{\ell} \ln \frac{1}{1-qz^\ell}, \end{aligned} \tag{4.6}$$

where the last equality holds since

$$\ln \frac{1}{1-qz^\ell} = \sum_i \frac{q^i}{i} z^{i\ell} = \sum_{k,\,\ell|k} \frac{\ell}{k} q^{k/\ell} z^k.$$

Then, $Q_1(z)$ has an isolated singularity of the algebraic-logarithm type at $z = 1/q$ and in the neighborhood of this point,

$$P(z)Q_1(z) \sim \frac{2}{(1-qz)^4}.$$

Since $\Gamma(4) = 6$, the transfer lemma (Theorem 1.3.3) implies $[z^n]P(z)Q_1(z) \sim q^n n^3/3$.

We focus now on $Q_2(z)$. Changing z to z/q in $Q_2(z)$ and using Eq. (1.5) that states $kI_k = q^k + O(q^{k/2})$, we obtain

$$\sum_{j\geq k} jI_j \left(\frac{z}{q}\right)^j = \sum_{j\geq k} \left(z^j + O\left(\frac{z^j}{q^{j/2}}\right)\right) = \frac{z^k}{1-z} + S_k(z),$$

where $S_k(z) = O\left(\frac{z^k}{q^{k/2}}\right)$. Thus,

$$Q_2\left(\frac{z}{q}\right) = \sum_{k\geq 1}\left(\frac{z^k}{1-z} + S_k(z)\right)^2 = \frac{z^2}{1+z}\frac{1}{(1-z)^3} + \frac{2\sum_{k\geq 1} z^k S_k(z)}{1-z} + \sum_{k\geq 1} S_k^2(z),$$

where the first term comes from

$$\sum_{k\geq 1} \frac{z^{2k}}{(1-z)^2} = \frac{1}{(1-z)^2}\left(\frac{1}{1-z^2} - 1\right) = \frac{1}{(1-z)^2}\frac{z^2}{1-z^2}.$$

Since $S_k(z)$ is uniform in every closed disk contained in the open disk $|z| < \sqrt{q}$, the generating function $Q_2(z/q)$ has dominant singularities at $z = 1/q$ and $z = -1/q$. The singularity in $z = 1/q$ has the greatest multiplicity, and therefore it is the dominant as we have seen in Theorem 2.4.1. Therefore, we have proven that

$$Q_2(z/q) \sim \frac{1}{2(1-z)^3}.$$

This implies that $[z^n]P(z)Q_2(z) \sim \frac{1}{2(1-z)^4}$. The transfer lemma (Theorem 1.3.3) shows $[z^n]P(z)Q_2(z) \sim q^n n^3/12$. Alternatively, a Tauberian theorem could have been used since $Q_2(z/q)$ has positive coefficients, and thus, the coefficients of $P(z/q)Q_2(z/q)$ are monotonic.

We have proven that $\overline{C} \sim n^3\left(\frac{1}{3} + \frac{1}{12}\right)$, and this completes the proof. □

4.3.2 Stopping at $n/2$

A natural stopping rule for the basic algorithm is at $n/2$, since at this step the remaining factor is either 1 or irreducible.

Theorem 4.3.2 *The expected cost of the basic DDF algorithm using the stopping rule at $n/2$ is asymptotic to*

$$\overline{DDF}_n^{(HD)} \sim \frac{5}{16}\left(\lambda(q)\tau_1 + \tau_2\right) n^3.$$

PROOF. The cost of this DDF is given by $(\lambda(q)\tau_1 + \tau_2)\sum_{i\le n/2} d_i^2$. Let us denote by $C^{(1)} = \sum_{i\le n/2} d_i^2$, and by D_1 the highest degree among the irreducible factors of f. Consider the difference $C^{(2)} = C - C^{(1)}$, where C is defined in Eq. (4.5). If the largest degree D_1 is not larger than $n/2$, necessarily $C^{(2)} = 0$. Otherwise, since there is only one possible irreducible factor of degree greater than $n/2$, $C^{(2)} = \left(D_1 - \left\lfloor \frac{n}{2} \right\rfloor\right) D_1^2$. Therefore, we have shown that

$$C^{(2)} = \begin{cases} \left(D_1 - \left\lfloor \frac{n}{2} \right\rfloor\right) D_1^2 & \text{if } D_1 > n/2; \\ 0 & \text{if } D_1 \le n/2. \end{cases}$$

The average value of $C^{(1)}$ will be deduced from the corresponding values for C and $C^{(2)}$. The average value of $C^{(2)}$ over all polynomials of degree n is given by

$$\mu = \sum_{n/2<k\le n} \Pr(D_1 = k)\left(k - \left\lfloor \frac{n}{2} \right\rfloor\right) k^2.$$

The probability $\Pr(D_1 = k)$ is derived from the generating function $\chi_k(z)$ of polynomials whose factors have all degree $\le k$ as

$$\Pr(D_1 = k) = \frac{1}{q^n}[z^n]\left(\chi_k(z) - \chi_{k-1}(z)\right), \qquad \chi_k(z) = \prod_{j=1}^{k}\left(\frac{1}{1-z^j}\right)^{I_j}.$$

Recall from Eq. (1.3) that $P(z) = \prod_{j\ge 1}(1-z^j)^{-I_j}$. When $k > n/2$, the n-th coefficient of $\chi_k(z) - \chi_{k-1}(z)$ is obtained from

$$\chi_k(z) - \chi_{k-1}(z) = P(z)\left(1 - (1-z^k)^{I_k}\right)\prod_{j>k}(1-z^j)^{I_j} = P(z)\left(I_k z^k + O(z^{n+1})\right)$$

Finally, we have proven that

$$\Pr(D_1 = k) = \frac{I_k}{q^k}, \quad \frac{n}{2} < k \le n.$$

We conclude that $\Pr(D_1 = k) \sim 1/k$, and that

$$\mu \sim \sum_{n/2<k\le n}\left(k - \left\lfloor \frac{n}{2} \right\rfloor\right) k \sim \frac{5}{48} n^3.$$

By Theorem 4.3.1, $C \sim \frac{5}{12} n^3$, so $C^{(1)}$ is asymptotic to $\frac{15}{48} n^3$, proving the theorem. Thus, the half-degree rule results in a savings of 25% asymptotically. □

4.3.3 Early abort strategy

A better strategy than the previous ones consists of stopping the main loop of DDF at the kth iteration, where $2k$ is greater than the degree of the remaining factor. At this moment, the remaining factor must be irreducible. This strategy is called the *early abort strategy.* In this subsection we analyze this variant of the DDF algorithm.

Theorem 4.3.3 *The expected cost of the "early-abort rule" DDF phase is asymptotic to*

$$\overline{DDF}_n^{(EA)} \sim \delta\,(\lambda(q)\tau_1 + \tau_2)\,n^3, \qquad \textit{where} \quad \delta = 0.2668903307\ldots,$$

$$\delta = \frac{5}{12} - \frac{1}{3}\int_0^\infty e^{-2x}\exp\left(-\int_x^\infty \frac{e^{-u}}{u}\,du\right)\frac{1-x^2}{x}\,dx.$$

PROOF. Let us denote by D_1 the degree of the highest irreducible factor of f, and by D_2 the degree of the second highest irreducible factor of f (set $D_2 = 0$ if f is the power of an irreducible polynomial). In accordance with our pilot algorithm in 4.1, we count only distinct factors of f. The condition for k being the early abort iteration is given by

- $k = \lfloor D_1/2 \rfloor + 1$ if $D_1 > 2D_2$;
- $k = D_2 + 1$ if $D_1 \le 2D_2$.

In other terms, we stop the process at step $k = \max\{\lfloor D_1/2 \rfloor, D_2\} + 1$. The analysis now has some analogy to that of integer factoring given by Knuth & Trabb-Pardo (1976). In their analysis the process is stopped at step $k = \max\{\lfloor \sqrt{D_1} \rfloor, D_2\} + 1$, D_1 being the largest prime factor of a number n, and D_2 its second largest prime factor.

The cost of the DDF with this stopping rule becomes $(\lambda(q)\tau_1 + \tau_2)\,C^{(3)}$, where

$$C^{(3)} = \sum_{k \le \max\{\lfloor D_1/2 \rfloor, D_2\}} d_k^2.$$

Consider the difference $C^{(4)} = C - C^{(3)}$, that is,

$$C^{(4)} = \begin{cases} \left(D_1 - \left\lfloor \frac{D_1}{2} \right\rfloor\right) D_1^2 & \text{if } D_1 > 2D_2; \\ (D_1 - D_2)D_1^2 & \text{if } D_1/2 \le D_2. \end{cases}$$

The generating function of polynomials for which there is only one irreducible factor of degree D_1 with $D_1 \geq 2D_2$ is given by

$$\phi_{D_1}(z) = \left[\prod_{1\leq \ell < D_1/2} \left(\frac{1}{1-z^\ell}\right)^{I_\ell}\right] I_{D_1}\left(\frac{z^{D_1}}{1-z^{D_1}}\right),$$

and the generating function of polynomials for which $D_2 < D_1 \leq 2D_2$ is given by

$$\psi_{D_1,D_2}(z) = \left[\prod_{1\leq \ell < D_2} \left(\frac{1}{1-z^\ell}\right)^{I_\ell}\right] \left[\left(\frac{1}{1-z^{D_2}}\right)^{I_{D_2}} - 1\right] \left[I_{D_1}\frac{z^{D_1}}{1-z^{D_1}}\right].$$

Observe that we do not need to take the case $D_1 = D_2$ into account since it contributes 0 to $C^{(4)}$. Therefore, the generating function of the sum of the parameter $C^{(4)}$ over all polynomials of degree n is

$$\Phi(z) = \sum_{D_1} \left(D_1 - \left\lfloor \frac{D_1}{2} \right\rfloor\right) D_1^2\, \phi_{D_1}(z) + \sum_{D_2 < D_1 \leq 2D_2} (D_1 - D_2) D_1^2\, \psi_{D_1,D_2}(z).$$

The analysis of this generating function near its positive dominant singularity $1/q$ is done by approximating sums with integrals (Euler-Maclaurin summation). First, we look at the behavior of the generating functions $\phi_{D_1}(z)$ and $\psi_{D_1,D_2}(z)$. Let us rewrite $\phi_{D_1}(z)$ as

$$\phi_{D_1}(z) = \frac{I_{D_1} z^{D_1}}{1-z^{D_1}} \frac{1}{1-qz} \prod_{\ell \geq D_1/2} \left(\frac{1}{1-z^\ell}\right)^{I_\ell}.$$

Expanding logarithms, approximating sums with integrals, and changing variables $z = e^{-t}/q$, we have as $t \to 0$

$$\begin{aligned}
\phi_{D_1}(z) &= \frac{I_{D_1} z^{D_1}}{1-z^{D_1}} \frac{1}{1-qz} \exp\left(-\sum_{\ell \geq D_1/2} I_\ell \log \frac{1}{1-z^\ell}\right) \\
&\sim \frac{e^{-tD_1}}{D_1} \frac{1}{t} \exp\left(-\int_{D_1/2}^{\infty} \frac{e^{-xt}}{x}\, dx\right) \\
&= \frac{e^{-D_1 t}}{tD_1} \exp\left(-\int_{tD_1/2}^{\infty} \frac{e^{-u}}{u}\, du\right).
\end{aligned}$$

Thus,

$$\begin{aligned}
\sum_{D_1} \left(D_1 - \left\lfloor \frac{D_1}{2} \right\rfloor\right) D_1^2\, \phi_{D_1}(z) &\sim \sum_{D_1} \frac{D_1^2}{2t} e^{-D_1 t} \exp\left(-\int_{tD_1/2}^{\infty} \frac{e^{-u}}{u}\, du\right) \\
&\sim \frac{1}{2t} \int_0^\infty x^2 e^{-tx} \exp\left(-\int_{tx/2}^{\infty} \frac{e^{-u}}{u}\, du\right) dx \\
&= \frac{4}{t^4} \int_0^\infty x^2 e^{-2x} \exp\left(-\int_{x}^{\infty} \frac{e^{-u}}{u}\, du\right) dx.
\end{aligned}$$

Analogously for $\psi_{D_1,D_2}(z)$, we obtain

$$\begin{aligned}\psi_{D_1,D_2}(z) &= \left(1-\left(1-z^{D_2}\right)^{I_{D_2}}\right)\frac{I_{D_1}z^{D_1}}{1-z^{D_1}}\frac{1}{1-qz}\prod_{\ell>D_2}\left(\frac{1}{1-z^\ell}\right)^{I_\ell}\\ &\sim I_{D_2}z^{D_2}\,I_{D_1}z^{D_1}\frac{1}{1-qz}\exp\left(-\sum_{\ell>D_2}I_\ell\log\frac{1}{1-z^\ell}\right)\\ &\sim \frac{e^{-(D_1+D_2)t}}{D_1D_2}\frac{1}{t}\exp\left(-\int_{tD_2}^{\infty}\frac{e^{-u}}{u}\,du\right).\end{aligned}$$

Thefore,

$$\sum_{D_2<D_1\le 2D_2}(D_1-D_2)D_1^2\,\psi_{D_1,D_2}(z)\sim\frac{1}{t}\int_{y<x<2y}(x-y)x^2\frac{e^{-(x+y)t}}{xy}\exp\left(-\int_{ty}^{\infty}\frac{e^{-u}}{u}\,du\right)dx\,dy.$$

Changing variables tx by x and ty by y, and applying Fubini's theorem, the above expression is equal to

$$\frac{1}{t^4}\int_0^\infty e^{-2y}\exp\left(-\int_y^\infty\frac{e^{-u}}{u}\,du\right)\frac{2+y-e^{-y}(2+3y+2y^2)}{y}\,dy.$$

Using the notation

$$\begin{aligned}c_1 &= \int_0^\infty x^2e^{-2x}\exp\left(-\int_x^\infty\frac{e^{-u}}{u}\,du\right)dx,\\ c_2 &= \int_0^\infty e^{-2y}\exp\left(-\int_y^\infty\frac{e^{-u}}{u}\,du\right)\frac{2+y-e^{-y}(2+3y+2y^2)}{y}\,dy,\end{aligned}$$

and $c_0=4c_1+c_2$, we have that, when $z\to 1^-$,

$$\Phi\left(\frac{z}{q}\right)\sim\frac{c_0}{(1-z)^4}.$$

Thus, since $\Gamma(4)=6$, the application of Theorem 1.3.3 implies that the expected value of $C^{(4)}$ over polynomials of degree n is asymptotic to $\frac{c_0}{6}n^3$. Let δ be the constant $\delta=5/12-c_0/6=0.2668903307\ldots$. Then, the expected value of the cost of DDF with the early abort strategy is asymptotic to $\delta\,(\lambda(q)\tau_1+\tau_2)\,n^3$. □

The above theorem implies that the cost when using the early abort strategy is about 36% less than the cost of basic DDF, and 15% less than the cost when using the previous stopping rule.

4.4 The output configuration of DDF

The DDF procedure does not completely factor a polynomial that has different irreducible factors of the same degree. However, we show in the following results that "most" of the factoring has been completed after DDF. First, the DDF factorization is a complete factorization with asymptotic probability greater than 1/2 (Theorem 4.4.1). Next, the number of calls to the subsequent phase of EDF, that is to say the number of degree values for which more than one factor occurs, is only $O(1)$. The sum of the degrees where this happens (the total degree of the fragments passed to EDF) is $O(\log n)$. However, this total degree has a fairly large variability so that the cost of EDF (to be analyzed in the next section) is comparatively small, but not completely negligible. Theorem 4.4.2 quantifies these phenomena. They are established here by means of a hybridization of singularity analysis and Darboux's method (Flajolet & Gourdon, personal communication). This is a general technique of independent interest that we explain in some detail in the following theorem.

Theorem 4.4.1 *The asymptotic probability of a complete distinct-degree factorization is*

$$c_q = \prod_{k\geq 1} \left(1 + \frac{I_k}{q^k - 1}\right) (1 - q^{-k})^{I_k}.$$

Some interesting values for c_q are: $c_2 \doteq 0.6656$, $c_3 \doteq 0.6123$, $c_5 \doteq 0.5861$, $c_{47} \doteq 0.5635$, $c_{257} \doteq 0.5618$, and $c_\infty = e^{-\gamma} \doteq 0.5614$, where γ is Euler's constant.

PROOF. We start sketching the analysis of the analogous problem for permutation. This will serve as an introduction to the hybridization method. In the limit case $q = \infty$, this study corresponds to the problem of polynomials with irreducible factors of different degrees (see Section 1.3.4).

The probability that a permutation of length n has all its cycles of different lengths is $[z^n]F(z)$, where the generating function $F(z)$ is susceptible to a variety of expressions:

$$\begin{aligned} F(z) &= \prod_{k=1}^{\infty} \left(1 + \frac{z^k}{k}\right) \\ &= \frac{1+z}{1-z} \prod_{k=2}^{\infty} \left(1 + \frac{z^k}{k}\right) e^{-z^k/k} \end{aligned}$$

$$= \left(\frac{1+z}{1-z}\exp\left(-\frac{1}{2}Li_2\left(z^2\right)\right)\right)\cdot\left(\prod_{k=2}^{\infty}\left(1+\frac{z^k}{k}\right)e^{-z^k/k+z^{2k}/(2k^2)}\right). \qquad (4.7)$$

Here $Li_2(z) := \sum_{\ell\geq 1} z^\ell/\ell^2$ is the classical *dilogarithm* function. Let us denote by $S(z)$ and $R(z)$ the first and second factor, respectively, in the bottom equality of Eq. (4.7), that is,

$$S(z) = \frac{1+z}{1-z}\exp\left(-\frac{1}{2}Li_2\left(z^2\right)\right) \quad \text{and} \quad R(z) = \prod_{k=2}^{\infty}\left(1+\frac{z^k}{k}\right)e^{-z^k/k+z^{2k}/(2k^2)}.$$

$S(z)$ satisfies the conditions of singularity analysis, while $R(z)$ is continuously differentiable (of class $\mathcal{C}^1$) on the closed unit disc $\overline{D}$, since it is of the form $e^{r(z)}$ where the coefficient $[z^n]r(z)$ is $O(n^{-3})$.

We thus have a situation where the generating function of interest is the product of a singular part $S(z)$ that satisfies strong analyticity properties, and of a function $R(z)$ of the Darboux type that is smooth enough on the closed unit disc $\overline{D}$ (see Theorems 1.3.3, and 1.3.4). We only need to justify the fact that dominant asymptotics of the coefficients $[z^n]F(z)$ can be extracted as though $R(z)$ was analytic on $\overline{D}$.

The singular expansions of $S(z)$ at ± 1 are readily found to be:

$$\begin{aligned} S_{+1}(z) &= 2e^{-\zeta(2)/2}\left(\frac{1}{1-z} - \log 2 + \frac{1}{2} - \log(1-z) + O((1-z)\log(1-z))\right) \\ S_{-1}(z) &= \frac{1}{4}e^{-\zeta(2)/2}(2(z+1) + O((z+1)\log(z+1)), \end{aligned}$$

where $\zeta(2) = \sum_{k=1}^{\infty}\frac{1}{k^2} = \frac{\pi^2}{6}$. The error terms are $\mathcal{C}^0$ on $\overline{D}$. In summary, we have found that

$$F(z) = R(z)\left(2e^{-\zeta(2)/2}\frac{1}{1-z} + t(z)\right),$$

for some $t(z)$ that is $\mathcal{C}^0$, and $R(z)$ that was already proved to be $\mathcal{C}^1$ on $\overline{D}$.

By using the expansion $R(z) = R(1) + O(z-1)$ at 1, with an error term that is now $\mathcal{C}^0$, one finds by an application of Darboux's method (see Theorem 1.3.4)

$$[z^n]F(z) = 2R(1)e^{-\zeta(2)/2} + o(1), \qquad R(1) := \prod_{k=2}^{\infty}\left(1+\frac{1}{k}\right)e^{-1/k+1/(2k^2)}.$$

Using the harmonic sum approximation in Eq. (1.4.1), we obtain

$$[z^n]F(z) = \prod_{k=1}^{\infty}\left(1+\frac{1}{k}\right)\exp\left(-\sum_{k\geq 1}\frac{1}{k}\right)\exp\left(\frac{1}{2}\sum_{k\geq 1}\frac{1}{k^2}\right)\exp(-\zeta(2)/2) + o(1)$$

$$\begin{aligned} &= \exp\left(\sum_{k\geq 1}\left(\ln\left(1+\frac{1}{k}\right)-\frac{1}{k}\right)\right)+o(1)\\ &= \exp\left(\lim_{k\to\infty}\ln(k+1)-H_k\right)+o(1)\\ &= e^{-\gamma}+o(1), \end{aligned}$$

where γ is the Euler's constant. This result was already established in the permutational model case by Greene & Knuth (1990) using a Tauberian argument.

The method used here is then a hybrid of singularity analysis and Darboux's method. It could be used to derive a complete asymptotic expansion, with primitive roots of unity that intervene with smaller and smaller weights and lead to fluctuating terms involving roots of unity. In this chapter, we assume its principle as granted. The consequence in the sequel is the following:

> *If a function can be written as the product of a smooth function and a function with algebraic-logarithmic singularities, then dominant asymptotics of its coefficients result by direct translation from the local behavior at the dominant positive singularity.*

We next turn to the case of polynomials over a finite field $\mathbb{F}_q$, q fixed. The generating function of polynomials with irreducible factors all of distinct degrees is

$$F(z)=\prod_{k\geq 1}\left(1+I_k(z^k+z^{2k}+z^{3k}+\cdots)\right)=\prod_{k\geq 1}\left(1+I_k\frac{z^k}{1-z^k}\right). \tag{4.8}$$

An equivalent form that reveals the pole-like singularity at $z = 1/q$ is obtained by multiplying each term of the product Eq. (4.8) by $(1-z^k)^{I_k}$,

$$F(z)=\frac{1}{1-qz}\prod_{k\geq 1}\left(1+I_k\frac{z^k}{1-z^k}\right)(1-z^k)^{I_k}.$$

Thus, as $z \to 1/q$, we have

$$F(z)\sim\frac{c_q}{1-qz},\qquad c_q=\prod_{k=1}^{\infty}\left(1+I_k\frac{1}{q^k-1}\right)(1-q^{-k})^{I_k}, \tag{4.9}$$

and the preceding discussion applies with the role of the dilogarithm function played by

$$\Lambda_2(z)=\sum_{k=1}^{\infty}\left(\frac{I_k}{q^k-1}\right)^2 z^k.$$

The hybrid method used for the permutation model then yields

$$[z^n]F\left(\frac{z}{q}\right) = c_q + o(1),$$

which, by Eq. (4.9), is our statement. □

Theorem 4.4.1 was independently obtained by Knopfmacher & Warlimont (1995) and by Flajolet, Gourdon & Panario (1996). Both papers are however rather different. We use generating functions as above but a Tauberian argument for the asymptotic analysis. Knopfmacher & Warlimont (1995) uses elementary techniques and deduces constructive bounds. The method developed here is less constructive but geared towards full asymptotic expansions. This method has been successfully used by Gourdon (1996) to solve the Golomb-Knuth conjecture (Knuth 1973b, Ex. 1.3.3.23) regarding the expectation of maximal cycle lengths in random permutations.

Theorem 4.4.2 *The number N_0 of degree values for which there is more than one irreducible factor in the polynomial produced by DDF has an average that is asymptotic to the constant*

$$\mu_0 = \sum_{k\geq 1}(1-q^{-k})^{I_k}\left((1-q^{-k})^{-I_k} - 1 - \frac{I_k q^{-k}}{1-q^{-k}}\right).$$

The total degree N_1 of the corresponding polynomials has expectation $\log n + O(1)$, *and standard deviation of approximate* $\sqrt{n}$.

PROOF. Given a family $\mathcal{F}$ of elements, we formally generates all multisets via the expression

$$\prod_{\omega\in\mathcal{F}} \frac{1}{1-\omega}.$$

The expression

$$1 + \sum_{\omega\in\mathcal{F}} \frac{\omega}{1-\omega} + u\left(\prod_{\omega\in\mathcal{F}} \frac{1}{1-\omega} - 1 - \sum_{\omega\in\mathcal{F}} \frac{\omega}{1-\omega}\right)$$

formally generates all multisets each affected by a coefficient u if there are different elements, and 1 otherwise. This applies to the class of irreducible polynomials of each degree n, taking $\mathcal{F} = \mathcal{I}_n$. Thus, the bivariate generating function of the number N_0 of degree values for which there are repeated elements is

$$P_0(z,u) = \prod_{k\geq 1}\left(1 + \frac{I_k z^k}{1-z^k} + u\left((1-z^k)^{-I_k} - 1 - \frac{I_k z^k}{1-z^k}\right)\right). \tag{4.10}$$

The logarithmic derivative with respect to u satisfies

$$\left.\frac{\partial P_0(z,u)}{\partial u}\right|_{u=1} = [z^n]\frac{1}{1-qz}\sum_{k\geq 1}(1-z^k)^{I_k}\left((1-z^k)^{-I_k}-1-\frac{I_k z^k}{1-z^k}\right). \tag{4.11}$$

By the general discussion of the hybrid singularity analysis and Darboux method, the quantity N_0 has an expectation that is asymptotic to the limit μ_0 of the right hand side in Eq. 4.11 as $z \to 1/q$.

For the sum N_1 of the degrees of these polynomials, an adaptation of Eq. (4.10) yields the bivariate generating function

$$P_1(z,u) = \prod_{k\geq 1}\left(\left(1+u^k\frac{z^k}{1-z^k}\right)^{I_k} - (u^k-1)I_k\frac{z^k}{1-z^k}\right).$$

We only discuss briefly the first moment of N_1. The mean value is $q^{-n}[z^n]R(z)$, where $R(z)$ equals $\left.\frac{\partial P_0(z,u)}{\partial u}\right|_{u=1}$. Thanks to the expansion $I_k = q^k/k + O(q^{k/2}/k)$, near $z = 1/q$, $R(z)$ is asymptotic to $(1-qz)^{-1}\log(1-qz)^{-1}$. Thus, the expectation of N_1 taken over polynomials of degree n is $q^{-n}[z^n]R(z) \sim \log n$. The second factorial moment of N_1 is obtained by a further differentiation of $P_1(z,u)$. □

The analysis of N_1 was given in Flajolet, Gourdon & Panario (1996), and Knopfmacher (1996) has independently obtained an estimate of the first two moments of N_0.

It should be clear that the hybrid asymptotic method has great flexibility. As a final illustration, we discuss a question by von zur Gathen. We consider the quantity $\widetilde{N}$ that is the largest degree for which two or more factors occur. The generating function of polynomials such that $\widetilde{N} \leq r$ is clearly,

$$F^{(r)}(z) = \prod_{k\leq r}(1-z^k)^{-I_k}\prod_{k>r}\left(1+I_k\frac{z^k}{1-z^k}\right).$$

Thus, the probability of $\widetilde{N} \leq r$ is asymptotic, as $n \to \infty$, to

$$c_q^{(r)} = \prod_{k>r}(1-q^{-k})^{-I_k}\left(1+I_k\frac{q^{-k}}{1-q^{-k}}\right), \tag{4.12}$$

and when $q \to \infty$, these constants tend to

$$c_\infty^{(r)} = e^{-\gamma}\prod_{k\leq r}\left(1+\frac{1}{k}\right)^{-1}e^{1/k}. \tag{4.13}$$

We have $1 - c_q^{(r)} = O(1/r)$ for all q, and some representative values with $q = \infty$ are:

$$c_\infty = 0.5614, \quad c_\infty^{(1)} = 0.7631, \quad c_\infty^{(2)} = 0.8387, \quad c_\infty^{(5)} = 0.9179, \quad c_\infty^{(10)} = 0.9549.$$

Thus, Eqs. (4.12), and (4.13) give the following picture in the asymptotic limit. A random polynomial has a small number, $O(1)$, of "colliding" degrees; the largest colliding degree has a probability distribution with a tail that decays like $O(1/r^2)$. Because of this slow tail decay, the largest colliding degree alone has a first moment that is $O(\sum_r r^{-1}) = O(\log n)$, and a second moment that is $O(\sum_r 1) = O(n)$. These observations seem to be consistent with the facts asserted in Theorem 4.4.2.

4.5 Equal-degree factorization (EDF)

In this section we concentrate on an algorithm for factoring a monic squarefree univariate polynomial f over a finite field $\mathbb{F}_q$ all of whose irreducible factors have the same (known) degree k. The probabilistic algorithm we present here is based on Cantor & Zassenhaus (1981). The analysis combines a recursive partioning problem akin to digital trees —also known as "tries" Knuth (1973a)— together with estimates on the degrees of irreducible factors of random polynomials (Knopfmacher & Knopfmacher 1993). The final result is that the global cost of EDF is quadratic, a sharp contrast with the cubic cost of DDF. For convenience, we first assume that q is odd, and relegate to Section. 4.5.4 the case of a characteristic equal to 2.

The EDF algorithm is described in Fig. 4.2 and we briefly recall its principle here (see also Section 3.2.3). Let b be a polynomial that is a product of j irreducible factors $f_1, \ldots, f_j$, each of degree k. The Chinese remainder theorem (see Theorem 1.4.4) implies the ring homomorphism

$$\mathbb{F}_q[x]/(b) \cong \mathbb{F}_q[x]/(f_1) \times \cdots \times \mathbb{F}_q[x]/(f_j).$$

Under this isomorphism, a random element h of $\mathbb{F}_q[x]/(b)$ is associated to a j-tuple $(h_1, \ldots, h_j)$, where each h_i is a random element of $\mathbb{F}_q[x]/(f_i)$.

We mention the splitting principle that eventually enables us to isolate the various f_i. Since each f_i is irreducible, the multiplicative group of each component $\mathbb{F}_q[x]/(f_i)$ is a

```
procedure EDF(b : polynomial, k : integer);
[b is a product of irreducibles of degree k]
        if degree(b) <= k then return(b) fi;
        h := randpoly(degree(b)-1);
1.      a := h^((q^k-1)/2)-1 mod b;
2.      d := gcd(a,b);
        return(EDF(d,k).EDF(b/d,k));
end;
```

Figure 4.2: The equal degree factorization algorithm (EDF).

field isomorphic to $\mathbb{F}_{q^k}$. Since this group is cyclic, there are the same number $(q^k-1)/2$ of squares and nonsquares. The test $h_i^{(q^k-1)/2} = 1$ discriminates the squares in this multiplicative group. Thus, taking a random h and computing $a := h^{(q^k-1)/2} - 1 \bmod b$, we see that $\gcd(a,b)$ "extracts" the product of all the f_i for which h is a square in $\mathbb{F}_q[x]/(f_i)$.

From the probabilistic standpoint, a component h_i that is random in $\mathbb{F}_q[x]/(f_i)$ has probability $\alpha = \frac{1}{2} - \frac{1}{2q}$ of being a square and the dual probability, $\beta = \frac{1}{2} + \frac{1}{2q}$, of being a nonsquare. The (small) difference between α and β is accounted for by the possibility of having noninvertible components.

In summary, irreducible factors of each degree can be extracted successively. For each degree where two or more such factors occur, the recursive splitting process will be launched. The analysis of the complete EDF phase (Section 4.5.3), then requires a purely combinatorial analysis of what takes place at each degree (Section 4.5.2) combined with an estimate of the probability that there are j irreducible factors of degree k in a random polynomial of degree n. These probabilities give interesting information on random polynomials and have been obtained by Knopfmacher & Knopfmacher (1993) whose results we recall in Section 4.5.1.

4.5.1 Irreducible factors of each degree

Let $\kappa_n(k)$ be the random variable counting the number of distinct irreducible factors of degree k in a random polynomial of degree n. We consider here k as fixed. The bivariate generating function for the number of irreducible polynomials of degree k is

$$\begin{aligned} Q_k(z,u) &= \left(1+u\frac{z^k}{1-z^k}\right)^{I_k}\prod_{n\neq k}(1-z^n)^{-I_n} \\ &= \frac{1}{1-qz}(1+(u-1)z^k)^{I_k}. \end{aligned}$$

The asymptotics as $n \to \infty$ are derived from the polar singularity at $z = 1/q$. The probability generating function of the distribution is, as $n \to \infty$,

$$(1+(u-1)q^{-k})^{I_k},$$

that is to say, the probability generating function of a binomial distribution $\mathcal{B}(I_k, q^{-k})$. As observed by Knopfmacher & Knopfmacher (1993), this asymptotic formula is exact for $n \geq kI_k$ (see Corollary 2.2.3 or Knopfmacher & Knopfmacher 1993 for the analogous theorem counting multiplicities).

Theorem 4.5.1 *The probability that there are j distinct irreducible factors of degree k in a random polynomial of degree n is, for n large enough ($n \geq kI_k$), given by the binomial distribution $\mathcal{B}(I_k, q^{-k})$, namely,*

$$\Pr\{\kappa_n(k)=j\} = \begin{cases} \binom{I_k}{j} q^{-kj}(1-q^{-k})^{I_k-j} & \text{if} \quad n \geq kI_k, \\ \binom{I_k}{j} q^{-kj}\sum_{\ell=0}^{[n/k]-j}(-1)^\ell q^{-k\ell}\binom{I_k-j}{\ell} & \text{if} \quad n < kI_k. \end{cases}$$

As q becomes large, the binomial probability distribution converges to a Poisson law of parameter $1/k$,

$$\Pr\{\kappa_n(k)=j\} = e^{-1/k}\frac{k^{-j}}{j!},$$

in accordance with the known distribution of cycle lengths in the random permutation model Shepp & Lloyd (1966).

4.5.2 Equal-degree and tries

The analysis of the EDF algorithm is essentially that of tries (Knuth 1973a, Flajolet & Steyaert 1982)

We can regard the EDF phase as an abstract splitting process as follows. Start with a group G formed of k individuals. By flipping coins, separate G randomly into two subgroups, G_0 and G_1, with the probabilities for each element to be sent to G_0 and G_1 being α and β. The process is repeated recursively until all elements have been isolated. Obviously, any such recursive execution is fully described by a binary tree. The probability induced by independent splittings is given by

$$\Pr\left\{|G_0| = k_0 \;\middle|\; |G| = k\right\} = \binom{k}{k_0}\alpha^{k_0}\beta^{k-k_0}. \tag{4.14}$$

This probability coincides with the one for binary tries (Knuth 1973a, Mahmoud 1992).

In general, two models for analyzing tries are considered. The first one is the *uniform model* where each element of G has probability $1/2$ of being sent to G_0, and probability $1/2$ of being sent to G_1. The second one is the *biased model* with probabilities α and β, $\alpha + \beta = 1$, as above. In our particular case, we have a biased model with probabilities $\frac{q-1}{2q}$ for a factor to be in polynomial d, and $\frac{q+1}{2q}$ for a factor to be in polynomial b/d (see the algorithm at the beginning of this section). Therefore, we focus on the biased case.

Amongst the many properties known, we mention:

- the expectation of the number of nodes in the tree (the number of splitting stages) is (see Knuth 1973a)

$$\frac{n}{H}(1 + \epsilon(n)) + o(n), \qquad H = \alpha \log_2 \frac{1}{\alpha} + \beta \log_2 \frac{1}{\beta},$$

  with $\epsilon(n)$ a fluctuating function of amplitude $< 10^{-5}$;

- the expectation of the height of the tree is (see Flajolet & Steyaert 1982)

$$\frac{2}{K}\log_2 n + O(1), \qquad K = \log_2(\alpha^2 + \beta^2)^{-1}.$$

Thus, the trees are fairly balanced and we may expect that the cost of an EDF phase should be close to that of a perfect splitting. The following theorem provides an explicit expression.

Theorem 4.5.2 *The expected cost of the EDF algorithm applied to any product of j irreducible factors of degree k is*

$$C_{j,k} = \left(\frac{1}{2\alpha\beta}j(j-1) + j\sum_{m\geq 0}\sum_{\ell=0}^{m}\binom{m}{\ell}\alpha^{m-\ell}\beta^{\ell}\left(1-(1-\alpha^{m-\ell}\beta^{\ell})^{j-1}\right)\right)(\mu_k\tau_1+\tau_2)k^2, \tag{4.15}$$

where $\mu_k = \lambda((q^k-1)/2) = \left\lfloor \log_2 \frac{q^k-1}{2}\right\rfloor + \nu\left(\frac{q^k-1}{2}\right) - 1$.

PROOF. The cost inside one recursive call of the EDF algorithm is dominated by the cost of computing $h^{(q^k-1)/2} - 1 \bmod b$. Indeed, calling C_i the cost of line number i, the cost of one recursive call of the EDF algorithm is $C_1 + C_2$. Since $\deg b = jk$, the cost of multiplying two polynomials of degree less than jk modulo b is $\tau_1 j^2k^2$, and the cost of a gcd between b and a polynomial of degree less than jk is $\tau_2 j^2k^2$. The number of products needed to compute $h^{(q^k-1)/2} \bmod b$ using the repeated squaring process is μ_k. We have shown that the cost of one recursive call of the EDF algorithm is $(\mu_k\tau_1+\tau_2)k^2j^2$. That is, essentially, $k^3j^2\log_2 q$.

It is convenient to regard the splitting process as a tree t with subtrees t_0, and t_1, and then consider a general cost function of the additive type,

$$C[t] = e_{|t|} + C[t_0] + C[t_1]. \tag{4.16}$$

The function $e_{|t|}$ is a (problem specific) "toll" function that depends on the size $|t|$ of t, with size being the number of nonempty external nodes of the tree. In the polynomial case, this size corresponds to the number j of irreducible factors (of degree k) of the associated polynomial. The toll function is j^2 for $j > 1$, and 0 for $j = 0, 1$ (we ignore for the moment the constant multiplicative factor $k^2(\mu_k\tau_1+\tau_2)$).

The subtree sizes obey the Bernoulli probability of Eq. (4.14). Also the subproblems described by t_0, and t_1 have, by design, the same characteristics as the whole tree. Thus, the expectation c_j of $C[t]$ over trees of size j satisfies the recurrence

$$c_j = e_j + \sum_{\ell=0}^{j}\binom{j}{\ell}\alpha^{\ell}\beta^{j-\ell}(c_\ell + c_{j-\ell}) = e_j + \sum_{\ell=0}^{j}\binom{j}{\ell}(\alpha^{\ell}\beta^{j-\ell} + \alpha^{j-\ell}\beta^{\ell})c_\ell.$$

This translates, in terms of exponential generating functions (see Section 1.3.1),

$$C(z) = \sum_j c_j\frac{z^j}{j!}, \qquad E(z) = \sum_j e_j\frac{z^j}{j!},$$

into the functional equation

$$C(z) = E(z) + e^{\beta z}C(\alpha z) + e^{\alpha z}C(\beta z). \tag{4.17}$$

The iteration of the functional equation (4.17) gives

$$C(z) = \sum_{\ell,m} \binom{m+\ell}{\ell} E(\alpha^m \beta^\ell z) e^{z(1-\alpha^m\beta^\ell)}. \tag{4.18}$$

This expression can be written as

$$C(z) = \sum_{j\geq 0}\sum_{\ell=0}^{j} \binom{j}{\ell} E(\alpha^{j-\ell}\beta^\ell z) e^{z(1-\alpha^{j-\ell}\beta^\ell)}. \tag{4.19}$$

Consider first the case $e_j = j(j-1)$. In this case, $E_1(z) = z^2e^z$. Substituting $E_1(z)$ into the last expression for $C(z)$ we get

$$C(z) = \sum_{j\geq 0}\sum_{\ell=0}^{j} \binom{j}{\ell} \alpha^{2(j-\ell)}\beta^{2\ell} z^2 e^z = z^2e^z \sum_{j\geq 0}(\alpha^2+\beta^2)^j = \frac{1}{2\alpha\beta} z^2e^z. \tag{4.20}$$

We have shown that if we consider $e_j = j(j-1)$ as the toll function, then $c_j = \frac{1}{2\alpha\beta}j(j-1)$.

In order to obtain the cost of the toll function j^2 when $j > 1$ and 0 otherwise, we must sum the cost when the toll function is $e_j = j - \delta_{j,1}$, for which $E_2(z) = z(e^z - 1)$. Substituting $E_2(z)$ in Eq. (4.19), we obtain

$$C(z) = z \sum_{m\geq 0}\sum_{\ell=0}^{m} \binom{m}{\ell} \alpha^{m-\ell}\beta^\ell \left(e^z - e^{z(1-\alpha^{m-\ell}\beta^\ell)}\right).$$

The generating function $C(z)$ associated with the cost of the EDF algorithm is

$$C(z) = \frac{1}{2\alpha\beta} z^2e^z + z \sum_{m\geq 0}\sum_{\ell=0}^{m} \binom{m}{\ell} \alpha^{m-\ell}\beta^\ell \left(e^z - e^{z(1-\alpha^{m-\ell}\beta^\ell)}\right),$$

and, finally

$$c_j = \frac{1}{2\alpha\beta} j(j-1) + j \sum_{m\geq 0}\sum_{\ell=0}^{m} \binom{m}{\ell} \alpha^{m-\ell}\beta^\ell \left(1 - (1-\alpha^{m-\ell}\beta^\ell)^{j-1}\right).$$

It remains to perform the product by $(\mu_k\tau_1 + \tau_2)k^2$ to complete the proof. □

4.5.3 Complete analysis

Completing the analysis of EDF only requires weighting the costs given by Lemma 4.5.2 by the probability $\Pr(\kappa_n(k) = j)$ of finding j irreducible factors of degree k given by Theorem 4.5.1. By Lemma 4.5.2, the cost is of the form $O(j^2k^3)$, and by Theorem 4.5.1, the probabilities are approximately $k^{-j}/j!$. It is then not hard to see that

$$\sum_{k,j} \left(j^2k^3\right) \cdot \left(\frac{k^{-j}}{j!}\right) = O(n^2).$$

The main result of this section gives a firm basis to this heuristic computation and determines the implied constant. In order to prove the theorem we need the following two lemmas.

Lemma 4.5.3 *When $k \to \infty$, we have for all j such that $kj \leq n$*

$$\Pr(\kappa_n(k) = j) = \frac{\binom{I_k}{j}}{q^{kj}} \left(1 + O(1/k)\right),$$

where the $O(1/k)$ is uniform in j. We have also the estimate

$$\Pr(\kappa_n(k) = j) = O\left(\frac{1}{j!k^j}\right).$$

PROOF. When $kj \leq n$, Theorem 4.5.1 yields

$$\Pr(\kappa_n(k) = j) = \frac{\binom{I_k}{j}}{q^{kj}}(1 + \gamma), \qquad \gamma = \sum_{\ell=1}^{N} (-1)^\ell \frac{\binom{I_k - j}{\ell}}{q^{k\ell}}, \qquad N = \min([n/k] - j, I_k - j).$$

When k is large, $\gamma = O(1/k)$, since

$$|\gamma| \leq \sum_{\ell=1}^{I_k - j} \frac{\binom{I_k - j}{\ell}}{q^{k\ell}} = (1 + q^{-k})^{I_k - j} - 1 \leq (1 + q^{-k})^{I_k} - 1 = O(1/k).$$

This proves the first estimate. As for the second, it suffices to write

$$\binom{I_k}{j} \leq \frac{I_k^j}{j!} = O\left(\frac{q^{kj}}{j!k^j}\right),$$

and to use the first estimate of the lemma. □

Lemma 4.5.4 *The average costs $C_{j,k}$ of Theorem 4.5.2 satisfy for all k*

$$C_{0,k} = C_{1,k} = 0, \qquad C_{2,k} = \frac{2}{\alpha\beta}\left(\mu_k\tau_1 + \tau_2\right)k^2,$$

and we have uniformly

$$C_{j,k} = O(j^2k^3).$$

PROOF. The first relations are direct applications of expression (4.15). For the estimate of $C_{j,k}$, we apply the inequality $1-(1-u)^{j-1} \leq (j-1)u$ to the expression (4.15)

$$\begin{aligned} C_{j,k} &\leq \left(\frac{j(j-1)}{2\alpha\beta} + j\sum_{m\geq 0}\sum_{\ell=0}^{m}\binom{m}{\ell}(j-1)\alpha^{2(m-\ell)}\beta^{2\ell}\right)\left(\mu_k\tau_1+\tau_2\right)k^2 \\ &= \frac{j(j-1)}{\alpha\beta}\left(\mu_k\tau_1+\tau_2\right)k^2, \end{aligned}$$

where the last equality holds from Eq. (4.20). Since $\mu_k = O(k)$, the result is proven. □

We are now ready to prove our main theorem of this section.

Theorem 4.5.5 *The expected cost of the EDF algorithm is asymptotic to*

$$C_n = \frac{\tau_1}{\alpha\beta}\sum_{k=1}^{\lfloor n/2\rfloor}\mu_k, \qquad \mu_k = \left\lfloor \log_2\frac{q^k-1}{2}\right\rfloor + \nu\left(\frac{q^k-1}{2}\right) - 1.$$

We have the relation

$$\frac{n(n+2)}{8}\log_2 q - \frac{n}{2} \leq \frac{\alpha\beta}{\tau_1}C_n \leq \frac{n(n+2)}{4}\log_2 q,$$

thus $C_n = \Theta(n^2)$.

PROOF. Let $\overline{EDF_k}$ be the expected cost of the EDF algorithm when the polynomial in the input has factors of degree k. The underlying idea in the proof is that the main contribution to $\overline{EDF_k}$ is due to the case when we have exactly two irreducible factors of degree k.

We have the relation

$$\overline{EDF_k} = \sum_j \Pr(\kappa_n(k) = j)\, C_{j,k},$$

where $C_{j,k}$ is the expected cost of the EDF algorithm when the polynomial in the input has j factors of degree k. When $2k \leq n$, Lemma 4.5.3 and Lemma 4.5.4 entail, as $k \to \infty$.

$$\overline{EDF_k} = C_{2,k}\frac{\binom{I_k}{2}}{q^{2k}}(1 + O(1/k)) + \sum_{j\geq 3} O\left(\frac{k^{-j}}{j!}j^2k^3\right) = \frac{\tau_1}{\alpha\beta}\mu_k + O(1).$$

When $2k > n$, we have $\overline{EDF_k} = 0$. Finally, the expected total cost of the EDF part in the factorization chain is

$$C_n = \sum_k \overline{EDF_k} = \frac{\tau_1}{\alpha\beta}\sum_{k=1}^{\lfloor n/2 \rfloor}\mu_k + O(n).$$

The lower and upper bounds on C_n are easily obtained from the inequalities

$$k\log_2 q - 1 \leq \mu_k \leq 2k\log_2 q.$$

□

It is natural to expect that $\nu((q^k-1)/2)$, the number of ones in the binary representation of $(q^k-1)/2$, behaves like that of a random number. This leads to the heuristic $\nu((q^k-1)/2) \sim \frac{k}{2}\log_2 q$. Thus, $\mu_k \sim \frac{3k}{2}\log_2 q$. Under this unproven assumption, we expect the following estimate to be true

$$C_n \sim \frac{3\tau_1}{16\alpha\beta}n^2\log_2 q,$$

where $\alpha = (q-1)/2q$, and $\beta = (q+1)/2q$.

4.5.4 Equal-degree factorization in characteristic 2

In Section 4.5, we have analyzed in detail the case of finite fields with odd characteristic. This section shows that similar results hold for the even case.

For the odd case, we have followed the algorithm in Cantor & Zassenhaus (1981). Their solution for the even case relies in factoring the polynomial in a quadratic extension. Ben-Or (1981) shows that this is not needed. His method is based on trace computations (see Section 3.2.3). This computations introduces only a small change in the EDF algorithm of Section 4.5. Let m be such that $q = 2^m$. In order to compute the traces, the line numbered ! must be replaced by

```
1.    a := h + h^2 + h^(2^2) + ... + h^(2^(km-1)) mod b;
```

We observe that the analysis for the odd case is valid for the even case. Indeed, the partitioning process is the same (with probabilities $\alpha = \beta = 1/2$). The cost of computing line 1 is also the same. In fact, the trace computations can be computed using km products of a polynomial containing j factors of degree k. This costs essentially $km(jk)^2 = k^3 j^2 \log q$, the same cost as in the odd case.

4.6 Algorithmic variants

The previous sections presented the analysis of a factorization chain including several variants for the DDF stage. In this section, we briefly mention other variants for the factorization algorithm.

Before commenting about several variants for the factorization chain, we summarize the results of the previous sections with a comparison between worst-case and average-case behavior.

Step	Worst-case	Average-case
ERF	$O(n^2)$	$\tau_2 n^2$
DDF	$O(n^3 \log q)$	$0.26688798\ldots\ (\lambda(q)\tau_1 + \tau_2)\, n^3$
EDF	$O(n^3 \log q)$	$\left(\frac{3}{4}\tau_1 \frac{q^2}{q^2-1} \log q \cdot n^2\right)\ \left(\frac{4}{3} + o(1)\right)$

The first natural variant is to consider the full squarefree factorization instead of our elimination of repeated factors. In this case, all generating functions are different to the ones in Section 4.3. This may suggest different asymptotic results. However, this is not the case. For instance, the generating function for the basic DDF (see Theorem 4.3.1) is, when considering full squarefree factorization,

$$P_k(z,u) = \prod_{j<k} \left(\frac{1}{1-z^j}\right)^{I_j} \prod_{j\geq k} \left(1 + u^j z^j + \frac{z^{2j}}{1-z^j}\right)^{I_j}.$$

The same asymptotic result as in Theorem 4.3.1 is obtained for this generating function with an analogous study. Of course, this was expected given our comments about the

influence of the first stage of the factorization chain in the total factorization process.

The previous variant increments the amount of work in the first stage of the algorithm with no asymptotic gain. When the field is large, the first stage does not contribute much in finding factors (see Theorem 4.2.1). In other words, almost all polynomials are squarefree when the size of the field is large. Although this stage is the cheapest one, it seems natural to try to reduce its amount of work. Thus, a second variant could be to execute only DDF and EDF. In this variant, DDF not only produces the polynomials for the EDF part but also returns a polynomial containing the nonsquarefree part of the original polynomial. Recursively, the algorithm factors this remaining factor. Again, the generating functions will be different with no gain in asymptotic terms. For instance, the generating function for the basic DDF is in this case

$$P_k(z,u) = \prod_{j<k} \left(\frac{1}{1-z^j}\right)^{I_j} \prod_{j\geq k} \left(\frac{1}{1-u^j z^j}\right)^{I_j}.$$

A similar analysis to the one in Theorem 4.3.1 gives the same dominant asymptotic result.

In addition to the above variants, one can study other algorithms for each stage. An account of papers including algorithm for the three stages in the factorization chain is in Section 3.2. The next chapter deals with a more substantial variation for the DDF stage. This guides us to the study of the distribution of irreducible factors in intervals.

Chapter 5

Polynomial Factorization and Analysis of Intervals

This chapter studies parameters related to how the degrees of the irreducible factors of a polynomial (of degree n) are distributed into partitions of the interval $[1, n]$. Several algorithms for computing the distinct-degree factorization of a polynomial over $\mathbb{F}_q$ of degree n break the interval $[1, n]$ into pieces (von zur Gathen and Shoup 1992, Kaltofen and Shoup 1995). Then, for each interval, the product of all irreducible factors whose degrees lie in the interval is computed, followed by the execution of the distinct-degree factorization for each of these products. As we will see, in terms of the algorithms that we consider in this chapter, an interval is costly for a polynomial when the polynomial has more than one of its irreducible factors whose degrees lie in the interval. We refer to these kind of intervals as *multi-factor* intervals for the polynomial. The understanding of the behavior of those factorization algorithms is related with the behavior of the distribution of the degrees of the irreducible factors of random polynomials into their multi-factor intervals. We focus on several questions about multi-factor intervals. For a family of partitions, we give the probability of a polynomial having no multi-factor intervals, the mean value of the number of multi-factor intervals for a polynomial, the mean value of the number of irreducible factors whose degrees lie in any of the multi-factor intervals of a polynomial, and the mean value of the total degree of the irreducible factors whose degrees lie in any of the multi-factor intervals of a polynomial.

5.1 Motivation and results

In Chapter 4, we provided the average-case analysis of Cantor & Zassenhaus' (1981) algorithm for factoring polynomials over finite fields. As we commented at the end of Section 3.2, the bottleneck of the factorization problem is the distinct-degree factorization (DDF). In this chapter, we focus only on this step of the factorization process.

Algorithms of the 90's use a blocking strategy (see Section 3.2.2). They split the interval $[1, n]$ into pieces, and for each interval, they compute the joint product of the irreducible factors of the original polynomial f whose degree lies in that interval. Then, one gcd between this joint product and f determines whether or not f contains more than one irreducible factor in that interval. A complete distinct-degree factorization is obtained using the distinct-degree algorithm for each interval such that the gcd ensures the presence of more than one irreducible factor.

In Section 5.2, we revisit the distinct-degree factorization algorithms of von zur Gathen & Shoup (1992), Kaltofen & Shoup (1995), Shoup (1996), and von zur Gathen & Gerhard (1996). The first three approaches split the interval $[1, n]$ into about $\sqrt{n}$ pieces of size $\sqrt{n}$ each. When dealing with random polynomials, this breaking strategy is not the best possible. Indeed, we know from Theorem 2.2.1 that the number of irreducible factors in a random polynomial of degree n tends to a Gaussian distribution with mean value $\log n$. Moreover, these $\log n$ factors are not equally distributed in the interval $[1, n]$. This follows from Theorem 4.5.1 (see also Knopfmacher & Knopfmacher 1993) since the expected value of the number of irreducible factors of degree k in a random polynomial is $1/k$. Thus, one expects to have more factors of lower degrees than of higher degrees.

Ideally, one would like to provide the best possible partition, i.e., the one that reduces the total cost of the algorithm. In particular, one would like to avoid collision of irreducible factors in intervals. In Section 5.3, we provide the first steps towards finding well-fitted partitions for factoring random polynomials. From the comments of the last paragraph, it seems natural to consider partitions with growing interval sizes. We cover a family of partitions with this property in this chapter. For each partition, we study several parameters related to distinct-degree factorization with growing interval sizes.

These parameters are: the number of multi-factor intervals for a polynomial, the number of irreducible factors that lie in any of the multi-factor intervals of a polynomial, and the total degree of irreducible factors that lie in any of the multi-factor intervals of a polynomial. We also study the probability of a polynomial having no multi-factor intervals for a given partition. Our studies are general for partitions with growing interval sizes; then, we specialize for specific partitions. In addition to this application to factoring algorithms with growing interval sizes, the study of the behavior of the distribution of irreducible factors of random polynomials in intervals is a natural question from a mathematical point of view. As far as we know, no answer to this theoretical question has been given in the past. We conclude this section briefly commenting on average-case analysis of these factorization algorithms, and how our work may help towards this goal.

5.2 Distinct-degree factorization with growing interval sizes

We start giving some definitions and introducing some notation required in the rest of the chapter. An *interval partition* of $[1, n]$ is a sequence of integers $0 = S_0 < S_1 < \cdots < S_m = n$. The *intervals* of the partition are the sets $\pi_j = \{S_{j-1} + 1, \ldots, S_j\}$, $1 \leq j \leq m$. The *length* of the partition is the number m of intervals. The *size* of the kth interval is the difference $d_k = S_k - S_{k-1}$. In this thesis, we only consider partitions such that $d_k = o(S_k)$. Therefore, $S_{k-1} = S_k - d_k \sim S_k$. This is the case of polynomially growing interval sizes, a case of particular interest when dealing with random polynomials. For random polynomials, we expect to have a decreasing number of irreducible factors as their degrees increase. So, it makes sense to use partitions with growing interval sizes in order to "balance" the computational work among different intervals.

The DDF algorithms of von zur Gathen & Shoup (1992), Kaltofen & Shoup (1995), Shoup (1996), and von zur Gathen & Gerhard (1996) break the interval $[1, n]$ into pieces following some interval partition S_k. First, they use a baby-step/giant-step technique to compute the modular powers $x^{q^i} \bmod f$ for several values of i. The giant step produces the powers for the points of the partition, $x^{q^{S_k}}$, while the baby step computes

the intermediate values for the first interval. After these computations are done, they proceed in two steps. A *coarse DDF* computes, for each interval, a *partial factorization* which contains all irreducible factors of the original polynomial with degrees belonging to the interval. If the partial factorization for interval $[S_{j-1}+1, S_j]$ has degree less than $2(S_{j-1}+1)$, then we conclude that the interval contains at most one factor, and that there is no need of further computations. Otherwise, a *fine DDF* is executed using the basic DDF algorithm of Section 3.2.2 for this partial factorization. Given a polynomial f, we say that an interval π is a *multi-factor* interval if more than one of the irreducible factors of f have degrees lying in π; otherwise, if f has zero or one irreducible factor with degrees in π, we say that π is a *trivial-factor* interval. There is a close relationship between executing a fine DDF for the partial factorization corresponding to an interval and the interval being multi-factor. Indeed, a fine DDF is performed for every multi-factor interval. Observe, however, that there might be other intervals with only one irreducible factor for which a fine DDF is also performed.

The costly step in these algorithms is the computation of the qth powers modulo a polynomial. For these computations, von zur Gathen & Shoup (1992) propose the "iterated Frobenius" algorithm (see Section 3.2.2). Kaltofen & Shoup (1995) and Shoup (1996) use repeated squaring for the baby step, and modular compositions (Brent & Kung 1978) for the giant step (modular compositions only for the practical version). Finally, von zur Gathen & Gerhard (1996) uses repeated squaring (this makes sense in their case since they are computing over $\mathbb{F}_2$ only). We note that in the worst-case scenario of these algorithms all qth powers are computed. However, many of the gcd computations will be 1, and therefore, the blocking strategy saves gcd computations.

The practical versions of the first three papers above consider constant interval sizes. More precisely, they split the interval $[1, n]$ into about $\sqrt{n}$ pieces of size $\sqrt{n}$ each. We restrict our attention to the practical versions in Kaltofen & Shoup (1995) and Shoup (1996). Let ℓ and m be integer numbers such that $\ell * m \approx n/2$. The baby step computes the polynomials $h_i = x^{q^i} \bmod f$, for $1 \leq i \leq \ell$, while the giant step produces the polynomials $H_j = x^{q^{j\ell}} \bmod f$ for $1 \leq j \leq m$. Therefore, there are $\ell + m$ compositions. Kaltofen & Shoup (1995) and Shoup (1996) choose $\ell = m \approx \sqrt{n/2}$. The next lemma

shows that this is the best way of choosing ℓ and m in order to reduce the total number $\ell + m$ of compositions.

Lemma 5.2.1 *Let ℓ and m be integer numbers such that $\ell * m = n/2$. The minimum for $\ell + m$ is achieved when $\ell = m = \sqrt{n/2}$.*

PROOF. The relation between arithmetic mean and geometric mean gives

$$\frac{\ell + m}{2} \geq \sqrt{m\ell} = \sqrt{n/2}.$$

Therefore, $\ell + m \geq \sqrt{2n}$. □

This lemma shows that both Kaltofen & Shoup (1995) and Shoup (1996) use an optimal number of modular compositions. However, this does not show that they use the modular compositions in the best possible moment of the factorization process. For instance, the polynomial f that is used for the modular compositions decreases in degree along the process. Therefore, the m compositions (in the giant step to compute the H_j) are less expensive than the previous ℓ ones (in the baby step). In addition, when dealing with average-case analysis of this algorithm, one needs less than m compositions for the giant step since on average the DDF step stops before $n/2$ iterations. This gives some room for further improvements in the average-case behavior of these algorithms.

We focus now on polynomially growing interval sizes. The coarse DDF and the fine DDF algorithms below are essentially taken from von zur Gathen & Gerhard (1996). Before we include the algorithms we need one more definition. An *interval polynomial* for an interval $\pi_j = \{S_{j-1} + 1, \ldots, S_j\}$ is a polynomial that is divisible by any irreducible factor whose degree lies in π_j. For example, by Theorem 3.2.1, $\prod_{i \in \pi_j} x^{q^i} - x$ is divisible by every irreducible polynomial in $\mathbb{F}_q[x]$ of degree dividing $i \in [S_{j-1} + 1, S_j]$. These are the interval polynomials in von zur Gathen & Shoup (1992). Kaltofen & Shoup (1995) and Shoup (1996) use the interval polynomial $\prod_{0 \leq i \leq S_j - S_{j-1}} x^{q^{S_j}} - x^{q^i}$ based on the following theorem similar to Theorem 3.2.1 (see Kaltofen & Shoup 1995, Lemma 1).

Theorem 5.2.2 *For nonnegative integers i, j, the polynomial $x^{q^i} - x^{q^j} \in \mathbb{F}_q[x]$ is divisible by precisely those irreducible polynomials in $\mathbb{F}_q[x]$ whose degree divides $i - j$.*

For the following algorithms we assume that all modular qth powers x^{q^i}, $1 \leq i \leq n$, have been previously computed. The cost of this precomputation depends on the method employed for doing these modular qth powers and the other operations in the algorithm (gcds, divisions, and so on). Since our main goal in this chapter is to study parameters related to these algorithms, we do not need to consider any of the possible costs. In the next section we concentrate on these parameters.

Algorithm *Coarse distinct-degree factorization*

Input: *A monic squarefree polynomial $f \in \mathbb{F}_q[x]$ of degree n.*

Output: *The polynomials $H_j = \prod_{i \in \pi_j} h_i$ for $1 \leq j \leq k$, where h_i is the product of all monic irreducible factors of f of degree i, $S_{j-1} + 1 \leq i \leq S_j$.*

$f^* := f$;
for j := 1 to k do
    Let g_j be an interval polynomial for π_j.
    Compute the remainder R_j of the division of g_j by f^*.
    $H_j := \gcd(R_j, f^*)$;
    $f^* := f^*/H_j$;
endfor;
return $H_1, \ldots, H_k$; if $f^* \neq 1$, then also return f^*;

Algorithm *Fine distinct-degree factorization*

Input: *A polynomial $H_j = \prod_{i \in \pi_j} h_i$, where h_i is the product of all monic irreducible factors of the polynomial f to be factored of degree i, $S_{j-1} + 1 \leq i \leq S_j$.*

Output: *The polynomials $h_i \in \mathbb{F}_q[x]$, $S_{j-1} + 1 \leq i \leq S_j$, with $h_i \neq 1$.*

$h^* := H_j$;
for i := $S_{j-1} + 1$ to S_j do
    $h_i := \gcd(h^*, x^{q^i} - x \bmod h^*)$;
    $h^* := h^*/h_i$;
endfor;
return $h_{S_{j-1}+1}, \ldots, h_{S_j}$;

5.3 Analysis of interval parameters for DDF

This section provides useful information on the parameters related to partitions of the interval $[1, n]$. The main results of this section are precise analyses of the mean value of the number of multi-factor intervals for a polynomial, the mean value of the number of irreducible factors of a polynomial whose degrees lie in any of its multi-factor intervals, and the mean value of the total degree of irreducible factors (of a polynomial) whose degrees lie in any of the multi-factor intervals for the polynomial. As usual in this thesis, the asymptotics are done with respect to n, the degree of the polynomial considered, with q, the size of the field, being fixed. Sometimes, however, we may also analyze asymptotic behavior with respect to q, in which case we state this explicitly.

5.3.1 Probability of a polynomial having no multi-factor intervals

When the partial factorization for interval $\pi_j = [S_{j-1}+1, S_j]$ has degree less than $2(S_{j-1}+1)$, it is clear that it contains at most one irreducible factor, and that there is no need for running a fine distinct-degree factorization for the interval. It is therefore of interest to study the probability that a random polynomial has at most one irreducible factor in each interval for a given interval partition. In other words, we want the probability that a polynomial has no multi-factor intervals for the interval partition. The next theorem answers this question for growing interval sizes.

Theorem 5.3.1 *Let $\pi_1, \pi_2, \ldots$ be the intervals of an interval partition $S_0, S_1, \ldots$ of $[1, n]$ such that $d_k = S_k - S_{k-1} = o(S_k)$, and $\sum_{k=1}^{\infty} \left(\frac{d_k}{S_k}\right)^2$ converges. Then, the probability that a polynomial has no multi-factor intervals in the given partition is asymptotic to*

$$\exp\left(-\frac{1}{2}\sum_{k=1}^{\infty}\left(\frac{d_k}{S_k}\right)^2\right).$$

PROOF. The generating function counting the number of polynomials that contain either zero or one irreducible factor per interval is

$$P(z) = \prod_{k\geq 1}\left(1 + \sum_{\ell\in\pi_k} I_\ell z^\ell\right).$$

The coefficient $[z^n]P(z)$ gives the number of polynomials of degree n with no more than one irreducible factor in each interval, i.e., the number of polynomials without multi-factor intervals in the interval partition. Since $(1-qz)^{-1} = \prod_{n=1}^{\infty}(1-z^n)^{-I_n} = \prod_{k\geq 1}\prod_{\ell\in\pi_k}(1-z^\ell)^{-I_\ell}$,

$$P(z) = \prod_{k\geq 1}\left(1+\sum_{\ell\in\pi_k} I_\ell z^\ell\right) = \frac{1}{1-qz}\prod_{k\geq 1}\left(\frac{1+\sum_{\ell\in\pi_k} I_\ell z^\ell}{\prod_{\ell\in\pi_k}(1-z^\ell)^{-I_\ell}}\right).$$

Let us call

$$Q(z) = \prod_{k\geq 1}\left(\frac{1+\sum_{\ell\in\pi_k} I_\ell z^\ell}{\prod_{\ell\in\pi_k}(1-z^\ell)^{-I_\ell}}\right).$$

In order to apply asymptotic analysis (Theorem 1.3.3), we have to prove that $Q(z)$ exists when $z \to 1/q$

$$Q\left(\frac{1}{q}\right) = \prod_{k\geq 1}\left(\frac{1+\sum_{\ell\in\pi_k} I_\ell q^{-\ell}}{\prod_{\ell\in\pi_k}(1-q^{-\ell})^{-I_\ell}}\right).$$

First note that since $I_\ell = q^\ell/\ell + O\left(q^{\ell/2}\right)$, a good approximation for I_ℓ/q^ℓ is $1/\ell$. Second, write $\prod_{\ell\in\pi_k}(1-q^{-\ell})^{-I_\ell} = \exp\left(\ln\prod_{\ell\in\pi_k}(1-q^{-\ell})^{-I_\ell}\right) = \exp\left(-\sum_{\ell\in\pi_k} I_\ell \ln(1-q^{-\ell})\right)$. Then, expanding the logarithm, we have

$$\exp\left(-\sum_{\ell\in\pi_k}\left(I_\ell\sum_{j\geq 1}-\frac{1}{jq^{j\ell}}\right)\right) = \exp\left(\sum_{\ell\in\pi_k}\left(\frac{I_\ell}{q^\ell}+\frac{I_\ell}{2q^{2\ell}}+\cdots\right)\right). \tag{5.1}$$

Using the approximation for I_ℓ/q^ℓ, we obtain

$$\prod_{\ell\in\pi_k}(1-q^{-\ell})^{-I_\ell} \sim \exp\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right). \tag{5.2}$$

The points of the partition are $S_1,\ldots,S_k,\ldots$, with $S_k - S_{k-1} = d_k$ being the size of the kth interval. Recall that $\pi_k = \{S_{k-1}+1,\ldots,S_k\}$. Then, for any $k \geq 1$,

$$\frac{d_k}{S_k} \leq \sum_{S_{k-1}<\ell\leq S_k}\frac{1}{\ell} \leq \frac{d_k}{S_{k-1}+1}.$$

By hypothesis, $d_k = S_k - S_{k-1} = o(S_k)$. Then $S_k \sim S_{k-1}$, and

$$\sum_{\ell\in\pi_k}\frac{1}{\ell} = \frac{d_k}{S_k}. \tag{5.3}$$

Thus, $1+\sum_{\ell\in\pi_k} I_\ell q^{-\ell} \sim 1+\frac{d_k}{S_k}$, and from Eqs. (5.2) and (5.3), $Q\left(\frac{1}{q}\right)$ approaches

$$\prod_{k\geq 1}\left(\left(1+\frac{d_k}{S_k}\right)\exp\left(-\frac{d_k}{S_k}\right)\right).$$

Using an exp-log form for the product and expanding the logarithm, we have

$$\exp\left(\sum_{k\geq 1}\left(\ln\left(1+\frac{d_k}{S_k}\right)+\left(-\frac{d_k}{S_k}\right)\right)\right)=\exp\left(\sum_{k\geq 1}\left(\left(\frac{d_k}{S_k}-\frac{d_k^2}{2S_k^2}+\cdots\right)+\left(-\frac{d_k}{S_k}\right)\right)\right).$$

Finally,

$$Q\left(\frac{1}{q}\right)\sim\exp\left(-\frac{1}{2}\sum_{k\geq 1}\frac{d_k^2}{S_k^2}\right).$$

By hypothesis $\sum_{k\geq 1}\frac{d_k{}^2}{S_k{}^2}$ converges. By Theorem 1.3.3, the probability of a polynomial having no multi-factor interval in the given partition is asymptotic to $\exp\left(-\frac{1}{2}\sum_{k\geq 1}\frac{d_k^2}{S_k^2}\right)$. □

S_k	$\frac{1}{2}\sum_{k\geq 1}\frac{d_k^2}{S_k^2}$	Prob
$k^{3/2}$	1.10936	.32977
k^2	1.42692	.24005
$k^{5/2}$	1.75938	.17215
k^3	2.09900	.12258
$k^{7/2}$	2.44200	.08699
k^4	2.78656	.06163
$k^{9/2}$	3.13184	.04364
k^5	3.47744	.03089

Table 5.1: Probability of a polynomial having no multi-factor intervals for partition S_k.

Table 5.1 presents the value of the probability of a polynomial having at most one irreducible factor per interval, that is having no multi-factor intervals, for some partitions under the hypothesis of Theorem 5.3.1.

The result in Eq. (5.2) is sufficient for the proof of Theorem 5.3.1. However, we can prove a stronger result. Indeed, we now prove that $\prod_{\ell\in\pi_k}(1-q^{-\ell})^{-I_\ell}=\Theta\left(\exp\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right)\right)$. More precisely, we have the following theorem.

Theorem 5.3.2 *Let $\pi_1, \pi_2, \ldots$ be the intervals of an interval partition $S_0, S_1, \ldots$ of $[1, n]$. For any prime power q,*

$$\left(1-\frac{1}{\sqrt{q}}\right)^{\frac{q}{q-1}} \cdot \exp\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right) \leq \prod_{\ell\in\pi_k}(1-q^{-\ell})^{-I_\ell} \leq \exp\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right).$$

When $q \to \infty$, we have

$$\prod_{\ell\in\pi_k}(1-q^{-\ell})^{-I_\ell} \sim \exp\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right).$$

PROOF. Let us start with the upper bound. First note that from Eq. (1.6), we obtain $I_\ell \leq (q^\ell - 1)/\ell$. Using Theorem 1.4.2 and that $0 < 1 - 1/q^\ell < 1$, we have

$$\begin{aligned}\prod_{\ell\in\pi_k}(1-q^{-\ell})^{-I_\ell} &\leq \prod_{\ell\in\pi_k}\left(1-\frac{1}{q^\ell}\right)^{-\frac{q^\ell-1}{\ell}} \\ &= \prod_{\ell\in\pi_k}\left(\left(1+\frac{1}{q^\ell-1}\right)^{q^\ell-1}\right)^{\frac{1}{\ell}} \leq \prod_{\ell\in\pi_k}\exp\left(\frac{1}{\ell}\right) = \exp\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right).\end{aligned}$$

We focus now on the lower bound. Using Eq. (5.1), expanding logarithms, considering Eq. (1.6) and that $\sum_{k=1}^{\infty}\frac{1}{kq^{k/2}} = -\ln\left(1-\frac{1}{\sqrt{q}}\right)$, we deduce

$$\begin{aligned}\prod_{\ell\in\pi_k}(1-q^{-\ell})^{-I_\ell} &= \exp\left(\sum_{\ell\in\pi_k}\left(\frac{I_\ell}{q^\ell}+\frac{I_\ell}{2q^{2\ell}}+\cdots\right)\right) \geq \exp\left(\sum_{\ell\in\pi_k}\frac{I_\ell}{q^\ell}\right) \\ &\geq \exp\left(\sum_{\ell\in\pi_k}\frac{\frac{q^\ell}{\ell}-\frac{q(q^{\ell/2}-1)}{(q-1)\ell}}{q^\ell}\right) \\ &= \exp\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right)\cdot\exp\left(\frac{q}{q-1}\sum_{\ell\in\pi_k}\frac{q^{\ell/2}-1}{\ell\, q^\ell}\right) \\ &\geq \exp\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right)\cdot\exp\left(-\frac{q}{q-1}\sum_{\ell\in\pi_k}\frac{1}{\ell\, q^{\ell/2}}\right) \\ &\geq \exp\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right)\cdot\exp\left(-\frac{q}{q-1}\sum_{\ell=1}^{\infty}\frac{1}{\ell q^{\ell/2}}\right) \\ &= \exp\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right)\cdot\left(\exp\left(\ln\left(1-\frac{1}{\sqrt{q}}\right)^{\frac{q}{q-1}}\right)\right) \\ &= \left(1-\frac{1}{\sqrt{q}}\right)^{\frac{q}{q-1}}\cdot\exp\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right).\end{aligned}$$

We consider fixed q, thus we have proven that $\prod_{\ell\in\pi_k}(1-q^{-\ell})^{-I_\ell} = \Theta\left(\exp\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right)\right)$.

Observe that when q is large (or when $q \to \infty$), $\left(1 - \frac{1}{\sqrt{q}}\right)^{\frac{q}{q-1}} \to 1$. Therefore, in that case, we have that $\prod_{\ell \in \pi_k}(1 - q^{-\ell})^{-I_\ell} \sim \exp\left(\sum_{\ell \in \pi_k} \frac{1}{\ell}\right)$. Indeed, this follows immediately from Eq. (5.1). □

5.3.2 Number of multi-factor intervals for a polynomial

Given an interval partition, the expected number of multi-factor intervals for a polynomial gives useful information on the number of fine distinct-degree factorizations that will be needed. The next theorem quantifies this expectation.

Theorem 5.3.3 *Let $\pi_1, \pi_2, \ldots$ be the intervals of an interval partition $S_0, S_1, \ldots$ of $[1, n]$ such that $d_k = S_k - S_{k-1} = o(S_k)$, and $\sum_{k=1}^{\infty} \left(\frac{d_k}{S_k}\right)^2$ converges. Then, the expected number of multi-factor intervals in $\pi_1, \pi_2, \ldots$ for a polynomial is asymptotic to*

$$\frac{1}{2}\sum_{k=1}^{\infty}\left(\frac{d_k}{S_k}\right)^2.$$

PROOF. The bivariate generating function corresponding to marking an interval π_k if it contains more than one irreducible factor can be derived marking all intervals, and then subtracting the cases of intervals with 0 and 1 irreducible factor. This approach gives the generating function

$$P(z,u) = \prod_{k \geq 1}\left(u \prod_{\ell \in \pi_k}\left(\frac{1}{1-z^\ell}\right)^{I_\ell} + (1-u)\left(1 + \sum_{\ell \in \pi_k} I_\ell z^\ell\right)\right).$$

The coefficient $[z^n u^k]P(z,u)$ gives the number of polynomials of degree n with k multi-factor intervals in the given partition. The mean value of the number of multi-factor intervals for a polynomial is obtained by differentiating $P(z,u)$ with respect to u, and then setting $u = 1$. The computation of the derivative of $P(z,u)$ with respect to u gives

$$\frac{\partial P(z,u)}{\partial u} = P(z,u)\left(\sum_{k \geq 1} \frac{\prod_{\ell \in \pi_k}(1-z^\ell)^{-I_\ell} - \left(1 + \sum_{\ell \in \pi_k} I_\ell z^\ell\right)}{u\prod_{\ell \in \pi_k}(1-z^\ell)^{-I_\ell} + (1-u)\left(1 + \sum_{\ell \in \pi_k} I_\ell z^\ell\right)}\right).$$

Evaluating the derivative at $u = 1$, we obtain the desired expected value

$$[z^n]\frac{1}{1-qz}Q(z), \qquad Q(z) = \sum_{k \geq 1} \frac{\prod_{\ell \in \pi_k}(1-z^\ell)^{-I_\ell} - \left(1 + \sum_{\ell \in \pi_k} I_\ell z^\ell\right)}{\prod_{\ell \in \pi_k}(1-z^\ell)^{-I_\ell}}.$$

Since $Q(z)$ is analytic for $|z| < 1/q$, we must show that $Q(z)$ exists when $z \to 1/q$, so the dominant singularity is at $z = 1/q$, in order to apply asymptotic analysis. We have

$$Q\left(\frac{1}{q}\right) = \sum_{k\geq 1} \frac{\prod_{\ell\in\pi_k}(1-q^{-\ell})^{-I_\ell} - (1+\sum_{\ell\in\pi_k} I_\ell q^{-\ell})}{\prod_{\ell\in\pi_k}(1-q^{-\ell})^{-I_\ell}}.$$

Using Eq. (5.2), $Q\left(\frac{1}{q}\right)$ approaches

$$\sum_{k=1}^{\infty} \frac{\exp\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right) - \left(1+\sum_{\ell\in\pi_k}\frac{1}{\ell}\right)}{\exp\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right)}. \tag{5.4}$$

This series of positive terms may converge or diverge depending on the interval partition. Approximating the denominator in Eq. (5.4) by 1 and expanding the numerator in Eq. (5.4), we obtain

$$\frac{\exp\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right) - \left(1+\sum_{\ell\in\pi_k}\frac{1}{\ell}\right)}{\exp\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right)} \sim \frac{1}{2}\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right)^2.$$

We conclude from Eqs. (5.3) and (5.4) that $Q\left(\frac{1}{q}\right)$ approaches to $\frac{1}{2}\sum_{k=1}^{\infty}\left(\frac{d_k}{S_k}\right)^2$. To complete the proof use Theorem 1.3.3.

□

Table 5.2 shows the expected number of multi-factor intervals for several partitions under the hypothesis of Theorem 5.3.1.

As an aside, for the practically important cases of q small, we could estimate

$$\sum_{k=1}^{N} \frac{\exp\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right) - \left(1+\sum_{\ell\in\pi_k}\frac{1}{\ell}\right)}{\exp\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right)},$$

for several values of N until this quantity stabilizes. We then pick this value as the value of $Q(\frac{1}{q})$.

5.3.3 Number of factors in any multi-factor interval for a polynomial

In this section, we present two analyses related to the number of multi-factor intervals for partitions. First we give the expected number of factors of a polynomial that lie in any of its multi-factor intervals. Then, we study the excess on the number of irreducible factors

S_k	$\frac{1}{2}\sum_{k\geq 1}\frac{d_k^2}{S_k^2}$
$k^{3/2}$	1.10936
k^2	1.42692
$k^{5/2}$	1.75938
k^3	2.09900
$k^{7/2}$	2.44200
k^4	2.78656
$k^{9/2}$	3.13184
k^5	3.47744

Table 5.2: Expected number of multi-factor intervals for a polynomial using partition S_k.

in multi-factor intervals for a polynomial, i.e., for each interval with $r > 1$ irreducible factors we count $r - 1$ of these factors. This parameter gives useful information on the number of extra factors that make the interval multi-factor.

Theorem 5.3.4 *Let $\pi_1, \pi_2, \ldots$ be the intervals of an interval partition $S_0, S_1, \ldots$ of $[1, n]$ such that $d_k = S_k - S_{k-1} = o(S_k)$, and $\sum_{k=1}^{\infty}\left(\frac{d_k}{S_k}\right)^2$ converges. Then, the expected number of irreducible factors whose degrees lie in any of the intervals $\pi_1, \pi_2, \ldots$ that are multi-factor for the polynomial is asymptotic to*

$$\sum_{k=1}^{\infty}\left(\frac{d_k}{S_k}\right)^2.$$

PROOF. The bivariate generating function counting the number of irreducible factors whose degrees lie in any multi-factor interval for a polynomial is

$$P(z,u) = \prod_{k\geq 1}\left(\prod_{\ell\in\pi_k}\left(\frac{1}{1-uz^\ell}\right)^{I_\ell} + \sum_{\ell\in\pi_k}(1-u)I_\ell z^\ell\right).$$

The coefficient $[z^n u^k]P(z,u)$ gives the number of polynomials of degree n with k irreducible factors lying in any of their multi-factor intervals. Differentiating $P(z,u)$ with respect to u gives

$$\frac{\partial P(z,u)}{\partial u} = P(z,u)\left(\sum_{k\geq 1}\frac{\sum_{\ell\in\pi_k}\left(\prod_{j\in\pi_k, j\neq\ell}(1-uz^j)^{-I_j}I_\ell z^\ell\right) - \sum_{\ell\in\pi_k}I_\ell z^\ell}{\prod_{\ell\in\pi_k}(1-uz^\ell)^{-I_\ell} + \sum_{\ell\in\pi_k}(1-u)I_\ell z^\ell}\right).$$

Evaluating the derivative in $u = 1$, we obtain

$$[z^n]\frac{1}{1-qz}Q(z), \qquad Q(z) = \sum_{k\geq 1} \frac{\sum_{\ell\in\pi_k}\left(\prod_{j\in\pi_k, j\neq\ell}(1-z^j)^{-I_j} I_\ell z^\ell\right) - \sum_{\ell\in\pi_k} I_\ell z^\ell}{\prod_{\ell\in\pi_k}(1-z^\ell)^{-I_\ell}}.$$

In the following we study the behavior of $Q(z)$ when z approaches $1/q$

$$Q\left(\frac{1}{q}\right) = \sum_{k\geq 1} \frac{\sum_{\ell\in\pi_k}\left(\prod_{j\in\pi_k, j\neq\ell}(1-q^{-j})^{-I_j} I_\ell q^{-\ell}\right) - \sum_{\ell\in\pi_k} I_\ell q^{-\ell}}{\prod_{\ell\in\pi_k}(1-q^{-\ell})^{-I_\ell}}.$$

Multiplying and dividing by $(1-q^{-\ell})^{-I_\ell}$ for each $\ell \in \pi_k$, we have

$$Q\left(\frac{1}{q}\right) = \sum_{k\geq 1} \frac{\left(\sum_{\ell\in\pi_k} \frac{I_\ell}{q^\ell(1-q^{-\ell})^{-I_\ell}}\right)\left(\prod_{\ell\in\pi_k}(1-q^{-\ell})^{-I_\ell}\right) - \sum_{\ell\in\pi_k} I_\ell q^{-\ell}}{\prod_{\ell\in\pi_k}(1-q^{-\ell})^{-I_\ell}}.$$

From the approximation in Eq. (5.2),

$$\sum_{\ell\in\pi_k} \frac{I_\ell}{q^\ell(1-q^{-\ell})^{-I_\ell}} \sim \sum_{\ell\in\pi_k} \frac{\frac{1}{\ell}}{\exp\frac{1}{\ell}}.$$

Therefore, $Q\left(\frac{1}{q}\right)$ approaches

$$\sum_{k=1}^{\infty}\left(\sum_{\ell\in\pi_k} \frac{1}{\ell\exp\frac{1}{\ell}} - \frac{\sum_{\ell\in\pi_k}\frac{1}{\ell}}{\exp\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right)}\right). \tag{5.5}$$

Expanding the exponential, we have

$$\sum_{\ell\in\pi_k} \frac{1}{\ell}\exp\left(-\frac{1}{\ell}\right) = \sum_{\ell\in\pi_k} \frac{1}{\ell}\left(1-\frac{1}{\ell}+\cdots\right) \sim \sum_{\ell\in\pi_k}\frac{1}{\ell}.$$

Simplifying each term in Eq. (5.5), we obtain

$$\sum_{\ell\in\pi_k} \frac{1}{\ell\exp\frac{1}{\ell}} - \frac{\sum_{\ell\in\pi_k}\frac{1}{\ell}}{\exp\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right)} \sim \sum_{\ell\in\pi_k}\frac{1}{\ell}\left(1-\exp\left(-\sum_{\ell\in\pi_k}\frac{1}{\ell}\right)\right) \sim \left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right)^2.$$

Finally, using Eq. (5.3), the series in Eq. (5.5) approaches to $\sum_{k=1}^{\infty}\left(\frac{d_k}{S_k}\right)^2$, and Theorem 1.3.3 completes the proof. □

Table 5.3 presents the expected number of irreducible factors with degrees lying in multi-factor intervals for several partitions under the hypothesis of Theorem 5.3.4.

Next, we focus on the excess of the number of irreducible factors with degrees in any multi-factor intervals. More formally, let the *excess* of irreducible factors of a polynomial f in an interval π_k be given by:

$$r_k = \begin{cases} 0 & \text{if } \pi_k \text{ is a trivial-factor interval for } f, \\ r-1 & \text{if } \pi_k \text{ is a multi-factor interval for } f \text{ with } r \text{ irreducible factors.} \end{cases}$$

The next theorem studies this parameter.

S_k	$\sum_{k\geq 1} \frac{d_k^2}{S_k^2}$
$k^{3/2}$	2.21871
k^2	2.85384
$k^{5/2}$	3.51876
k^3	4.19800
$k^{7/2}$	4.88400
k^4	5.57312
$k^{9/2}$	6.26368
k^5	6.95488

Table 5.3: Expected number of factors in multi-factor intervals using partition S_k.

Theorem 5.3.5 *Let $\pi_1, \pi_2, \ldots$ be the intervals of an interval partition $S_0, S_1, \ldots$ of $[1, n]$ such that $d_k = S_k - S_{k-1} = o(S_k)$, and $\sum_{k=1}^{\infty} \left(\frac{d_k}{S_k}\right)^2$ converges. Then, the expected sum of the excesses of irreducible factors of a polynomial that lie in any of its multi-factor intervals is asymptotic to*

$$\frac{1}{2}\sum_{k=1}^{\infty}\left(\frac{d_k}{S_k}\right)^2.$$

PROOF. Let $\mathcal{I}$ be the collection of all monic irreducible polynomials in $\mathbb{F}_q$, $\omega \in \mathcal{I}$ an irreducible factor, and u the marking variable for the parameter r_k. Formally, the collection of polynomials marking the parameter r_k can be written as

$$\prod_{k\geq 1}\left(1 + \sum_{\omega,|\omega|\in\pi_k} \omega + \prod_{\omega,|\omega|\in\pi_k} \left(u\omega^2 + u^2\omega^3 + \cdots\right)\right),$$

where $|\omega|$ is the degree of $\omega \in \mathcal{I}$. Observe that we did not mark trivial-factor intervals (that is, π_k with 0 or 1 irreducible polynomial is represented by $1 + \sum_{\omega,|\omega|\in\pi_k} \omega$), but we marked with u^{r-1} whenever factors in π_k appear $r > 2$ times. We write the above formal expression as

$$\prod_{k\geq 1}\left(1 + \sum_{\omega,|\omega|\in\pi_k} \omega + \frac{1}{u}\left(\prod_{\omega,|\omega|\in\pi_k} \frac{1}{1-u\omega} - 1 - u\sum_{\omega,|\omega|\in\pi_k} \omega\right)\right).$$

Let z be a formal variable. The substitution $\omega \mapsto z^{|\omega|}$ produces the bivariate generating function for the parameter r_k, that is

$$P(z,u) = \prod_{k\geq 1}\left(1 + \sum_{\ell\in\pi_k} I_\ell z^\ell + \frac{1}{u}\left(\prod_{\ell\in\pi_k}\left(\frac{1}{1-uz^\ell}\right)^{I_\ell} - 1 - u\sum_{\ell\in\pi_k} I_\ell z^\ell\right)\right).$$

The coefficient $[z^n u^k]P(z,u)$ gives the number of polynomials of degree n, where k is the sum of the excesses of irreducible factors for all multi-factor intervals. Differentiating $P(z,u)$ with respect to u gives $\frac{\partial P(z,u)}{\partial u} = P(z,u)\,Q(z,u)$, where $Q(z,u)$ is equal to

$$\sum_{k\geq 1} \frac{\left(\sum_{\ell\in\pi_k}\left(\prod_{\substack{j\in\pi_k\\ j\neq\ell}} \frac{I_\ell z^\ell}{(1-uz^j)^{I_j}}\right) - \sum_{\ell\in\pi_k} I_\ell z^\ell\right) u - \prod_{\ell\in\pi_k}(1-uz^\ell)^{-I_\ell} + 1 + u\sum_{\ell\in\pi_k} I_\ell z^\ell}{\left(1 + \sum_{\ell\in\pi_k} I_\ell z^\ell + \frac{1}{u}\left(\prod_{\ell\in\pi_k}\left(\frac{1}{1-uz^\ell}\right)^{I_\ell} - 1 - u\sum_{\ell\in\pi_k} I_\ell z^\ell\right)\right) u^2}.$$

Evaluating the derivative in $u = 1$, we obtain $[z^n]\frac{1}{1-qz}Q(z)$, where

$$Q(z) = \sum_{k\geq 1} \frac{\left(\sum_{\ell\in\pi_k}(1-z^\ell)^{I_\ell} I_\ell z^\ell\right)\prod_{\ell\in\pi_k}(1-z^\ell)^{-I_\ell} + 1 - \prod_{\ell\in\pi_k}(1-z^\ell)^{-I_\ell}}{\prod_{\ell\in\pi_k}(1-z^\ell)^{-I_\ell}}.$$

In the following, we study the behavior of $Q(z)$ when z approaches $1/q$

$$Q\left(\frac{1}{q}\right) = \sum_{k\geq 1}\left(\prod_{\ell\in\pi_k}(1-q^{-\ell})^{I_\ell} + \sum_{\ell\in\pi_k}(1-q^{-\ell})^{I_\ell} I_\ell q^{-\ell} - 1\right).$$

From the approximation in Eq. (5.2),

$$\prod_{\ell\in\pi_k}(1-q^{-\ell})^{I_\ell} \sim \exp\left(\sum_{\ell\in\pi_k} -\frac{1}{\ell}\right),$$

and,

$$\sum_{\ell\in\pi_k}\frac{I_\ell}{q^\ell}(1-q^{-\ell})^{I_\ell} \sim \sum_{\ell\in\pi_k}\left(\frac{1}{\ell}\exp\left(-\frac{1}{\ell}\right)\right).$$

Thus, $Q\left(\frac{1}{q}\right)$ approaches

$$\sum_{k=1}^{\infty}\left(\exp\left(\sum_{\ell\in\pi_k} -\frac{1}{\ell}\right) + \sum_{\ell\in\pi_k}\frac{1}{\ell}\exp\left(-\frac{1}{\ell}\right) - 1\right). \tag{5.6}$$

Expanding the exponentials, and simplifying each term in Eq. (5.6), we obtain

$$\exp\left(\sum_{\ell\in\pi_k} -\frac{1}{\ell}\right) + \sum_{\ell\in\pi_k}\left(\frac{1}{\ell}\exp\left(-\frac{1}{\ell}\right)\right) - 1 \sim 1 - \sum_{\ell\in\pi_k}\frac{1}{\ell} + \frac{1}{2}\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right)^2 + \cdots + \sum_{\ell\in\pi_k}\frac{1}{\ell} - 1.$$

S_k	$\frac{1}{2}\sum_{k\geq 1}\frac{d_k^2}{S_k^2}$
$k^{3/2}$	1.10936
k^2	1.42692
$k^{5/2}$	1.75938
k^3	2.09900
$k^{7/2}$	2.44200
k^4	2.78656
$k^{9/2}$	3.13184
k^5	3.47744

Table 5.4: Expected number of factors in excess for all multi-factor intervals for S_k.

Therefore,

$$Q\left(\frac{1}{q}\right) \sim \frac{1}{2}\sum_{k=1}^{\infty}\left(\sum_{\ell\in\pi_k}\frac{1}{\ell}\right)^2.$$

Finally, using Eq. (5.3), the series in Eq. (5.6) approaches to $\frac{1}{2}\sum_{k=1}^{\infty}\left(\frac{d_k}{S_k}\right)^2$, and Theorem 1.3.3 completes the proof. □

Table 5.4 shows the expected sum of the excesses of irreducible factors in all of the multi-factor intervals for several partitions under the hypothesis of Theorem 5.3.5.

5.3.4 Total degree of factors in all multi-factor intervals

The cost of the different stages in the factorization algorithms depends on the size q of the field, and on the degree of the polynomial being considered. In particular, the cost of the fine distinct-degree factorization algorithm depends on the degree of the polynomial being passed to the algorithm. This reducible polynomial has as degree the sum of the degrees of its irreducible factors in the interval. Therefore, information on the total degree of irreducible factors lying in any of the multi-factor intervals for a polynomial is useful for estimating the total cost of these algorithms. We study this total degree in the following theorem.

Theorem 5.3.6 *Let $j > 1$ be a real number, $S_k = k^j$ an interval partition of $[1, n]$ with intervals $\pi_1, \pi_2, \ldots$, and $d_k = S_k - S_{k-1}$. Then, the expected total degree of irreducible factors that lie in any of the multi-factor intervals of a polynomial, when considering the intervals $\pi_1, \pi_2, \ldots$, is asymptotic to*

$$\frac{j^2}{(j-1)2^{1-\frac{1}{j}}} n^{1-\frac{1}{j}}.$$

PROOF. The bivariate generating function counting the total degree of irreducible factors in any of the multi-factor intervals for a polynomial is

$$P(z,u) = \prod_{k\geq 1}\left(\prod_{\ell\in\pi_k}\left(\frac{1}{1-u^\ell z^\ell}\right)^{I_\ell} + \sum_{\ell\in\pi_k}(1-u^\ell)I_\ell z^\ell\right).$$

The coefficient $[z^n u^k]P(z,u)$ gives the number of polynomials of degree n, where k is the total degree of the irreducible factors lying in any multi-factor interval for the polynomial. Differentiating $P(z,u)$ with respect to u, we have $\frac{\partial P(z,u)}{\partial u} = P(z,u)Q(z,u)$, with

$$Q(z,u) = \sum_{k\geq 1}\frac{\left(\sum_{\ell\in\pi_k}\ell I_\ell u^{\ell-1}z^\ell(1-u^\ell z^\ell)\right)\prod_{\ell\in\pi_k}(1-u^\ell z^\ell)^{-I_\ell} - \sum_{\ell\in\pi_k}\ell I_\ell u^{\ell-1}z^\ell}{\prod_{\ell\in\pi_k}(1-u^\ell z^\ell)^{-I_\ell} + \sum_{\ell\in\pi_k}(1-u^\ell)I_\ell z^\ell}.$$

Evaluating the derivative in $u = 1$, we obtain

$$[z^n]\frac{1}{1-qz}Q(z), \quad Q(z) = \sum_{k\geq 1}\frac{\sum_{\ell\in\pi_k}\frac{\ell I_\ell z^\ell}{(1-z^\ell)}\left(\prod_{\ell\in\pi_k}(1-z^\ell)^{-I_\ell}\right) - \sum_{\ell\in\pi_k}\ell I_\ell z^\ell}{\prod_{\ell\in\pi_k}(1-z^\ell)^{-I_\ell}}.$$

Let us write

$$Q(z) = \sum_{k\geq 1}\left(\sum_{\ell\in\pi_k}\frac{\ell I_\ell z^\ell}{(1-z^\ell)} - \sum_{\ell\in\pi_k}\ell I_\ell z^\ell\prod_{\ell\in\pi_k}(1-z^\ell)^{I_\ell}\right).$$

We set $z = \frac{t}{q}$ so that $t \to 1^-$ as $z \to \frac{1}{q}$. First, we approximate $\ell I_\ell z^\ell$ by t^ℓ, and $\frac{1}{1-z^\ell}$ by 1. Then, $\sum_{\ell\in\pi_k}\ell I_\ell z^\ell \sim \sum_{\ell\in\pi_k} t^\ell$. Second, using the exp-log trick, we write $\prod_{\ell\in\pi_k}(1-z^\ell)^{I_\ell} = \exp\left(\sum_{\ell\in\pi_k} I_\ell \ln(1-z^\ell)\right)$, and approximating $\ln(1-z^\ell)$ by $-z^\ell$ we obtain

$$\prod_{\ell\in\pi_k}(1-z^\ell)^{I_\ell} \sim \exp\left(\sum_{\ell\in\pi_k} -I_\ell z^\ell\right) = \exp\left(-\sum_{\ell\in\pi_k}\frac{t^\ell}{\ell}\right).$$

Then,

$$Q\left(\frac{t}{q}\right) \sim \sum_{k=1}^{\infty}\left(\sum_{\ell\in\pi_k} t^\ell - \left(\sum_{\ell\in\pi_k} t^\ell\right)\exp\left(-\sum_{\ell\in\pi_k}\frac{t^\ell}{\ell}\right)\right).$$

This can be written as

$$\sum_{k=1}^{\infty}\left(\left(\sum_{\ell\in\pi_k} t^{\ell}\right)\left(1-\exp\left(-\sum_{\ell\in\pi_k}\frac{t^{\ell}}{\ell}\right)\right)\right).$$

We now prove that the exponent

$$\sum_{\ell\in\pi_k}\frac{t^{\ell}}{\ell}\to 0, \tag{5.7}$$

and therefore, we can approximate $1-\exp\left(-\sum_{\ell\in\pi_k}\frac{t^{\ell}}{\ell}\right)$ by $\sum_{\ell\in\pi_k}\frac{t^{\ell}}{\ell}$. Indeed, since $t\to 1^{-}$, and $\ell\geq S_{k-1}+1$ for $\ell\in\pi_k=\{S_{k-1}+1,\ldots,S_k\}$, we have

$$0\leq\sum_{\ell\in\pi_k}\frac{t^{\ell}}{\ell}\leq\sum_{\ell\in\pi_k}\frac{1}{\ell}\leq\frac{d_k}{S_{k-1}+1}.$$

For the partitions we are considering, $S_k=k^j$, $d_k=S_k-S_{k-1}\sim j\,k^{j-1}=o(S_k)$. Then, $S_{k-1}=S_k-d_k\sim S_k$, and therefore $\frac{d_k}{S_{k-1}}\sim\frac{d_k}{S_k}\to 0$. Thus Eq. (5.7) holds, and using the approximation for $1-\exp\left(-\sum_{\ell\in\pi_k}\frac{t^{\ell}}{\ell}\right)$, we obtain

$$Q\left(\frac{t}{q}\right)\sim\sum_{k=1}^{\infty}\left(\left(\sum_{\ell\in\pi_k} t^{\ell}\right)\left(\sum_{\ell\in\pi_k}\frac{t^{\ell}}{\ell}\right)\right).$$

Since $\ell>S_{k-1}$ for $\ell\in\pi_k$, and $S_{k-1}\sim S_k$, we have

$$Q\left(\frac{t}{q}\right)\sim\sum_{k=1}^{\infty}\left(\frac{\left(\sum_{\ell\in\pi_k} t^{\ell}\right)^2}{S_k}\right).$$

Given that $t\to 1^{-}$, $\sum_{\ell\in\pi_k} t^{\ell}=t^{S_{k-1}+1}\left(1+\cdots+t^{d_k-1}\right)\sim t^{S_k}\,d_k$. Finally, we conclude

$$Q\left(\frac{t}{q}\right)\sim\sum_{k=1}^{\infty}\frac{t^{2S_k}\,d_k^2}{S_k}. \tag{5.8}$$

Everything we have shown up to this moment holds for any interval partitions S_k with $d_k=o(S_k)$. Now we specialize the discussion for partition $S_k=k^j$, for a fixed real number $j>1$. In this case, $d_k=S_k-S_{k-1}=k^j-(k-1)^j\sim j\,k^{j-1}$, and the series in Eq. (5.8) approaches

$$\sum_{k=1}^{\infty}\frac{t^{2k^j}\,\left(j\,k^{j-1}\right)^2}{k^j}=\sum_{k=1}^{\infty} j^2\,t^{2k^j}\,k^{j-2}.$$

Let us set $t=e^{-h}$ so that $h\to 0^{+}$ when $t\to 1^{-}$. Approximating the summation by the integral, we have

$$Q\left(\frac{e^{-h}}{q}\right)\sim\int_1^{\infty} j^2\,e^{-2hk^j}\,k^{j-2}\,dk.$$

Considering $u = 2hk^j$, we get $k = \left(\frac{u}{2h}\right)^{\frac{1}{j}}$, $dk = \frac{1}{j}\left(\frac{u}{2h}\right)^{\frac{1}{j}}\frac{du}{u}$, and when $k \to 1^+$, $u \to 2h \sim 0$. Thus,

$$\begin{aligned} Q\left(\frac{e^{-h}}{q}\right) &\sim \int_0^\infty j^2 \frac{1}{j} e^{-u} \left(\frac{u}{2h}\right)^{1-\frac{2}{j}} \left(\frac{u}{2h}\right)^{\frac{1}{j}} \frac{du}{u} \\ &= j \int_0^\infty e^{-u} \left(\frac{u}{2h}\right)^{1-\frac{1}{j}} \frac{du}{u} = \frac{j}{(2h)^{1-\frac{1}{j}}} \int_0^\infty e^{-u} u^{\frac{1}{j}} du \\ &= \frac{j}{(2h)^{1-\frac{1}{j}}} \Gamma\left(1-\frac{1}{j}\right) \sim \frac{j}{2^{1-\frac{1}{j}}} \Gamma\left(1-\frac{1}{j}\right) \left(\frac{1}{1-t}\right)^{1-\frac{1}{j}}, \end{aligned}$$

where the last approximation holds since when $h \to 0^+$, $1-t \to 0^+$, and Γ is the Gamma function. This implies that

$$Q\left(\frac{t}{q}\right) \sim \frac{j}{2^{1-\frac{1}{j}}} \Gamma\left(1-\frac{1}{j}\right) \left(\frac{1}{1-t}\right)^{1-\frac{1}{j}}.$$

Finally,

$$\frac{1}{1-qz} Q(z) = \frac{1}{1-t} Q\left(\frac{t}{q}\right) \sim \Gamma\left(1-\frac{1}{j}\right) \frac{j}{2^{1-\frac{1}{j}}} \left(\frac{1}{1-t}\right)^{2-\frac{1}{j}}.$$

We transfer to coefficients using Theorem 1.3.3 obtaining

$$[z^n] \frac{1}{1-qz} Q(z) \sim \frac{\Gamma\left(1-\frac{1}{j}\right)}{\Gamma\left(2-\frac{1}{j}\right)} \frac{j}{2^{1-\frac{1}{j}}} n^{1-\frac{1}{j}}.$$

Since $\Gamma(1+x) = x!$, we simplify $\frac{\Gamma(1-\frac{1}{j})}{\Gamma(2-\frac{1}{j})}$ to $\frac{1}{1-\frac{1}{j}}$. Therefore, we conclude

$$[z^n] \frac{1}{1-qz} Q(z) \sim \frac{j^2}{j-1} \frac{1}{2^{1-\frac{1}{j}}} n^{1-\frac{1}{j}}.$$

This completes the proof. □

For example, for the partition $S_k = k^2$, i.e. $j = 2$, the expected value of the total degree of the irreducible factors in any of the multi-factor intervals of a polynomial is asymptotic to $\frac{4}{\sqrt{2}}\sqrt{n}$. For the partition $S_k = k^3$, this value is asymptotic to $\frac{4.5}{\sqrt[3]{4}} n^{\frac{2}{3}}$, and for the partition $S_k = k^{\frac{3}{2}}$, this value is asymptotic to $4.5\sqrt{2}\, n^{\frac{1}{3}}$.

Observe that all the studies in this section, with the exception of the total degree in multi-factor intervals, gave constant results. It could be interesting to study standard deviations and other moments for each of the derived parameters. This work promises to be very technical, especially for the total degree of irreducible factors in multi-factor intervals. In consideration of the patience of the reader that keep us company, and a la premura del tiempo, we will not develop those studies in this thesis.

S_k	Total degree in multi-factor intervals
$k^{3/2}$	$4.5\sqrt{2}\,n^{\frac{1}{3}}$
k^2	$\frac{4}{\sqrt{2}}\sqrt{n}$
k^3	$\frac{4.5}{\sqrt[3]{4}}\,n^{\frac{2}{3}}$
$k^\alpha(\alpha > 1)$	$\frac{\alpha^2}{\alpha-1}\,\frac{1}{2^{1-\frac{1}{\alpha}}}\,n^{1-\frac{1}{\alpha}}$

Table 5.5: Expected total degree of factors in multi-factor intervals for partition S_k.

5.3.5 Relation to average-case analysis of growing interval sizes factoring algorithms

We briefly comment on the relation between our results and the factorization algorithms of Section 5.2. A look at the theorems in the previous section shows that in order to reduce the number of collisions in intervals of irreducible factors of randomly chosen polynomials (Theorem 5.3.4), it is convenient to consider partitions $S_k = k^j$ with $j > 1$, and as small as possible. In fact, this holds not only for the number of irreducible factors but also for the number of multi-factor intervals (Theorems 5.3.1 and 5.3.3), and for the total degree of irreducible factors in multi-factor intervals (Theorem 5.3.6). This implies that in the limit we have the partition with intervals of size 1. In terms of the DDF algorithm, this leads to the basic DDF algorithm (see Section 3.2.2). However, the smaller j is, the larger is the length of the partition. So, in the case of small j, we will have less work at the *fine* level, and more work at the *coarse* level of the algorithm. These observations introduce an interesting tradeoff for choosing the best interval partition $S_k = k^j$ for the factorization algorithms. The worst-case behavior of the algorithm in Section 5.2 for polynomial factorization over $\mathbb{F}_2$ in which powers are computed via repeated squaring was analyzed by von zur Gathen & Gerhard (1996). They showed that under these hypotheses the best partition is $S_k = k^2$. It seems possible to study the average-case behavior of the algorithm for this partition following closely our average-case analysis of Chapter 4.

Chapter 6

Other Polynomial Problems in Finite Fields

This chapter focuses on several problems involving polynomials over finite fields. They all deal directly or indirectly with constructions of extension fields, but they are also of independent interest. Depending on the problem, our counting techniques of irreducible polynomials may vary in their flavor.

First, we concentrate on tests for irreducible polynomials over finite fields. We revisit the algorithm by Rabin (1980) giving an improved variant of this algorithm. We give a precise analysis of the probability that a random polynomial of degree n contains no irreducible factors of degree less than $O(\log n)$. This probability is naturally related to Ben-Or's (1981) algorithm for testing irreducibility of polynomials. We also present a fast variation of Ben-Or's method. We then compute the probability of a polynomial being irreducible when it has no irreducible factors of low degree. This probability is useful in the analysis of various algorithms for factoring polynomials over finite fields. We present an experimental comparison of these irreducibility methods when testing random polynomials.

Second, we establish a construction of sparse (up to a constant number of terms) irreducible polynomials over any finite field for infinitely many degrees.

Third, we improve the lower bound on the Euler totient function for polynomials over finite fields. This lower bound is useful for solving systems of sparse linear equations and

for establishing a lower bound for the density of normal elements.

Finally, we prove that there exists a positive lower bound for the density of normal elements over any field for infinitely many degrees. We show that this is not true for all degrees. This bound is used in randomized algorithms to construct normal elements in finite fields.

Most of the results in this chapter are from the joint papers Gao & Panario (1995, 1997). In this thesis we introduce Ben-Or's variant, we extend the experimental section, and give more detailed, sometimes different, proofs of the theorems.

6.1 Motivation and results

For a prime power q and an integer $n \geq 2$, let $\mathbb{F}_{q^n}$ be the extension field of degree n of $\mathbb{F}_q$. Extensions of finite fields (especially $\mathbb{F}_2$) are important in implementing cryptosystems and error correcting codes. As seen in Section 1.2.1, we know that one way of constructing extensions of finite fields is via an irreducible polynomial over the ground field with degree equal to the degree of the extension. Therefore, finding irreducible polynomials and testing the irreducibility of polynomials are central problems in finite fields.

A probabilistic algorithm for finding irreducible polynomials that works well in practice is presented in Rabin (1980). The central idea is to take polynomials at random and test them for irreducibility. As usual, let I_n be the number of irreducible polynomials of degree n over a finite field $\mathbb{F}_q$. Lower and upper bounds for I_n are given in Theorem 1.3.2 (Lidl & Niederreiter 1983, p. 142, Ex. 3.26 & 3.27):

$$\frac{q^n}{n} - \frac{q(q^{n/2}-1)}{(q-1)n} \leq I_n \leq \frac{q^n - q}{n}. \tag{6.1}$$

This means that a fraction $1/n$ of the polynomials of degree n is irreducible, and so we find on average one irreducible polynomial of degree n after n tries. This observation leads naturally to a randomized algorithm for constructing irreducible polynomials, provided that we have a way to test for irreducibility. In Sections 6.2, we center on tests for irreducibility. Let $f \in \mathbb{F}_q[x]$, $\deg f = n$, be a polynomial to be tested for irreducibility. Let $p_1, \ldots, p_k$ be the distinct prime divisors of n. In practice, there are two general approaches for this problem:

- Butler (1954): f is irreducible if and only if $\dim \ker(\Phi - I) = 1$, where Φ is the Frobenius map on $\mathbb{F}_q[x]/(f)$ that sends $h \in \mathbb{F}_q[x]/(f)$ to $h^q \in \mathbb{F}_q[x]/(f)$, and I is the identity map on $\mathbb{F}_q[x]/(f)$;

- Rabin (1980): f is irreducible if and only if $\gcd(f, x^{q^{n/p_i}} - x) = 1$ for all $1 \leq i \leq k$, and $x^{q^n} - x \equiv 0 \bmod f$.

The former method is one of the components of Berlekamp's (1967) algorithm for factoring polynomials (see Section 3.3). Other irreducible tests can be found in von zur Gathen & Shoup (1992), Shoup (1995), and von zur Gathen & Gerhard (1996).

We concentrate on Rabin's test, and a variant presented in Ben-Or (1981). In Sections 6.2.1, we revisit Rabin's irreducibility algorithm. We state a variant of this algorithm that allows a $\log n$ factor saving (Theorem 6.2.2). In Section 6.2.2, we focus on Ben-Or's algorithm. This leads us to study the behavior of *rough* polynomials, i.e., polynomials without irreducible factors of low degrees (Theorems 6.2.3 and 6.2.5). The analysis is expressed as an asymptotic form in n, the degree of the polynomial to be tested for irreducibility. First, we fix a finite field $\mathbb{F}_q$, and then we study asymptotics on q. As was noted in Section 1.3.4, probabilistic properties of polynomials over finite fields frequently have a shape that resembles corresponding properties of the cycle decomposition of permutations to which they reduce when the size of the field goes to infinity. An instance of this is derived for the probability that a polynomial of degree n over $\mathbb{F}_q$ contains no factors of degree m, $1 \leq m \leq O(\log n)$, when $q \to +\infty$. This probability relates naturally with Ben-Or's algorithm. We then use this information to derive a fast variation of Ben-Or's algorithm. The probability of a polynomial being irreducible when it has no irreducible factors of low degree provides useful information for factoring polynomials over finite fields (see for instance, von zur Gathen & Gerhard 1996, §6). We provide the probability of a polynomial being irreducible when it has no irreducible factors of degree at most $O(\log n)$ (Theorem 6.2.8).

In Section 6.2.3, we provide a running time comparison among the algorithms discussed in Sections 6.2.1 and 6.2.2. A comparison between the theoretical and the experimental results for each of the algorithms is given for several field sizes and polynomial

degrees.

Very sparse irreducible polynomials are useful for several applications: pseudorandom number generators using feedback shift registers (Golomb 1982), discrete logarithm over $\mathbb{F}_{2^n}$ (Coppersmith 1984, Odlyzko 1985), and efficient arithmetic in finite fields (Shoup 1994). However, few results are known about these polynomials beyond binomials and trinomials (see Menezes *et al.* 1993, Chapter 3, and the references there; Shparlinski 1993). In Section 6.3, we present a construction of irreducible polynomials over $\mathbb{F}_q$ of degree n with up to $O(1)$ nonzero terms (not necessarily the lowest coefficients), for infinitely many degrees n (Theorem 6.3.2). In fact, our construction can be seen as a generalization of Shparlinski (1993), although it was independently developed.

Let $f \in \mathbb{F}_q[x]$ be a monic polynomial of degree n, and let $\Phi(f)$ denote the Euler totient function for polynomials, that is, the number of polynomials in $\mathbb{F}_q[x]$ of degree $< n$ that are relatively prime to f. Let $\kappa(f) = \Phi(f)/q^n$. We are interested in good lower bounds for $\kappa(f)$. In Section 6.4, we present previous results for lower bounds on $\kappa(f)$, and slightly improve them (Theorem 6.4.1). This kind of lower bound is useful for solving sparse linear equations (Wiedemann 1986), and for constructing normal elements over finite fields using randomized algorithms (von zur Gathen & Giesbrecht 1990, Giesbrecht 1993). We recall from Section 1.2.2 that an element $\alpha \in \mathbb{F}_{q^n}$ is *normal* over $\mathbb{F}_q$ if and only if its conjugates $\alpha, \alpha^q, \ldots, \alpha^{q^{n-1}}$ are linearly independent over $\mathbb{F}_q$. When α is normal, $(\alpha, \alpha^q, \ldots, \alpha^{q^{n-1}})$ is a basis, and it is called the *normal basis* generated by α. Ore (1934) proves that the number of normal elements in $\mathbb{F}_{q^n}$ over $\mathbb{F}_q$ is $\Phi(x^n - 1)$. The density of normal elements in $\mathbb{F}_{q^n}$ over $\mathbb{F}_q$ is then $\kappa(x^n - 1) = \Phi(x^n - 1)/q^n$.

To construct a normal element in $\mathbb{F}_{q^n}$, one simple method is to draw an element from $\mathbb{F}_{q^n}$ uniformly at random, test if it is normal, and repeat until a normal element is obtained. The density $\kappa(x^n - 1)$ is the probability of a random element being normal. Thus one expects to draw $\lceil 1/\kappa(x^n - 1) \rceil$ elements to get a normal one. A good lower bound for $\kappa(x^n - 1)$ gives a good upper bound for the expected number of draws (or samplings) of the algorithm. In Section 6.5, we prove that $\kappa(x^n - 1) \geq C > 0$ for $n = p_1^{e_1} p_2^{e_2} \cdots p_t^{e_t}$ where p_i are fixed primes, e_i vary, and C is a constant independent of the e_i's (Theorems 6.5.4 and 6.5.5). Unfortunately, this is not true for general n. Indeed,

we show an upper bound on $\kappa(x^n - 1)$ for infinitely many values of n that goes to 0 as n approaches infinity (Theorem 6.5.6). This upper bound is almost tight with the lower bound of Section 6.4.

Finally, we recall from Section 1.2.1 the cost of the operations over polynomial that appear in the algorithms of this chapter. For the sake of brevity, we consider exclusively FFT based arithmetic; similar results hold for classical arithmetic. Let us write $M(n) = n \log n \log\log n$. The cost of multiplying two polynomials of degree at most n using "fast" arithmetic (Schönhage & Strassen 1971, Schönhage 1977, Cantor & Kaltofen 1991) can be taken as $\tau_1 M(n)$, for a constant τ_1. The cost of a gcd between two polynomials of degree at most n can be taken as $\tau_2 M(n) \log n$ operations in $\mathbb{F}_q$, for a constant τ_2. Let h be a polynomial over $\mathbb{F}_q$ of degree less than n. Then, the number of products needed to compute $h^q \bmod f$ using the repeated squaring method (Knuth 1981, p. 441–442) is $C_q = \lfloor \log_2 q \rfloor + \nu(q)$, with $\nu(q)$ the number of ones in the binary representation of q. Therefore, the cost of computing $h^q \bmod f$ by this method is $\tau_1 C_q M(n)$ operations in $\mathbb{F}_q$ using FFT based methods.

6.2 Irreducibility tests for polynomials

6.2.1 An improvement on Rabin irreducibility test

In this section, we revisit Rabin's test, and we present a variant of his method.

Algorithm: Rabin Irreducibility Test

Input: A monic polynomial $f \in \mathbb{F}_q[x]$ of degree n,
and $p_1, \ldots, p_k$ all the distinct prime divisors of n .

Output: Either "f is irreducible" or "f is reducible".

```
for j := 1 to k do
    n_j := n/p_j;
for i := 1 to k do
    g := gcd(f, x^{q^{n_i}} − x mod f);
```

```
        if g ≠ 1, then 'f is reducible' and STOP;
    endfor;
    g := x^{q^n} − x mod f;
    if g = 0, then 'f is irreducible'
              else 'f is reducible';
```

The correctness of Rabin's algorithm is based on the following theorem (see Rabin 1980, p. 275, Lemma 1).

Theorem 6.2.1 *Let $p_1, \ldots, p_k$ be all the prime divisors of n, and denote $n_i = n/p_i$, for $1 \leq i \leq k$. A polynomial $f \in \mathbb{F}_q[x]$ of degree n is irreducible in $\mathbb{F}_q[x]$ if and only if gcd $(f, x^{q^{n_i}} - x \bmod f) = 1$, for $1 \leq i \leq k$, and f divides $x^{q^n} - x$.*

The basic idea of this algorithm is to compute $x^{q^{n_i}} \bmod f$ independently for each value $n_1, \ldots, n_k$ by repeated squaring, and then to take the corresponding gcd. The worst-case analysis given in Rabin (1980) is $O(nM(n)\log n \log q)$ operations in $\mathbb{F}_q$. However, it can be shown that $O(nM(n)\log\log n \log q)$ is an upper bound on the number of operations in $\mathbb{F}_q$ for this algorithm. Indeed, first note that the number of distinct prime factors of n is at most $\log n$. In fact, if n is the product of the ℓ smallest distinct prime factors, $n \geq 2^\ell$, and then $\ell \leq \log n$. The cost of the k exponentiations is

$$\sum_{i=1}^{k} \frac{n}{p_i} \log q\, M(n) \leq nM(n) \log q \sum_{i=1}^{\log n} \frac{1}{p_i} \leq nM(n) \log q\, H_{\log n},$$

where $H_{\log n} = \sum_{i=1}^{\log n} \frac{1}{i}$ is the harmonic sum. By Theorem 1.4.1, $H_{\log n} = \log\log n + \gamma + O(\frac{1}{\log n})$. Therefore, the cost of the k exponentiations is $O(nM(n)\log\log n \log q)$, which dominates the cost $O(M(n)\log^2 n)$ of computing k gcd's. Finally, the total cost of Rabin's algorithm is $O(nM(n)\log\log n \log q)$.

As an improvement, we propose the following variant for the computation of $x^{q^{n_i}} - x \bmod f$, for $1 \leq i \leq k$.

Algorithm: Variant of Rabin Irreducibility Test

Input: A monic polynomial $f \in \mathbb{F}_q[x]$ of degree n,
and $p_1, \ldots, p_k$ all the distinct prime divisors of n .

Output: Either "f is irreducible" or "f is reducible".

n_0 := 0; h_0 := x;
for j := 1 to k do
    n_j := n/p_j;
sort($n_1, \ldots, n_k$); (* Assume $n_1 < n_2 < \ldots < n_k$. *)
for i := 1 to k do
    h_i := $h_{i-1}^{q^{n_i - n_{i-1}}}$ mod f;
    g := gcd(f, $h_i - x$);
    if $g \neq 1$, then 'f is reducible' and STOP;
endfor;
g := $h_k^{q^{n-n_k}} - x$ mod f;
if $g = 0$, then 'f is irreducible'
        else 'f is reducible';

Theorem 6.2.2 *The above algorithm correctly tests for polynomial irreducibility, and uses $O(nM(n)\log q)$ operations in $\mathbb{F}_q$.*

PROOF. The correctness of the algorithm follows from the correctness of the power computations. We prove that $h_i = x^{q^{n_i}} \bmod f$, for $1 \leq i \leq k$, by induction on k.
Basis: when $k = 1$,

$$h_1 \equiv h_0^{q^{n_1 - n_0}} \equiv h_0^{q^{n_1}} \equiv x^{q^{n_1}} \bmod f.$$

Inductive step: for some k, $h_i = x^{q^{n_i}} \bmod f$, $1 \leq i \leq k$. Then,

$$h_{k+1} \equiv h_k^{q^{n_{k+1} - n_k}} \equiv \left(x^{q^{n_k}}\right)^{q^{n_{k+1} - n_k}} \equiv x^{q^{n_k} \cdot q^{n_{k+1} - n_k}} \equiv x^{q^{n_{k+1}}} \bmod f.$$

With this variant, in the worst-case, the number of polynomial multiplications in Rabin's algorithm to compute all powers using repeated squaring is

$$n_1 \log q + (n_2 - n_1)\log q + \cdots + (n - n_k)\log q = n \log q.$$

Hence, the cost of k exponentiations is $O(nM(n)\log q)$. To complete the analysis of the algorithm, we have to count the cost of computing the k gcd's. Since the number of distinct prime factors of n is at most $\log n$, the cost of taking all the gcd's in the algorithm is $O(M(n)\log^2 n)$, and the total cost of this variant is $O(nM(n)\log q)$. □

An average-case estimate for the cost of this algorithm is an open problem. It may be possible to give this estimate by studying the probability that a random polynomial of degree n contains an irreducible factor of degree n_i and no factors of degree dividing n_j, for $j < i$, but we have not explored this possibility further.

6.2.2 Ben-Or irreducibility test

In this section, we study the following variant of the Rabin test presented in Ben-Or (1981).

Algorithm: Ben-Or Irreducibility Test
Input: A monic polynomial $f \in \mathbb{F}_q[x]$ of degree n.
Output: Either "f is irreducible" or "f is reducible".

```
for i := 1 to n/2 do
    g := gcd(f, x^{q^i} − x mod f);
    if g ≠ 1, then 'f is reducible' and STOP;
endfor;
'f is irreducible';
```

The correctness of Ben-Or's procedure is based on Theorem 3.2.1, which we restate here.

Theorem. *For $i \geq 1$, the polynomial $x^{q^i} - x \in \mathbb{F}_q[x]$ is the product of all monic irreducible polynomials in $\mathbb{F}_q[x]$ whose degree divides i.*

Indeed, Ben-Or's algorithm computes $x^{q^i} \bmod f$, and $\gcd(f, x^{q^i} - x)$ for $i = 1, \ldots, \frac{n}{2}$. The polynomial is reducible if and only if one of the gcd's is different from 1.

In the worst case, this algorithm computes $\frac{n}{2}$ times a qth power and a gcd of polynomials of degree at most n. Recalling the cost of these operations from Section 6.1, the worst-case behavior of Ben-Or's algorithm is $O(nM(n)\log(qn))$, using FFT based multiplication algorithms. Thus, it has a worst behavior than both Rabin's and our variant algorithms. The main reason for the efficiency of this algorithm is that the average number of irreducible factors of degree k of a random polynomial of degree n approaches $1/k$ as n tends to infinity (see Theorem 2.2.2, and Knopfmacher & Knopfmacher 1993). Thus, one expects to have an irreducible factor of smaller degree in a random polynomial very often, and Ben-Or's algorithm quickly discards these polynomials.

Rabin's and Ben-Or's methods for constructing irreducible polynomials are based on trial and error of *random* polynomials. What makes Ben-Or's irreducibility test interesting is its average-case behavior when testing random polynomials. Ben-Or already estimates the average-case behavior of his algorithm. Let $s(f)$ be the expected value of the smallest degree among the irreducible factors of f; then the expected cost of his algorithm is $O(s(f)M(n)\log(qn))$. Ben-Or (1981, Theorem 2) derives an $O(\log n)$ estimate for $s(f)$. In fact, he relates the factorial decomposition of polynomials with the cyclic decomposition of permutations. The result follows from the study of the expected length of the shortest cycle in a random permutation (Shepp & Lloyd 1966). Bach & Shallit (1996, Ex. 7.32, p. 359) formalize this big-Oh estimate. No precise analysis of the implied constant is known.

Polynomials without irreducible factors of low degree make Ben-Or's irreducibility test to execute a large number of iterations. The probability that a random polynomial of degree n contains no factors of low degree gives meaningful information on the behavior of this algorithm. We call a polynomial *m-rough* if it has no irreducible factors of degrees $\leq m$. In the following, we focus on the distribution of rough polynomials for any field $\mathbb{F}_q$. Then, we consider the permutation model case, that is, we make q approach infinity. In this case, our result agrees with the corresponding one of permutations with no cycles of length m or less (for instance, see Sedgewick & Flajolet 1996, Theorem 6.3).

In the next theorem, we provide estimates for the function $g_q(m) = \prod_{k=1}^{m}\left(1-\frac{1}{q^k}\right)^{I_k}$. These estimates will be useful not only for studying the distribution of rough polynomials

but also for establishing lower bounds for the Euler totient function (Section 6.4), and for the density of normal elements (Section 6.5).

Theorem 6.2.3 *Let* $g_q(m) = \prod_{k=1}^{m} \left(1 - \frac{1}{q^k}\right)^{I_k}$. *Then, for any prime power* q *and positive integer* m,

$$\frac{1}{em} \leq \exp\left(-\sum_{k=1}^{m} \frac{1}{k}\right) \leq g_q(m) \leq \left(1 - \frac{1}{\sqrt{q}}\right)^{-\frac{q}{q-1}} \cdot \exp\left(-\sum_{k=1}^{m} \frac{1}{k}\right).$$

When $q \to \infty$, *we have*

$$g_q(m) = \prod_{k=1}^{m} \left(1 - \frac{1}{q^k}\right)^{I_k} \to e^{-H_m} \sim \frac{e^{-\gamma}}{m}, \tag{6.2}$$

where γ *is the Euler's constant, and* $e^{-\gamma} = 0.56416\ldots$

PROOF. Taking logarithms,

$$g_q(m) = \prod_{k=1}^{m} \exp \ln \left(1 - \frac{1}{q^k}\right)^{I_k} = \prod_{k=1}^{m} \exp\left(I_k \ln\left(1 - \frac{1}{q^k}\right)\right).$$

Expanding logarithms, considering that $\ln(1+x) \leq x$ for $x > -1$, that $\sum_{k=1}^{\infty} \frac{1}{kq^{k/2}} = -\ln\left(1 - \frac{1}{\sqrt{q}}\right)$, and using Eq. (6.1), we have

$$\begin{aligned}
g_q(m) &= \prod_{k=1}^{m} \exp\left(-I_k \sum_{j\geq 1} \frac{1}{jq^{jk}}\right) \\
&\leq \prod_{k=1}^{m} \exp\left(-\frac{I_k}{q^k}\right) = \exp\left(-\sum_{k=1}^{m} \frac{I_k}{q^k}\right) \leq \exp\left(-\sum_{k=1}^{m} \frac{\frac{q^k}{k} - \frac{q(q^{k/2}-1)}{(q-1)k}}{q^k}\right) \\
&\leq \exp\left(-\sum_{k=1}^{m} \frac{1}{k}\left(1 - \frac{q}{q-1}\frac{q^{k/2}-1}{q^k}\right)\right) \\
&= \exp\left(-\sum_{k=1}^{m} \frac{1}{k}\right) \exp\left(\frac{q}{q-1} \sum_{k=1}^{m} \frac{q^{k/2}-1}{kq^k}\right) \\
&\leq \exp\left(-\sum_{k=1}^{m} \frac{1}{k}\right) \cdot \left(\exp\left(\sum_{k=1}^{\infty} \frac{1}{kq^{k/2}}\right)\right)^{\frac{q}{q-1}} \\
&= \exp\left(-\sum_{k=1}^{m} \frac{1}{k}\right) \cdot \left(\exp\left(-\ln\left(1 - \frac{1}{\sqrt{q}}\right)\right)\right)^{\frac{q}{q-1}} \\
&= \left(1 - \frac{1}{\sqrt{q}}\right)^{-\frac{q}{q-1}} \cdot \exp\left(-\sum_{k=1}^{m} \frac{1}{k}\right).
\end{aligned}$$

We have derived a precise upper bound for $g_q(m) = \prod_{k=1}^{m} \left(1 - \frac{1}{q^k}\right)^{I_k}$ in terms of q, the size of the field, and of the parameter m. For the lower bound, first note that from Eq. (6.1), we obtain $I_k \leq (q^k - 1)/k$. Using Theorem 1.4.2 and that $0 < 1 - 1/q^k < 1$, we have

$$\begin{aligned} g_q(m) &= \prod_{k=1}^{m} \left(1 - \frac{1}{q^k}\right)^{I_k} \geq \prod_{k=1}^{m} \left(1 - \frac{1}{q^k}\right)^{\frac{q^k-1}{k}} \\ &= \prod_{k=1}^{m} \left(\left(1 + \frac{1}{q^k - 1}\right)^{q^k-1}\right)^{-\frac{1}{k}} \geq \prod_{k=1}^{m} \exp\left(-\frac{1}{k}\right) = \exp\left(-\sum_{k=1}^{m} \frac{1}{k}\right). \end{aligned}$$

Let H_m be the harmonic sum, i.e., $H_m = \sum_{k=1}^{m} 1/k$. Since q is fixed, we have proven that $\prod_{k=1}^{m} \left(1 - \frac{1}{q^k}\right)^{I_k} \in \Theta(e^{-H_m})$.

The lower bound $1/(em)$ for $g_q(m)$ is easily derived since

$$H_m \leq 1 + \int_1^m 1/x \, dx = 1 + \ln m,$$

and thus $e^{-H_m} \geq 1/(em)$. Finally, when $q \to \infty$, $\left(1 - \frac{1}{\sqrt{q}}\right)^{-\frac{q}{q-1}} \to 1$. Therefore, in that case,

$$g_q(m) = \prod_{k=1}^{m} \left(1 - \frac{1}{q^k}\right)^{I_k} \sim e^{-H_m}.$$

Using the well-known approximation of the harmonic sum in Theorem 1.4.1, $H_m = \ln m + \gamma + O(\frac{1}{m})$, we have

$$e^{-H_m} \sim \frac{e^{-\gamma}}{m}, \quad m \to \infty. \tag{6.3}$$

This result is in accordance with the corresponding one on permutations with no cycles of length m or less (see Sedgewick & Flajolet 1996, Theorem 6.3). □

In order to study the distribution of rough polynomials, we need the following result from Odlyzko (1996, Theorem 10.8). Recall that a function is *meromorphic* in an open set if it is analytic in the set with the possible exception of a countable number of poles.

Theorem 6.2.4 *Suppose that $f(z)$ is meromorphic in an open set containing $|z| \leq r$, that is analytic at $z = 0$ and on $|z| = r$, and that the only poles of $f(z)$ in $|z| < r$ are at $\rho_1, \ldots, \rho_k$, each of multiplicity 1. Suppose further that*

$$\max_{|z|=r} |f(z)| \leq w,$$

and that $r - \rho_j \geq \delta$ *for some* $\delta > 0$ *and* $1 \leq j \leq k$. *Then,*

$$\left| [z^n] f(z) + \sum_{j=1}^{k} r_j \rho_j^{-n-1} \right| \leq w r^{-n} + \delta^{-1} r^{-n} \sum_{j=1}^{k} |r_j|, \tag{6.4}$$

where r_j *is the residue of* $f(z)$ *at* ρ_j.

We now have all the elements to study the distribution of rough polynomials.

Theorem 6.2.5 *Denote by* $P_q(n,m)$ *the probability of a random monic polynomial of degree* n *over* $\mathbb{F}_q$ *being* m*-rough. Then when* $n \to \infty$,

$$P_q(n,m) = \prod_{k=1}^{m} \left(1 - \frac{1}{q^k}\right)^{I_k} (1 + o(1)),$$

uniformly for q *and* $1 \leq m \leq O(\log n)$.

PROOF. Let $\mathcal{I}$ be the collection of all monic irreducible polynomials in $\mathbb{F}_q$. Formally, the collection of all monic polynomials with all irreducible factors with degree $> m$ is

$$\mathcal{P}_m = \prod_{\omega \in \mathcal{I},\ |\omega| > m} (1 + \omega + \omega^2 + \cdots) = \prod_{\omega \in \mathcal{I},\ |\omega| > m} (1 - \omega)^{-1}.$$

Let z be a formal variable, and $|\omega|$ the degree of $\omega \in \mathcal{I}$. The substitution $\omega \mapsto z^{|\omega|}$ produces the generating function $P_m(z)$ of polynomials with all irreducible factors with degree $> m$

$$P_m(z) = \prod_{\omega \in \mathcal{I},\ |\omega| > m} \left(1 - z^{|\omega|}\right)^{-1} = \prod_{k > m} \left(1 - z^k\right)^{-I_k} = \frac{1}{1 - qz} \prod_{k=1}^{m} (1 - z^k)^{I_k}.$$

As an aside, we note that this generating function can be derived using a more combinatorial approach. Indeed, recalling that $\frac{1}{1-z^k} = \sum_{i=0}^{\infty} (z^k)^i$, and considering I_k convolutions of $\frac{1}{1-z^k}$, we obtain

$$\left(1 - z^k\right)^{-I_k} = \sum_{i=0}^{\infty} \binom{I_k + i - 1}{i} z^{ki}, \tag{6.5}$$

which precisely counts the number of polynomials of degree ki with i irreducible factors of degree k. By Eq. (6.5), to obtain $P_m(z)$, we have to consider the product when k varies from 1 to m, i.e.,

$$P_m(z) = \prod_{k > m} \sum_{i=0}^{\infty} \binom{I_k + i - 1}{i} z^{ki} = \prod_{k > m} \left(1 - z^k\right)^{-I_k} = \frac{1}{1 - qz} \prod_{k=1}^{m} (1 - z^k)^{I_k}.$$

Note that m may vary when $n \to +\infty$. Thus, the singularity may become large and we can not apply the transfer lemmas in Flajolet & Odlyzko (1990), and Odlyzko (1996), as we did in Corollary 2.3.2.

As usual, $[z^n]P_m(z)$ represents the coefficient of z^n in $P_m(z)$. Observe that $P_q(n,m) = [z^n]P_m(z)/q^n$. In order to estimate $P_q(n,m)$, we apply Theorem 6.2.4. Suppose that $m \leq c \ln n$ for some constant $c > 0$, and let b be a constant such that $1 < b < e^{1/c}$. Take $r = b/q > 1/q$. The only singularity that $P_m(z)$ presents is a pole of order 1 at $z = 1/q$. Therefore, $P_m(z)$ is meromorphic in an open set containing $|z| \leq r$, analytic at $z = 0$ and on $|z| = r$, and we can apply Theorem 6.2.4. The residue of $P_m(z)$ at $z = 1/q$ is

$$-\frac{1}{q}\prod_{k=1}^{m}(1-q^{-k})^{I_k}.$$

As in Theorem 6.2.3, we denote the product $\prod_{k=1}^{m}(1-q^{-k})^{I_k}$ by $g_q(m)$. Taking $w = \max_{|z|=r}|P_m(z)|$ and $0 < \delta \leq r - \frac{1}{q}$, Eq. (6.4) gives

$$\left|[z^n]P_m(z) + \left(-\frac{1}{q}g_q(m)\left(\frac{1}{q}\right)^{-n-1}\right)\right| \leq w\,r^{-n} + \left(r-\frac{1}{q}\right)^{-1} r^{-n}\,\frac{1}{q}\,g_q(m).$$

Hence,

$$\begin{aligned} |P_q(n,m) - g_q(m)| &\leq \left(w + \left(r-\frac{1}{q}\right)^{-1}\frac{1}{q}\,g_q(m)\right)(rq)^{-n} \\ &= \left(w + \frac{1}{b-1}\,g_q(m)\right)/\,b^n \leq \left(w+\frac{1}{b-1}\right)/\,b^n, \end{aligned} \tag{6.6}$$

as $0 \leq g_q(m) \leq 1$. Since $b > 1$ is a constant independent of m, q and n, we only need to estimate w in term of n. When $|z| = r = b/q$, we have $|1-qz| \geq b-1$, and $|1-z^k| \leq 1+r^k$. Considering that $I_k\,r^k \leq q^k\,r^k = b^k$, we obtain

$$\begin{aligned} |P_m(z)| &= \frac{1}{|1-qz|}\prod_{k=1}^{m}|1-z^k|^{I_k} \leq \frac{1}{b-1}\prod_{k=1}^{m}(1+r^k)^{I_k} \\ &= \frac{1}{b-1}\prod_{k=1}^{m}\exp\left(I_k\ln(1+r^k)\right) \leq \frac{1}{b-1}\prod_{k=1}^{m}\exp(I_k\,r^k) \\ &\leq \frac{1}{b-1}\prod_{k=1}^{m}\exp(b^k) = \frac{1}{b-1}\exp\left(\sum_{k=1}^{m}b^k\right) \leq \frac{1}{b-1}\exp\left(\frac{b^{m+1}}{b-1}\right) \\ &\leq \frac{1}{b-1}\exp\left(\frac{b}{b-1}b^{c\ln n}\right) = \frac{1}{b-1}\exp\left(\frac{b}{b-1}n^{c\ln b}\right). \end{aligned}$$

Recalling that $w = \max_{|z|=r} |P_m(z)|$, we have $w \leq \frac{1}{b-1} \exp\left(\frac{b}{b-1} n^{c \ln b}\right)$. Thus, $w/b^n \leq \frac{1}{b-1} \exp\left(\frac{b}{b-1} n^{c \ln b} - n \ln b\right)$, and finally, from Eq. (6.6) we obtain

$$\left(w + \frac{1}{b-1}\right) / b^n \leq \frac{2}{b-1} \exp\left(\frac{b}{b-1} n^{c \ln b} - (\ln b)\, n\right).$$

By Theorem 6.2.3,

$$g_q(m) \geq \frac{1}{em} \geq \frac{1}{ec \ln n}.$$

Therefore,

$$\left| \frac{P_q(n,m)}{g_q(m)} - 1 \right| \leq \frac{2}{b-1} ec \ln n \exp\left(\frac{b}{b-1} n^{c \ln b} - (\ln b)\, n\right). \tag{6.7}$$

As $\ln b > 0$ and $c \ln b < 1$, the right-hand side of Eq. (6.7) approaches 0 as $n \to \infty$. Since the quantity on the right-hand side of Eq. (6.7) is independent of q and m, we conclude that $P_q(n,m)/g_q(m)$ approaches 1 uniformly for q and $m \leq c \ln n$, i.e., $m \leq O(\log n)$, when $n \to \infty$. □

A natural theoretical question is whether the interval of validity of Theorem 6.2.5 can be moved beyond $O(\log n)$. The study of singularities in the proof of the theorem does not apply outside this range. However, we could use the saddle point bound (see Odlyzko 1996, § 12.1). This method extends the range up to about $O(n/\log^2 n)$. We do not follow this study in this thesis.

Now, we concentrate on a more practical issue, that is, how to use the above results to improve Ben-Or's algorithm. Theorems 6.2.3 and 6.2.5 imply that with probability tending to 1 when n goes to infinity, Ben-Or's algorithm finds an irreducible factor of degree less than $O(\log n)$ and stops. In such a case, we can derive the expected cost of the number of operations in Ben-Or's algorithm. Recall that the cost of multiplying two polynomials of degree at most n can be taken as $\tau_1 M(n)$ operations in $\mathbb{F}_q$; the cost of a gcd between two polynomials of degree at most n can be taken as $\tau_2 M(n) \log n$ operations in $\mathbb{F}_q$; and the cost of computing $h^q \bmod f$ by the repeated squaring method is $\tau_1 C_q M(n)$ operations in $\mathbb{F}_q$, for constants τ_1, τ_2, and $C_q = \lfloor \log_2 q \rfloor + \nu(q)$, with $\nu(q)$ the number of ones in the binary representation of q.

Theorem 6.2.6 *If $f \in \mathbb{F}_q[x]$ of degree n has an irreducible factor of degree less than $O(\log n)$, then the expected number of operations used by Ben-Or's algorithm is*

$$(C_q\tau_1 + \tau_2 \log n)\ M(n) \left(1 + \sum_{m=1}^{\lfloor \log n \rfloor} \frac{\prod_{k=1}^{m}(1-q^{-k})^{I_k}}{1-\prod_{k=1}^{\lfloor \log n \rfloor}(1-q^{-k})^{I_k}}\right).$$

When $q \to \infty$, this expected number of operations is asymptotic to

$$\left[(C_q\tau_1 + \tau_2 \log n)\ M(n)\, e^{-\gamma} \log\log n\right](1 + o(1)),$$

where γ is Euler's constant, and $e^{-\gamma} = 0.56416\ldots$

PROOF. The most expensive step in the algorithm is the computation of $\gcd\,(f, x^{q^i} - x \bmod f)$, for $i = 1, \ldots, n/2$. Note that this computation costs the same for any value of i. Indeed, we can compute $x^{q^i} - x \bmod f$ from $h = x^{q^{i-1}} \bmod f$ by performing the following steps: compute $h^q \bmod f$, store it as the new h, and compute $h - x = x^{q^i} - x \bmod f$. Therefore, the cost of any iteration step is $(C_q\tau_1 + \tau_2 \log n)\ M(n)$.

Let A_n be the collection of polynomials over $\mathbb{F}_q$ of degree n with some irreducible factor of degree at most $\lfloor \log n \rfloor$, and let X be a random variable indicating for a polynomial $f \in A_n$ the smallest degree among its irreducible factors. Using Theorem 6.2.5, for any $1 \le m \le \lfloor \log n \rfloor$,

$$\text{Prob}\,[\,X > m\,] = \frac{P_q(n,m)}{1 - P_q(n, \lfloor \log n \rfloor)} \sim \frac{\prod_{k=1}^{m}(1-q^{-k})^{I_k}}{1-\prod_{k=1}^{\lfloor \log n \rfloor}(1-q^{-k})^{I_k}}.$$

The expectation of X can be estimated as follows:

$$\begin{aligned} \mathrm{E}(X) &= \sum_{m \ge 1} m \cdot \text{Prob}\,[\,X = m\,] = \sum_{m \ge 0} \text{Prob}\,[\,X > m\,] \\ &= \text{Prob}\,[\,X > 0\,] + \sum_{m \ge 1} \text{Prob}\,[\,X > m\,] \sim 1 + \sum_{m=1}^{\lfloor \log n \rfloor} \frac{\prod_{k=1}^{m}(1-q^{-k})^{I_k}}{1-\prod_{k=1}^{\lfloor \log n \rfloor}(1-q^{-k})^{I_k}}. \end{aligned}$$

This states the first part of the theorem.

When $q \to \infty$, the estimates for $g_q(m) = \prod_{k=1}^{m}(1-q^{-k})^{I_k}$ in formula (6.2) provide the approximation

$$1 + \sum_{m=1}^{\lfloor \log n \rfloor} \frac{\prod_{k=1}^{m}(1-q^{-k})^{I_k}}{1-\prod_{k=1}^{\lfloor \log n \rfloor}(1-q^{-k})^{I_k}} \sim \frac{1}{1 - \frac{e^{-\gamma}}{\lfloor \log n \rfloor}} \sum_{m=1}^{\lfloor \log n \rfloor} \frac{e^{-\gamma}}{m} \sim e^{-\gamma} \log\log n.$$

Therefore, when $q \to \infty$, the expected number of operations in the first $\log n$ iterations of Ben-Or's algorithm is asymptotic to

$$\left[(C_q\tau_1 + \tau_2 \log n)\ M(n)\, e^{-\gamma} \log\log n\right](1 + o(1)).$$

This proves the second part of the theorem. □

A computational comparison of $g_q(m)$, $P_q(n,m)$ and their ratio when $m = \log n$ and $q = 2$ shows that the convergence of $P_q(n,m)$ to $g_q(m)$ is very fast. Moreover, as n grows, $P_q(n,m)$ quickly decreases. For instance, for a random polynomial of degree 40, there is a probability of more than 0.9 of having a factor of degree at most 5. This is another explanation for the efficiency of Ben-Or's algorithm. Indeed, it is enough to search for irreducible polynomials of degree at most $O(\log n)$ in order to have a high probability of finding a factor.

This fact immediately suggests the following algorithm: use Ben-Or's process up to $O(\log n)$ iterations and our Rabin's variant from that point up to $n/2$ iterations. In principle, this algorithm combines the advantage of Ben-Or's method (discarding quickly many polynomials) with our fast exponentiation computation for large powers (saving many power and gcd computations).

Algorithm: Variant of Ben-Or Irreducibility Test

Input: A monic polynomial $f \in \mathbb{F}_q[x]$ of degree n, $c \in \mathbb{N}$, and $p_1, \ldots, p_k$ all the distinct prime divisors of n .

Output: Either "f is irreducible" or "f is reducible".

```
for i := 1 to c log n do
    g := gcd(f, x^{q^i} − x mod f);
    if g ≠ 1, then 'f is reducible' and STOP;
endfor;
n_0 := 0;   n_{k+1} := n;
for j := 1 to k do
    n_j := n/p_j;
```

```
sort(n_1, ..., n_k);  (* Assume n_1 < n_2 < ... < n_k.  *)
search(c log n in n_0, n_1, ..., n_{k+1});  (* Assume n_{j-1} ≤ c log n < n_j.  *)
h_{j-1} := x^{q^{n_{j-1}}};  (* Value already computed.  *)
for i := j to k do
    h_i := h_{i-1}^{q^{n_i - n_{i-1}}} mod f;
    g := gcd(f, h_i - x);
    if g ≠ 1, then 'f is reducible' and STOP;
endfor;
g := h_k^{q^{n-n_k}} - x mod f;
if g = 0, then 'f is irreducible'
          else 'f is reducible';
```

Theorem 6.2.7 *The above algorithm correctly tests for polynomial irreducibility, and uses $O(nM(n)\log q)$ operations in $\mathbb{F}_q$.*

PROOF. The correctness of the algorithm follows from the correctness of both Ben-Or's and our variant of Rabin's algorithm. Theorems 6.2.6 and 6.2.2 imply that the total cost of this variant is $O(nM(n)\log q)$. □

We have proven that our Ben-Or's variant has the same worst case behavior as our Rabin's variant, and better worst case behavior than both Rabin's and Ben-Or's algorithms. On average, it is expected that our Ben-Or's variant outperforms Ben-Or's algorithm since it has the same advantages for the lowest iterations and better behavior for the largest iterations. Section 6.2.3 shows experimental results comparing these four algorithms.

For the remainder of this section we concentrate on the probability that a polynomial is irreducible, if it has no irreducible factors of low degree. In Section 3.2, we mentioned that many (but not all) algorithms for factoring polynomials over finite fields comprise the following three stages: *squarefree factorization*, *distinct-degree factorization*, and *equal-degree factorization.* As things now stand, distinct-degree factorization is the bottleneck of the polynomial factorization problem (see Section 3.2.2). This step of the factorization

process works as follows: at any point k, all the irreducible factors of degree up to k have been found, and all the irreducible factors of degree greater than k remain to be determined from a certain factor g.

A natural way of improving the distinct-degree factorization step is by testing the irreducibility of the remaining factor g. Unfortunately, in the worst-case asymptotic scenario, the cost of testing for irreducibility is about the same as the cost of running a distinct-degree factorization algorithm. An alternative to overcome this problem is given in von zur Gathen & Gerhard (1996, §6). The central idea is to run the irreducibility test and the DDF algorithm in parallel, feeding the former with partial information obtained by the latter (see the details in von zur Gathen & Gerhard 1996). Therefore, the probability of a monic polynomial being irreducible when it has no irreducible factors of low degree provides useful information in the above process. In the following, we derive an asymptotic formula for this probability.

Theorem 6.2.8 *Let $P_q^I(n,m)$ be the probability that a polynomial of degree n over $\mathbb{F}_q$ be irreducible if it has no factors of degree less than or equal to m, for $1 \leq m \leq O(\log n)$. Then, as n, m and q approach infinity,*

$$P_q^I(n,m) \sim e^{\gamma} \frac{m}{n},$$

where γ is Euler's constant.

PROOF. This probability can be estimated considering the subset of irreducible polynomials of degree n over $\mathbb{F}_q$ inside the set of polynomials of degree n over $\mathbb{F}_q$ without irreducible factors of degree less than or equal to m, $1 \leq m \leq O(\log n)$. Using Eq. (6.1), Theorems 6.2.3 and 6.2.5, when n, m and q approach infinity, we obtain

$$P_q^I(n,m) = \frac{I_n}{q^n P_q(n,m)} \sim \frac{\frac{1}{n}}{\frac{e^{-\gamma}}{m}} = e^{\gamma} \frac{m}{n}.$$

□

6.2.3 Experimental results

In this section, we describe an implementation of the algorithms discussed in Section 6.2.1 and Section 6.2.2. We provide a running time comparison of the algorithms for pseu-

n	Rabin	Rabin's Variant	Ben-Or	Ben-Or's Variant	Numb Irr.
105	0.4222	0.2507	0.1079	0.1492	4
210	1.7507	1.4746	0.4746	0.5698	5
330	4.8919	5.5111	0.5909	0.6080	2
420	6.8269	5.4793	1.3412	1.4039	4
510	11.0085	10.6902	2.2830	2.3705	5
630	21.4921	16.4704	3.1587	3.2365	4
715	38.8364	43.8522	4.2694	4.7244	1
805	27.3420	38.1876	2.8294	3.1118	2
910	38.1161	43.9048	5.5062	4.6912	6
1020	48.9353	47.0265	4.4709	4.9800	2

Table 6.1: Average time in seconds for testing $3 * n$ polynomials over $\mathbb{F}_2$ of degree n with many prime divisors.

dorandom polynomials on a Sun Sparc 20 computer. The algorithms were implemented using basic operations from a C++ software package due to Shoup. This package contains classes for finite fields and polynomials over finite fields with implementations for basic operations such as multiplication, taking gcd, and so on (for a description of the software see Shoup 1996).

The results of our experiments are reported in Tables 6.2 through 6.5. We tested all algorithms with the same random polynomials over the fields $\mathbb{F}_2$, $\mathbb{F}_{1021}$, and $\mathbb{F}_p$, where $p = 1267650600228229401496703205361$ is a prime with 100 bits. In all cases, $3 * n$ polynomials of degree n were tested. In the case of $\mathbb{F}_2$, we ran our variant of Ben-Or's algorithm exactly as Ben-Or's algorithm during $c \log n$ iterations for $c = 20$.

Rabin's algorithm and both of our variants behave better when the degree n of the polynomial being tested for irreducibility has many divisors. Thus, for each field, we consider two tables depending on the number of distinct factors of the degree n. We

n	Rabin	Rabin's Variant	Ben-Or	Ben-Or's Variant	Numb Irr.
101	7.7986	7.8217	<u>1.3399</u>	1.7359	3
256	47.8763	35.3997	<u>4.9414</u>	5.5182	3
331	128.1350	88.9919	<u>10.5972</u>	12.2931	1
431	167.2980	86.1609	19.1129	<u>18.6009</u>	6
512	223.0000	193.8140	<u>16.3704</u>	16.9219	1

Table 6.2: Average time in seconds for testing $3 * n$ polynomials over $\mathbb{F}_2$ of degree n with few prime divisors.

distinguish both cases in the tables by the phrases "n with many prime divisors", and "n with few prime divisors". The worst-case scenario for all these algorithms happens when testing irreducible polynomials. We include a column with the number of irreducible polynomials that were tested for each degree. We underline the winner for every input.

Now, we summarize the results of this implementation. It is important to note that these results are not conclusive in the sense that another implementation may imply different conclusions. However, we do not expect dramatic variations in the comparison of the algorithms. It can be seen from the tables that Ben-Or's algorithm consistently gives the better results. However, our variant has, essentially, an equivalent behavior over $\mathbb{F}_2$. In addition, it has the possibility of a tune up via the constant c embedded in the algorithm. Thus, it is possible that over this field our variant will be more advantageous. More experimental results varying the parameter c in our variant are needed in order to have a more definitive conclusion. It is clear that Rabin's algorithm and our variant of this algorithm have the worst performance. However, our variant improves the standard Rabin method. Few or many factors of n do not seem to affect the consistency of the algorithms. These timings suggest that either Ben-Or's algorithm or our variant (over $\mathbb{F}_2$) has a much better average time behavior than others. Ben-Or's performance is good over any field, and on any input size n. In terms of worst-case cost, though, our variant

n	Rabin	Rabin's Variant	Ben-Or	Ben-Or's Variant	Numb Irr.
101	101.7920	101.2480	6.9438	15.8482	2
105	92.0317	45.5302	7.9587	14.1556	3
210	350.3300	189.5840	26.8778	74.6841	3
256	567.5790	445.7750	26.8099	81.6810	6
330	1089.9000	831.7970	52.7192	160.4000	4

Table 6.3: Average time in seconds for testing $3 * n$ polynomials of degree n over $\mathbb{F}_{1021}$.

n	Rabin	Rabin's Variant	Ben-Or	Ben-Or's Variant	Numb Irr.
101	1534.880	1570.620	132.726	291.604	3
105	774.054	613.267	137.565	312.377	5

Table 6.4: Average time in seconds for testing $3 * n$ polynomials of degree n over $\mathbb{F}_p$, where $p = 1267650600228229401496703205361$ is a prime with 100 bits.

behaves much better that Ben-Or's algorithm.

6.3 Construction of sparse irreducible polynomials

The problem of constructing sparse irreducible polynomials has been receiving some attention from a theoretical algebraic point of view (for instance, see Lidl & Niederreiter 1983, §3.3 and §3.5, and Menezes *et al.* 1993, Chapter 3). A well-known open problem is to provide a construction of irreducible polynomials over $\mathbb{F}_2$ of degree n with at most $O(\log n)$ nonzero terms in its lowest coefficients. These polynomials are useful in the discrete logarithm problem (Coppersmith 1984, Odlyzko 1985).

Another potential application of these very sparse polynomials is the faster computa-

tion of arithmetical operations in extension fields generated by these polynomials. Shoup (1994) points out that if $\mathbb{F}_{2^n} = \mathbb{F}_2[x]/(f)$ with $f = x^n + g$ irreducible and $g \in \mathbb{F}_2[x]$ of small degree, say $\deg g \leq 2\log n$, then exponentiation in $\mathbb{F}_{2^n}$ can be achieved with $O(n^2 \operatorname{loglog} n)$ operations in $\mathbb{F}_2$ and storage for $O(n/\log n)$ elements from $\mathbb{F}_{2^n}$. Experimental results show polynomials f with this characteristic exist for $n \leq 1000$ taking $\deg g \leq 2 + \log_2 n$. However, Gao, von zur Gathen & Panario (1995b) (see also Section 1.2.2) show that exponentiation in $\mathbb{F}_{2^n}$ can be done in $O(n^2 \operatorname{loglog} n)$ operations in $\mathbb{F}_2$ without requiring a presentation of $\mathbb{F}_{2^n}$ in terms of such sparse polynomials.

Construction of sparse polynomials is also important from a practical prospective. For instance, when testing the algorithms of Section 6.2, we needed irreducible polynomials of various degrees for estimating the worst case time of these algorithms (since an irreducible polynomial is a costly input for irreduciblity tests). Sparse irreducible polynomials have the advantage of being compactly represented.

Now, we focus our attention on the results regarding the construction of sparse polynomials. Shparlinski (1993, Theorem 1) proved that for any $N \in \mathbb{N}$ one can find an irreducible polynomial $f \in \mathbb{F}_p$ of degree n, where $n = N + O\left(N \exp\left(-(\log\log N)^{\frac{1}{2}-\epsilon}\right)\right)$. In fact, he gave a construction of irreducible polynomials with degrees of the form $4 \cdot 3^k \cdot 5^\ell$ over $\mathbb{F}_2$, $4 \cdot 2^k \cdot 5^\ell$ over $\mathbb{F}_3$, and $2 \cdot 2^k \cdot 3^\ell$ over $\mathbb{F}_p$, for any prime $p > 3$, and k, l nonnegative integers.

In this section, we give another step in solving this open problem. We provide, for any finite field, an explicit construction of irreducible polynomials of degree n with up to constant nonzero terms (not necessarily the lowest coefficients), for infinitely many degrees n. This construction can be viewed as a generalization of Shparlinski's result, although it was independently developed. We first established the following theorem (see Lidl & Niederreiter 1983, p. 124, Theorem 3.75).

Theorem 6.3.1 *Let $t \geq 2$ be an integer and consider $a \in \mathbb{F}_q^*$. Then, the binomial $x^t - a$ is irreducible in $\mathbb{F}_q[x]$ if and only if the following two conditions are satisfied:*

(i) each prime factor of t divides the order e of a in $\mathbb{F}_q^*$, but not $(q-1)/e$; and

(ii) $q \equiv 1 \mod 4$, if $t \equiv 0 \mod 4$.

We now state the main result of this section.

Theorem 6.3.2 *Let $p_1, \ldots, p_k$ be any fixed primes and $n = mp_1^{e_1} \ldots p_k^{e_k}$ where m is the multiplicative order of q modulo $p_1 \cdots p_k$, and $e_1, \ldots, e_k$ are arbitrary nonnegative integers. Then, over any finite field $\mathbb{F}_q$ whose characteristic is distinct from $p_1, \ldots, p_k$, there is an irreducible polynomial of degree n with at most $2m + 1 \leq 2(p_1 - 1) \cdots (p_k - 1) + 1$ nonzero terms.*

PROOF. If $p_1, \ldots, p_k$ are all odd, then the multiplicative order of q modulo $p_1 \cdots p_k$ is $m = (p_1 - 1) \ldots (p_k - 1)$, and let $\ell = m$. If one of p_i, say p_1, is 2 and $q \equiv 3 \bmod 4$, then let $\ell = \operatorname{lcm}(2, m)$. Then, $p_i | (q^\ell - 1)$ for $1 \leq i \leq \ell$, and $4 | (q^\ell - 1)$ if p_1 is even.

Let β be an element in $\mathbb{F}_{q^\ell}$ that is not a p_ith power in $\mathbb{F}_{q^\ell}$ for $1 \leq i \leq k$. The minimal polynomial $M(x)$ of β over $\mathbb{F}_q$ has degree ℓ. Then, by Theorem 6.3.1,

$$x^{p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}} - \beta$$

is irreducible in $\mathbb{F}_{q^\ell}$ for all nonnegative integers $e_1, \ldots, e_k$. Then, the evaluation of $M(x)$ in $x^{p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}}$

$$M\left(x^{p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}}\right) \tag{6.8}$$

is irreducible over $\mathbb{F}_q$ of degree $\ell p_1^{e_1} p_2^{e_2} \ldots p_k^{e_k} = n$. The polynomial in Eq. (6.8) has at most $\ell + 1 \leq 2m + 1$ nonzero terms. □

Irreducible polynomials from the construction of Theorem 6.3.2 are usually very sparse. In fact, the number of nonzero terms depends only on the prime factors of n, and if we fix them and let the exponents grow arbitrarily then these polynomials have only $O(1)$ nonzero terms. This can be seen from the following applications of the previous theorem.

Example 6.3.3 *Suppose $q \equiv 1 \bmod 4$ and $n = 2^k$. Take $a \in \mathbb{F}_q$ to be any quadratic nonresidue. Then, by Theorem 6.3.2, $x^{2^k} - a$ is irreducible over $\mathbb{F}_q$ for all $k \geq 0$. For instance,*

1. *if $q = p \equiv \pm 3 \bmod 8$ is a prime, then take $a = 2$;*

2. *if $q = p \equiv \pm 5 \bmod 12$ is a prime, then take $a = 3$;*

3. if $q = p \equiv \pm 2 \bmod 5$ is a prime, then take $a = 5$.

These results are from Ireland & Rosen (1990, Proposition 5.1.3 for the first two, and Theorem 2, p. 54, for the third).

Example 6.3.4 *Take $q = 2$. Theorem 6.3.2 yields the following families of irreducible polynomials over $\mathbb{F}_2$ for all $k, \ell, m, n \geq 0$:*

$$x^{2\cdot 3^k} + x^{3^k} + 1$$
$$x^{3\cdot 7^k} + x^{7^k} + 1$$
$$x^{4\cdot 3^k \cdot 5^\ell} + x^{3^k \cdot 5^\ell} + 1$$
$$x^{6\cdot 3^k \cdot 7^\ell} + x^{3^k \cdot 7^\ell} + 1$$
$$x^{10\cdot 3^k \cdot 11^\ell \cdot 31^m} + x^{3\cdot 3^k \cdot 11^\ell \cdot 31^m} + 1$$
$$x^{12\cdot 3^k \cdot 5^\ell \cdot 7^m \cdot 13^n} + x^{8\cdot 3^k \cdot 5^\ell \cdot 7^m \cdot 13^n} + x^{2\cdot 3^k \cdot 5^\ell \cdot 7^m \cdot 13^n} + x^{3^k \cdot 5^\ell \cdot 7^m \cdot 13^n} + 1.$$

For other explicit constructions of irreducible polynomials, see Menezes *et al.* (1993, Chapter 3), and Gao & Mullen (1994).

6.4 Lower bounds on the Euler totient function for polynomials

For a monic $f \in \mathbb{F}_q[x]$, define by $|f|$

$$|f| = \#\big(\mathbb{F}_q[x]/(f)\big) = q^{\deg f}.$$

Let $\Phi(f)$ denote the Euler totient function for polynomials, that is, the number of polynomials in $\mathbb{F}_q[x]$ of degree $< \deg f$ that are relatively prime to f. Then, similarly to the Euler function for integers, we have (Lidl & Niederreiter 1983, p. 122, Lemma 3.69)

$$\Phi(f) = |f| \prod_{g|f,\, g \text{ irr.}} (1 - |g|^{-1}), \tag{6.9}$$

where the product ranges over the monic irreducible factors g of f. Suppose that $f \in \mathbb{F}_q[x]$ has degree n, its distinct irreducible factors are $f_1, \ldots, f_r \in \mathbb{F}_q[x]$, and it decomposes into

$$f = f_1^{e_1} f_2^{e_2} \cdots f_r^{e_r}. \tag{6.10}$$

Let d_i denote the degree of f_i for $i = 1, \ldots, r$. Then Eq. (6.9) can be written as

$$\Phi(f) = q^n \prod_{i=1}^{r} \left(1 - \frac{1}{q^{d_i}}\right). \tag{6.11}$$

We focus on lower bounds for $\kappa(f) = \Phi(f)/q^n = \prod_{i=1}^{r}\left(1 - \frac{1}{q^{d_i}}\right)$, where $f \in \mathbb{F}_q[x]$ is monic of degree n. This kind of lower bound is useful for solving sparse linear equations (Wiedemann 1986), and for constructing normal elements over finite fields using randomized algorithms (von zur Gathen & Giesbrecht 1990, Giesbrecht 1993).

Several authors give lower bounds on $\kappa(f)$:

- Wiedemann (1986): $\kappa(f) \geq \frac{1}{6(1+\log_q n)}$.
- von zur Gathen & Giesbrecht (1990): $\kappa(f) > \frac{1}{16 \log_q n}$, and $\kappa(f) \geq \frac{1}{34}$ if $n \leq q^2$.
- Frandsen (1991): $\kappa(f) \geq (1 - \frac{1}{q})\frac{1}{e(1+\log_q n)}$, and $\kappa(f) \geq \frac{1}{2}$ if $q \geq 2n(n-1)$.
- Giesbrecht (1993): $\kappa(f) \geq \frac{1}{5 \log_q n}$ if $n > q$, and $\kappa(f) \geq \frac{1}{e}$ if $n \leq q$.

We slightly improve the lower bound on $\kappa(f)$ following the proof of Frandsen (1991).

Theorem 6.4.1 *For any $f \in \mathbb{F}_q[x]$ of degree n with $f(0) \neq 0$, if $n \geq q$ then*

$$\kappa(f) \geq \frac{1}{e^{\gamma + \frac{1}{2(1+\log_q n)}}(1 + \log_q n)} > \frac{1}{e^{0.83}(1 + \log_q n)},$$

where $\gamma = 0.577216\cdots$ is the Euler constant, and if $n < q$ then $\kappa(f) > 1/e$.

PROOF. We use the notation in Eqs. (6.10) and (6.11). For any positive integer m, $1 - 1/q^m \geq (1 - 1/q)^m$, when $q \geq 2$. Since $d_i \geq 1$ for $i = 1, \ldots, r$, and $\sum_{i=1}^{r} d_i \leq n$, we have

$$\kappa(f) = \prod_{i=1}^{r}\left(1 - \frac{1}{q^{d_i}}\right) \geq (1 - 1/q)^{\sum_{i=1}^{r} d_i} \geq (1 - 1/q)^n.$$

If $n < q$, Theorem 1.4.2 implies

$$(1 - 1/q)^n \geq (1 - 1/q)^{q-1} = ((1 + 1/(q-1))^{q-1})^{-1} \geq e^{-1}.$$

It remains to prove the case of $n \geq q$. For $1 \leq k \leq n$, let

$$\tau_k = \#\{i : d_i = k,\ 1 \leq i \leq r\}.$$

Then

$$\sum_{k=1}^{n} k\tau_k = n \qquad \text{and} \qquad \kappa(f) = \prod_{k=1}^{n}\left(1-\frac{1}{q^k}\right)^{\tau_k}.$$

For $k \geq 2$, let I_k be the number of monic irreducible polynomials in $\mathbb{F}_q[x]$ of degree k. Let I_1 be the number of irreducible factors of degree 1 different from x. Then for any $k \geq 1$,

$$\tau_k \leq I_k \leq \frac{q^k-1}{k}, \tag{6.12}$$

where the second inequality follows from Eq. (6.1). (Alternatively, the second inequality follows from the fact that each irreducible polynomial in $\mathbb{F}_q[x]$ of degree k divides $x^{q^k}-x$.) Define $u = \lfloor \log_q n \rfloor$. Then, the lower bound for I_k in Eq. (6.1) implies

$$\sum_{k=1}^{u+1} kI_k \geq q^{u+1} - 1 \geq q^{\log_q n} = n = \sum_{k=1}^{n} k\tau_k. \tag{6.13}$$

(Alternatively, the first inequality follows from the fact that the irreducible factors of $x^{q^{u+1}-1}-1$ are distinct, not equal to x, and have degrees $\leq u+1$.)

We claim that

$$\kappa(f) = \prod_{k=1}^{n}\left(1-\frac{1}{q^k}\right)^{\tau_k} \geq \prod_{k=1}^{u+1}\left(1-\frac{1}{q^k}\right)^{I_k}. \tag{6.14}$$

Indeed, it follows from Eqs. (6.12) and (6.13) that

$$(u+1)\sum_{k=1}^{u+1}(I_k-\tau_k) \geq \sum_{k=1}^{u+1} k(I_k-\tau_k) \geq \sum_{k=u+2}^{n} k\tau_k \geq (u+1)\sum_{k=u+2}^{n}\tau_k,$$

and hence

$$\sum_{k=1}^{u+1}(I_k-\tau_k) \geq \sum_{k=u+2}^{n}\tau_k. \tag{6.15}$$

We have

$$\prod_{k=1}^{u+1}\left(1-\frac{1}{q^k}\right)^{I_k}\left(\prod_{k=1}^{u+1}\left(1-\frac{1}{q^k}\right)^{\tau_k}\right)^{-1} = \prod_{k=1}^{u+1}\left(1-\frac{1}{q^k}\right)^{I_k-\tau_k} \leq \prod_{k=1}^{u+1}\left(1-\frac{1}{q^{u+1}}\right)^{I_k-\tau_k}.$$

Using Eq. (6.15),

$$\prod_{k=1}^{u+1}\left(1-\frac{1}{q^{u+1}}\right)^{I_k-\tau_k} \leq \left(1-\frac{1}{q^{u+1}}\right)^{\sum_{k=1}^{u+1}(I_k-\tau_k)} \leq \left(1-\frac{1}{q^{u+1}}\right)^{\sum_{k=u+2}^{n}\tau_k}.$$

Finally,

$$\left(1-\frac{1}{q^{u+1}}\right)^{\sum_{k=u+2}^{n}\tau_k} \leq \prod_{k=u+2}^{n}\left(1-\frac{1}{q^{u+1}}\right)^{\tau_k} \leq \prod_{k=u+2}^{n}\left(1-\frac{1}{q^k}\right)^{\tau_k}.$$

This proves Eq. (6.14). Thus, considering Eq. (6.12),

$$\kappa(f) = \prod_{k=1}^{n}\left(1-\frac{1}{q^k}\right)^{\tau_k} \geq \prod_{k=1}^{u+1}\left(1-\frac{1}{q^k}\right)^{I_k} \geq \prod_{k=1}^{u+1}\left(1-\frac{1}{q^k}\right)^{\frac{q^k-1}{k}}.$$

Using Theorem 1.4.2 and that $0 < 1 - 1/q^k < 1$, we have

$$\begin{aligned}\kappa(f) &\geq \prod_{k=1}^{m}\left(1-\frac{1}{q^k}\right)^{\frac{q^k-1}{k}} = \prod_{k=1}^{m}\left(\left(1+\frac{1}{q^k-1}\right)^{q^k-1}\right)^{-\frac{1}{k}} \\ &\geq \prod_{k=1}^{m}\exp\left(-\frac{1}{k}\right) = \exp\left(-\sum_{k=1}^{m}\frac{1}{k}\right).\end{aligned}$$

For the approximation of $H_m = \sum_{k=1}^{m} 1/k$, we can use Theorem 1.4.1,

$$H_m = \ln m + \gamma + \frac{1}{2m} - \frac{1}{12m^2} + O(m^{-4}),$$

where $\gamma = 0.577216\cdots$ is the Euler constant. We finally obtain

$$\kappa(f) \geq \left(\frac{1}{e}\right)^{\ln(1+u)+\gamma+\frac{1}{2(1+u)}} \geq \frac{1}{e^{\gamma+\frac{1}{2(1+\log_q n)}}(1+\log_q n)} > \frac{1}{e^{0.83}(1+\log_q n)},$$

since when $n \geq q$, $\gamma + \frac{1}{2(1+\log_q n)} \leq \gamma + 1/4 < 0.83$. This completes the proof. □

We should mention that the inequality in (6.14) is not stated explicitly in Frandsen (1991), though it is used implicitly by intuitive arguments. Our contribution is in the rigorous proof of the inequality in (6.14), as well as in the improvement in the lower bound.

6.5 Density of normal elements

The most important result about density of normal elements is given in Ore (1934, Theorem 12). Below, we state this result as in Lidl & Niederreiter (1983), p. 124, Theorem 3.73 (see also Menezes *et al.* 1993, Chap. 4).

Theorem 6.5.1 *In $\mathbb{F}_{q^n}$ there exists exactly $\Phi(x^n-1)$ elements ζ such that $(\zeta, \zeta^q, \zeta^{q^2}, \ldots, \zeta^{q^{n-1}})$ is a basis of $\mathbb{F}_{q^n}$ over $\mathbb{F}_q$.*

Theorem 6.5.1 proves that the number of normal elements in $\mathbb{F}_{q^n}$ over $\mathbb{F}_q$ is $\Phi(x^n-1)$. The density of normal elements in $\mathbb{F}_{q^n}$ over $\mathbb{F}_q$ is then $\kappa(x^n-1) = \Phi(x^n-1)/q^n$.

Theorem 6.4.1 gives a lower bound on the density of normal elements. The question is: can we prove better lower bounds for the special polynomial $x^n - 1$? In this section, we give both positive and negative answers to this question. On the one hand, we show that, for infinitely many values of n, $\kappa(x^n - 1) \geq C > 0$, for some constant C. On the other hand, for infinitely many n, we show an upper bound on $\kappa(x^n - 1)$ that goes to 0 as n approaches infinity. This upper bound is almost tight with the lower bound in Theorem 6.4.1. First we state some useful lemmas.

Lemma 6.5.2 *Let $\tau(d)$ denote the order of q modulo d, and let $\phi(d)$ be the Euler totient function for integers. Then,*

$$\kappa(x^n - 1) = \prod_{d|n} \left(1 - q^{-\tau(d)}\right)^{\frac{\phi(d)}{\tau(d)}}. \tag{6.16}$$

PROOF. See von zur Gathen & Giesbrecht (1990, p. 552). □

Lemma 6.5.3 *For any integer $m \geq 2$ and real number $0 < a \leq 1/2$,*

$$\prod_{k=1}^{\infty} (1 - a^{m^k}) \geq 1 - a^{m-1} \geq 1 - a. \tag{6.17}$$

PROOF. $\prod_{k=1}^{\infty}(1 - a^{m^k}) \geq 1 - \sum_{k=1}^{\infty} a^{m^k} \geq 1 - \sum_{k=0}^{\infty} a^{m+k} \geq 1 - \frac{a^m}{1-a} \geq 1 - a^{m-1}$. □

6.5.1 Case of prime powers

We first study the behavior of $\kappa(x^n - 1)$ when n is the power of a prime. This result motivated our work.

Theorem 6.5.4 *Let p be an odd prime factor of $q - 1$, and let e be its exponent in the prime decomposition of $q - 1$, i.e., $q - 1 = p^e Q$ and $p \nmid Q$, for some Q. Let $n = p^m$, for arbitrarily large m. Then*

$$\kappa(x^n - 1) \geq \begin{cases} \left(1 - \frac{1}{q}\right)^{p^{e-1}+p^{e-1}(p-1)} & \text{if } e < m, \\ \left(1 - \frac{1}{q}\right)^{p^e} & \text{if } e \geq m, \end{cases}$$

implying $\kappa(x^n - 1) > 0$.

PROOF. The divisors of $n = p^m$ are p^k for $1 \leq k \leq m$. Since $q = 1 + p^e Q$ with $p \nmid Q$, the order of q modulo p^k satisfies

$$\tau(p^k) = \begin{cases} 1 & \text{if } k \leq e, \\ p^{k-e} & \text{if } k > e. \end{cases}$$

When $e < m$, Lemmas 6.5.2 and 6.5.3 imply that

$$\begin{aligned} \kappa(x^n - 1) &= \prod_{k=1}^{e} \left(1 - \frac{1}{q}\right)^{p^{k-1}(p-1)} \prod_{k=e+1}^{m} \left(1 - \frac{1}{q^{p^{k-e}}}\right)^{\frac{p^{k-1}(p-1)}{p^{k-e}}} \\ &= \left(1 - \frac{1}{q}\right)^{(p-1)\sum_{k=1}^{e} p^{k-1}} \prod_{k=1}^{m-e} \left(1 - \frac{1}{q^{p^k}}\right)^{p^{e-1}(p-1)} \\ &\geq \left(1 - \frac{1}{q}\right)^{p^e - 1} \left(\prod_{k=1}^{\infty} \left(1 - \frac{1}{q^{p^k}}\right)\right)^{p^{e-1}(p-1)} \\ &\geq \left(1 - \frac{1}{q}\right)^{p^e - 1} \left(1 - \frac{1}{q}\right)^{p^{e-1}(p-1)} = C > 0. \end{aligned}$$

When $m \leq e$,

$$\kappa(x^n - 1) = \left(1 - q^{-1}\right)^{p^m} \geq \left(1 - q^{-1}\right)^{p^e} > 0.$$

This completes the proof. □

6.5.2 Case of fixed prime factors

Theorem 6.5.4 provides a positive lower bound for $\kappa(x^n - 1)$ in the particular case when n contains only one prime factor. In this section, we generalize that result proving that $\kappa(x^n - 1)$ is bounded away from zero when n has fixed prime factors.

Theorem 6.5.5 *Let $n = p_1^{e_1} p_2^{e_2} \cdots p_t^{e_t}$, where $p_1, p_2, \ldots, p_t$ are fixed distinct primes, and $e_1, e_2, \ldots, e_t \geq 0$ are arbitrary integers. Then $\kappa(x^n - 1) \geq C > 0$, where C is a constant independent of $e_1, e_2, \ldots, e_t$.*

PROOF. Recall $\tau(p_i)$ is the order of q modulo p_i for $1 \leq i \leq t$. Therefore, $p_i \mid \left(q^{\tau(p_i)} - 1\right)$. Let v_i and Q_i be integer numbers such that $q^{\tau(p_i)} - 1 = p_i^{v_i} Q_i$ and $p_i \nmid Q_i$, for $1 \leq i \leq t$. Then

$$\tau(p_i^\ell) = \begin{cases} \tau(p_i) & \text{if } \ell \leq v_i, \\ p_i^{\ell - v_i} \tau(p_i) & \text{if } \ell > v_i. \end{cases}$$

This means that $\tau(p_i^\ell) = p_i^{\max\{\ell - v_i, 0\}} \tau(p_i)$.

An arbitrary divisor d of n is of the form $d = p_1^{d_1} p_2^{d_2} \cdots p_t^{d_t}$, with $0 \le d_i \le e_i$ for $1 \le i \le t$. Clearly, $\tau(d) = \operatorname{lcm}(\tau(p_1^{d_1}), \tau(p_2^{d_2}), \ldots, \tau(p_t^{d_t}))$, and is divisible by $\prod_{i=1}^{t} p_i^{\max\{d_i - v_i, 0\}}$. Thus

$$\tau(d) \ge \prod_{i=1}^{t} p_i^{\max\{d_i - v_i, 0\}}. \tag{6.18}$$

By Eq. (6.16), we have

$$\kappa(x^n - 1) = \prod_{\substack{d = p_1^{d_1} \cdots p_t^{d_t} \\ 0 \le d_i \le e_i,\ 1 \le i \le t}} \left(1 - q^{-\tau(d)}\right)^{\frac{\phi(d)}{\tau(d)}} \ge \prod_{\substack{d = p_1^{d_1} \cdots p_t^{d_t} \\ 0 \le d_i \le \infty,\ 1 \le i \le t}} \left(1 - q^{-\tau(d)}\right)^{\frac{\phi(d)}{\tau(d)}}.$$

Recalling that ϕ is multiplicative, $\phi(d) = \phi(p_1^{d_1}) \cdots \phi(p_t^{d_t}) = \prod_{i=1}^{t} p_i^{d_i - 1}(p_i - 1)$, and using Eq. (6.18), we obtain

$$\frac{\phi(d)}{\tau(d)} \le \prod_{i=1}^{t} p_i^{v_i - 1}(p_i - 1) = C_1,$$

with C_1 a constant. Then

$$\kappa(x^n - 1) \ge \left(\prod_{\substack{d = p_1^{d_1} \cdots p_t^{d_t} \\ 0 \le d_i \le \infty,\ 1 \le i \le t}} \left(1 - q^{-\tau(d)}\right) \right)^{C_1}.$$

By Eq. (6.18) and Lemma 6.5.3,

$$\prod_{\substack{d = p_1^{d_1} \cdots p_t^{d_t} \\ 0 \le d_i \le \infty,\ 1 \le i \le t}} \left(1 - q^{-\tau(d)}\right) \ge \prod_{0 \le d_i \le \infty,\ 1 \le i \le t} \left(1 - q^{-\prod_{i=1}^{t} p_i^{\max\{d_i - v_i, 0\}}}\right).$$

The last expression can be rewritten as

$$\begin{aligned}
&\prod_{\substack{0 \le d_t \le v_t \\ 0 \le d_i \le \infty,\ 1 \le i \le t-1}} \left(1 - q^{-\prod_{i=1}^{t-1} p_i^{\max\{d_i - v_i, 0\}}}\right) \prod_{\substack{d_t > v_t \\ 0 \le d_i \le \infty,\ 1 \le i \le t-1}} \left(1 - q^{-\prod_{i=1}^{t-1} p_i^{\max\{d_i - v_i, 0\}} p_t^{d_t - v_t}}\right) \\
\ge\ &\prod_{0 \le d_i \le \infty,\ 1 \le i \le t-1} \left(1 - q^{-\prod_{i=1}^{t-1} p_i^{\max\{d_i - v_i, 0\}}}\right)^{v_t + 1} \prod_{0 \le d_i \le \infty,\ 1 \le i \le t-1} \left(1 - q^{-\prod_{i=1}^{t-1} p_i^{\max\{d_i - v_i, 0\}}}\right) \\
=\ &\prod_{0 \le d_i \le \infty,\ 1 \le i \le t-1} \left(1 - q^{-\prod_{i=1}^{t-1} p_i^{\max\{d_i - v_i, 0\}}}\right)^{v_t + 2}.
\end{aligned}$$

Repeating this process successively for $d_{t-1}, d_{t-2}, \ldots, d_1$, we obtain

$$\kappa(x^n - 1) \ge \left(1 - q^{-1}\right)^{C_1 (v_1 + 2) \cdots (v_t + 2)} = \left(1 - q^{-1}\right)^{\prod_{i=1}^{t} p_i^{v_i - 1}(p_i - 1)(v_i + 2)} = C > 0$$

This completes the proof. □

The estimates in the above proof are very rough. One can use finer estimates to get a bigger constant C. However, we do not intend to do so in this thesis. In addition, we note that, by Theorem 6.5.5, there exists an infinite tower of finite fields whose densities of normal elements are all larger than a constant. Thus, to find a normal element in any of these fields requires only a fixed number of samplings.

6.5.3 Upper bounds

The results in Theorem 6.5.5 may suggest that $\kappa(x^n - 1) \geq C > 0$ for all n for some constant C. In this section, we show that this is not true. Indeed, we show that the lower bound on the number of normal elements given by Theorem 6.4.1 is almost tight for infinitely many n.

By the proof of Theorem 6.4.1, $\kappa(f)$ is small if f contains all the linear factors, quadratic factors, and so on, up to all the irreducible polynomials of degree m. For such f, Giesbrecht (1993, Theorem 3.13) proves that

$$\Phi(f)/|f| \leq \frac{1}{e^{H_m}} \left(1 + \frac{1}{\sqrt{q}}\right)^{q/(q-1)}. \tag{6.19}$$

Thus $\kappa(f)$ approaches 0 as m goes to infinity. In the following theorem, we prove that a result similar to Eq. (6.19) holds even when f is restricted to polynomials of the type $x^n - 1$. We state the upper bound explicitly in terms of the degree n.

Theorem 6.5.6 *For any integer $m > 1$, let $n = (q-1)(q^2-1)\cdots(q^m-1)$. Let $c_q = \frac{q}{(q-1)(\sqrt{q}-1)}$. Then*

$$\frac{1}{e^{0.83}(1 + \log_q n)} \leq \kappa(x^n - 1) \leq \frac{1}{e^{\gamma - c_q}\sqrt{1 + \log_q n}}.$$

PROOF. The lower bound is in Theorem 6.4.1.

For a divisor d of n, Theorem 1.4.2 guarantees that

$$e \leq \left(1 - q^{-\tau(d)}\right)^{-q^{\tau(d)}}.$$

From

$$\left(1-q^{-\tau(d)}\right)^{\frac{\phi(d)}{\tau(d)}} = \left(\left(\left(1-q^{-\tau(d)}\right)^{-q^{\tau(d)}}\right)^{\frac{\phi(d)}{q^{\tau(d)}\tau(d)}}\right)^{-1},$$

we derive

$$\left(1-q^{-\tau(d)}\right)^{\frac{\phi(d)}{\tau(d)}} \leq \left(\frac{1}{e}\right)^{\frac{\phi(d)}{q^{\tau(d)}\tau(d)}}.$$

Therefore, considering Eq. (6.16), we obtain

$$\kappa(x^n-1) \leq \left(\frac{1}{e}\right)^{\sum_{d|n}\frac{\phi(d)}{q^{\tau(d)}\tau(d)}}. \tag{6.20}$$

We develop a lower bound for the above exponent. First note that

$$\sum_{d|n}\frac{\phi(d)}{q^{\tau(d)}\tau(d)} \geq \sum_{k=1}^{m}\sum_{\substack{d|q^k-1\\ \tau(d)=k}}\frac{\phi(d)}{k\,q^k},$$

since the divisors d in the second sum runs only over part of the divisors of n. For fixed k, the number of elements in $\mathbb{F}_{q^k}$ that have degree exactly k over $\mathbb{F}_q$ is

$$\sum_{\substack{d|q^k-1\\ \tau(d)=k}}\phi(d) = kI_k.$$

Therefore,

$$\sum_{d|n}\frac{\phi(d)}{q^{\tau(d)}\tau(d)} \geq \sum_{k=1}^{m}\sum_{\substack{d|q^k-1\\ \tau(d)=k}}\frac{\phi(d)}{k\,q^k} = \sum_{k=1}^{m}\frac{I_k}{q^k}.$$

By Theorem 1.3.2,

$$I_k \geq \frac{q^k}{k} - \frac{q(q^{k/2}-1)}{k(q-1)}.$$

Thus, from Theorem 1.4.1,

$$\begin{aligned}\sum_{d|n}\frac{\phi(d)}{q^{\tau(d)}\tau(d)} &\geq \sum_{k=1}^{m}\frac{I_k}{q^k} \geq \sum_{k=1}^{m}\frac{1}{k} - \frac{q}{q-1}\sum_{k=1}^{m}\frac{q^{k/2}-1}{kq^k}\\ &\geq H_m - \frac{q}{q-1}\sum_{k=1}^{m}\frac{1}{kq^{k/2}} \geq H_m - \frac{q}{q-1}\sum_{k=1}^{\infty}\left(\frac{1}{\sqrt{q}}\right)^k\\ &= H_m - \frac{q}{(q-1)(\sqrt{q}-1)} \geq \ln m + \gamma - c_q.\end{aligned}$$

Since $n = (q-1)(q^2-1)\cdots(q^m-1)$, we have

$$\log_q n = \sum_{i=1}^{m}\log_q(q^i-1) \leq \sum_{i=1}^{m} i \leq m^2-1.$$

Then,

$$\ln m \geq \frac{1}{2}\ln(1+\log_q n).$$

Therefore,

$$\sum_{d|n}\frac{\phi(d)}{q^{\tau(d)}\tau(d)} \geq \ln m + \gamma - c_q \geq \frac{1}{2}\ln(1+\log_q n) + \gamma - c_q.$$

Finally, using Eq. (6.20)

$$\kappa(x^n-1) \leq \left(\frac{1}{e}\right)^{\sum_{d|n}\frac{\phi(d)}{q^{\tau(d)}\tau(d)}} \leq \left(\frac{1}{e}\right)^{\frac{1}{2}\ln(1+\log_q n)+\gamma-c_q} = \frac{1}{e^{\gamma-c_q}\sqrt{1+\log_q n}}.$$

This completes the proof. □

Chapter 7

Conclusion and Further Research

In this thesis, we systematically apply a combinatorial methodology based on generating functions and asymptotic analysis to several problems that deal with univariate polynomials over finite fields. We conclude this thesis with a list of problems for further research. This list includes possible improvements to the problems studied in this thesis, as well as other problems for which this framework may be useful.

1. Analysis of factorization algorithms of the 1990's

Chapter 5 presented a detailed analysis of the distribution of irreducible factors of a random polynomial in intervals associated with their degrees. We motivated that chapter explaining the relation of this analysis with recent algorithms for factoring polynomials (von zur Gathen & Shoup 1992, Kaltofen & Shoup 1995, Shoup 1996, von zur Gathen & Gerhard 1996). It seems possible to combine the use of the techniques of Chapter 4 and results in the line of Chapter 5 in order to analyze these algorithms. This approach would introduce several other parameters to be studied. In addition, we would have to consider fast arithmetic, as all these algorithms do, instead of the classical one.

2. Average case analysis of irreducibility tests

In Chapter 6, we studied polynomials of degree n with no factors of degree smaller than $O(\log n)$. We mentioned that this could be extended up to $O(\frac{n}{\log^2 n})$ using the saddle point bound. We do not know how to extend this analysis for all the interval up to $\frac{n}{2}$. This information would make possible the complete analysis of Ben-Or's algorithm. Gourdon (1996) studies the largest degree in irreducible factors of a random polynomial.

It may be possible to adapt, even though not trivially, his technique to the study of the smallest degree, which would be sufficient to fully analyze Ben-Or's algorithm. This approach would provide not only the smallest degree, but also the distribution of the rth smallest degree for any positive integer r. We mention that this study could be done in the more general setting of random decomposable structures.

The average-case analysis of Rabin's algorithm is also an interesting open problem. It requires the study of irreducible factors of a random polynomial of degree n with degrees dividing n. This is related to classical number-theoretical functions.

3. Other problems

Finding roots of polynomials over finite fields has received some attention in the past (Berlekamp 1970, Rabin 1980, Ben-Or 1981, van Oorschot & Vanstone 1989). It has applications in cryptography (Chor & Rivest 1988, Lenstra 1991). This problem seems approachable by a treatment similar to the one in Chapter 4.

In Chapter 4, we commented on the index calculus method for computing discrete logarithms over extension fields and on its relation to counting smooth polynomials (Odlyzko 1985, 1994). Other algorithms for computing discrete logarithms in $\mathbb{F}_{2^n}$ are presented in Blake *et al.* (1984), and Coppersmith (1984). The rigorous proof of the cost of these algorithms rely on unproven assumptions that involve counting pairs of smooth polynomials with no common factors. These problems seems to be tractable with an analysis of multivariate generating functions.

In Chapter 2, we mentioned the study of polynomials with preassigned coefficients, a topic we did not cover in this thesis. In this case, we prefix up to $n/2$ coefficients of a polynomial of degree n, and we take the other coefficients at random. Following the papers by Cohen (1972) and others, we could do asymptotics on the size q of the field, and study properties such as smoothness, roughness, and so on. These involves similar techniques to the ones presented in this thesis applied to the permutational model.

In Chapter 6, we mentioned the randomized construction of normal elements given by von zur Gathen & Giesbrecht (1990). The study of the average-case behavior of this algorithm seems to be possible by using the same techniques presented in this thesis.

Flajolet & Golin (1994) study the mergesort recurrence in detail. The technique used

involves Mellin transform, a method that we did not apply in this thesis. There is a clear relation, given by the divide-and-conquer process, between mergesort and Karatsuba-Offman algorithm. A study of the latter algorithm could shed some light on the comparison among polynomial multiplication algorithms. We note that this analysis may possibly extend to Strassen's matrix multiplication algorithm.

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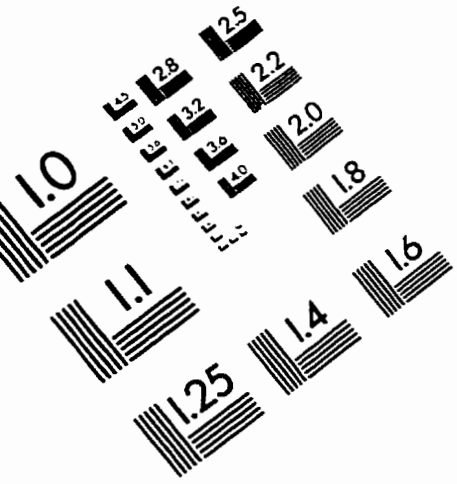
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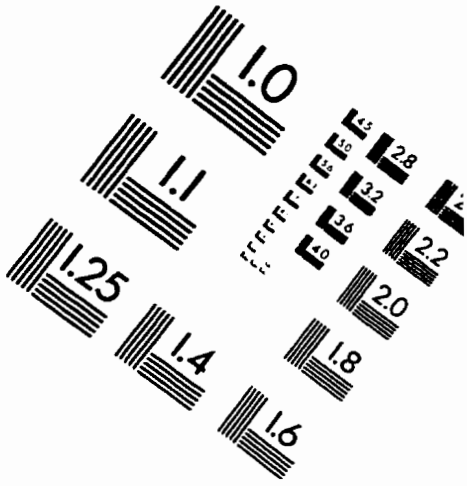
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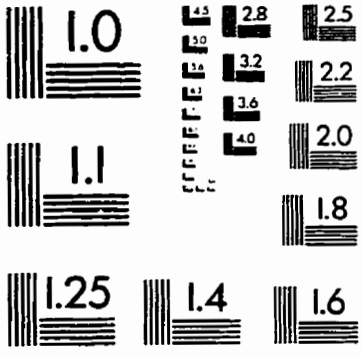
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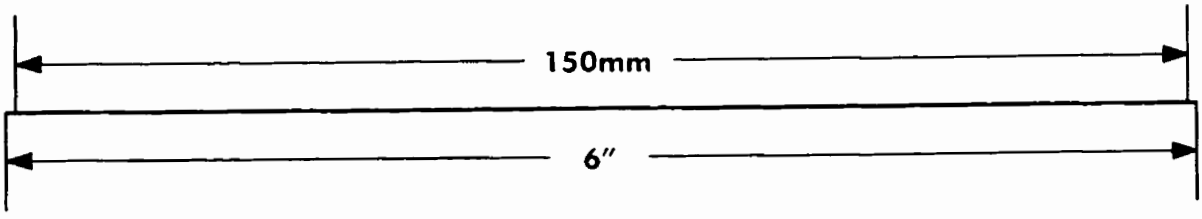
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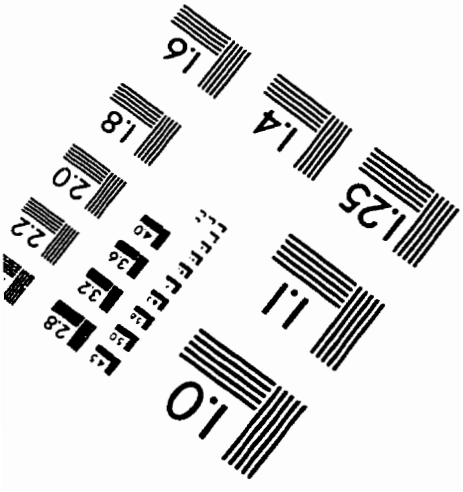
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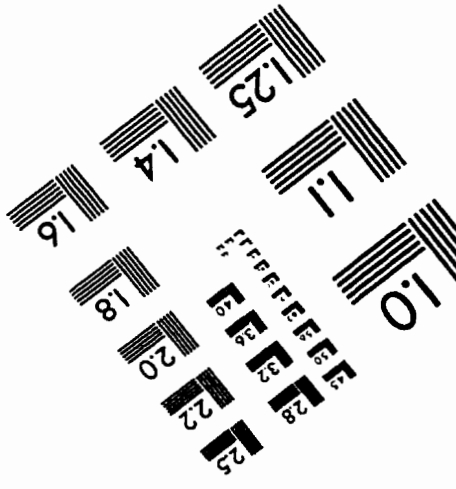












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